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Project Management

Project Documentation

OliveTrack

Requirements and Architecture

Olive Harvest Management Platform

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Contents

1 Introduction **1**

1.1 Purpose of the Document 1

1.2 Project Context 1

1.3 Objectives 1

1.4 Sources and Document Status 2

2 Users, Personas and Permissions **3**

2.1 Stakeholders 3

2.2 User Personas 3

2.3 System Actors 4

2.4 Access Permissions 5

3 Scope and Feature Overview **7**

3.1 Included Scope 7

3.2 Excluded Scope 7

3.3 MVP Proposal Pending Approval 8

3.4 Feature Overview 9

3.5 Assumptions and Dependencies 9

4 Use Cases **11**

4.1 Scope and Review Status 11

4.2 Shared Definitions 11

4.3 Use Case Catalogue 12

4.4 UC-01 — Register as an Owner 13

4.5 UC-02 — Log In and Log Out 14

4.6 UC-03 — Recover Account Access 16

4.7 UC-04 — Manage Manager Access and Permissions 17

4.8 UC-05 — Manage Properties and Parcels 19

4.9 UC-06 — Manage Harvest Campaigns 22

4.10 UC-07 — Manage Worker Records 26

4.11 UC-08 — Plan Work 29

4.12 UC-09 — Consult Work Schedules 32

4.13 UC-10 — Record Attendance and Actual Work Periods 35

4.14 UC-11 — Correct Work Records 39

4.15 UC-12 — Consult Worker Documents 43

4.16 UC-13 — Upload and Update Worker Documents 46

4.17 UC-14 — Review and Sign a Document 50
4.18 UC-15 — Record Transport and Olive Deliveries 53
4.19 UC-16 — Record Weighing Data and Adjustments 56
4.20 UC-17 — Record Analysis Results 60
4.21 UC-18 — Manage Cooperative-Reported Ticket Values 63
4.22 UC-19 — Consult Campaign Indicators 67
4.23 UC-20 — Generate Worker Documents 71

5 Functional Requirements **73**
5.1 Requirements Catalogue 73
5.2 Mapping to Use Cases 92

6 Business Rules and Record States **99**
6.1 Business Rules 99
6.2 Record States and Transitions 108
6.3 Corrections and Change History 117
6.4 Calculation Rules 123

7 Non-Functional Requirements **133**
7.1 Performance 133
7.2 Usability and Accessibility 133
7.3 Security 133
7.4 Privacy 133
7.5 Reliability and Recovery 133
7.6 Availability and Offline Continuity 133

8 Solution Architecture **134**
8.1 System Context 134
8.2 Components and Responsibilities 134
8.3 Main Interaction Flows 134
8.4 Offline Operation and Synchronisation 134
8.5 Deployment and Operation 134
8.6 Architecture Decisions 134

9 Data Architecture **135**
9.1 Conceptual Data Model 135
9.2 Logical Data Model 135
9.3 Data Dictionary 135

9.4 Integrity Constraints 135
9.5 Data Lifecycle and History 135
9.6 Local and Central Data 135
9.7 Physical Data Model 135

10 Validation, Decisions and Open Questions **136**
10.1 Acceptance Criteria 136
10.2 Requirements Traceability 136
10.3 Decision Log 136
10.4 Open Questions 136

1 Introduction

1.1 Purpose of the Document

This document defines the proposed requirements and architecture of OliveTrack, a platform for managing olive harvest operations.

It establishes a shared understanding of the problem, intended users, functional scope, business rules and expected system behaviour. It also connects the use cases to the functional requirements, quality requirements and architectural decisions developed by the team.

The document supports both product development and project management. An explicit description of the expected behaviour helps the team identify missing work, define implementation and validation tasks, allocate responsibilities and prepare effort estimates.

This document complements the Statement of Work and the other artefacts required by the Project Management course. It does not independently amend the agreed project boundaries or replace the project's work plan, task assignments or acceptance records.

1.2 Project Context

Olive harvest management involves information that is produced by different people and becomes available at different times.

Before fieldwork, the owner needs to decide which workers are expected, where they will work and, where known, their schedules and tasks. During or after the working day, actual attendance, work periods, tasks and breaks must be recorded.

Harvested olives are transported and delivered to a cooperative or another receiving destination. Weighing tickets provide information about the delivered load. Analysis results and monetary values may arrive later.

Worker documents, including contracts and insurance records, must also be accessible. Some documents require a signing interaction. Insurance information is particularly important when work takes place in the field.

When this information is distributed across paper records and separate files, it becomes harder to retrieve, reconcile and summarise the campaign's results. Differences between planned and actual work, missing analysis results and ambiguous weight values can also lead to incorrect conclusions.

OliveTrack proposes a shared digital environment for the owner's business. It supports properties with optional parcels, harvest campaigns covering one or more areas, worker records, operational entry, document consultation and campaign indicators.

Field connectivity may be unreliable. Essential field operations therefore require prepared offline access and local entry, followed by synchronisation when connectivity returns. Administrative operations may require connectivity, as specified in the individual use cases.

The project is being developed within the Project Management course. The proposed scope must remain compatible with the academic schedule, available resources and agreed delivery objectives.

1.3 Objectives

The proposed solution pursues the following objectives:

1. **Organise the harvest context.** Maintain identifiable properties, optional parcels and

campaigns without mixing operational records from different areas or harvest periods.

2. **Separate planning from actual work.** Support daily and multi-day planning, while requiring explicit confirmation of attendance and actual work.
3. **Record work consistently.** Distinguish machinery operation from general work, account for explicitly recorded breaks and prevent overlapping actual periods for the same worker.
4. **Track deliveries and source information.** Connect transport, delivery, weighing, analysis and monetary information without inventing missing measurements or applying deductions twice.
5. **Support worker-document workflows.** Provide controlled access to uploaded documents, prepared offline insurance consultation and the agreed signing interaction.
6. **Provide understandable indicators.** Present campaign results with clear units, reporting scope, data completeness and provisional or final status where applicable.
7. **Support work in the field.** Allow essential operations to continue with prepared authorised data when a network connection is unavailable.
8. **Support project planning and validation.** Provide a traceable basis for requirements, implementation tasks, estimates, acceptance checks and team review.

Detailed measurable targets are addressed in the non-functional requirements. The objectives above do not constitute verified performance or operational results.

1.4 Sources and Document Status

The current document draws on:

- The *OliveTrack Team Guide*, dated 9 October 2026, describing the business context, illustrative examples and unresolved questions.
- The existing Statement of Work, persona descriptions, general architecture, data architecture and non-functional requirements proposals supplied to the team.
- Initial responses supplied by the teammate acting as an intermediary with the client.
- The subsequent requirements review, including the use-case discussions and proposed resolutions recorded during document development.

These sources do not all have the same approval status. An illustrative example, team preference or design assumption must not be presented as a client-approved business rule.

The document is currently a working draft. The use cases consolidate the team's current understanding and proposals. They remain subject to team review and client validation where necessary.

The pending decision register in Chapter 10 identifies the unresolved scope, business and implementation questions and the proposed roles needed for validation.

Conflicting source statements are resolved through an explicit decision, rather than silently selecting one interpretation.

Use cases and rules cite the decision IDs; complete question wording, status and proposed validation responsibilities are maintained in Chapter 10.

2 Users, Personas and Permissions

2.1 Stakeholders

OliveTrack affects people who use the application, people whose information is managed and parties that supply information or assess the project.

Farm Owner and Client. Explains the operational needs, validates business behaviour and uses the application to manage the harvest and inspect its results.

Field Manager. A potential delegated operational user. Its inclusion remains to be confirmed. If included, the manager acts within the permissions granted by the owner.

Workers. Their profiles, attendance, work periods and documents are managed. Workers may participate in signing without holding application accounts.

Cooperative or Receiving Organisation. Supplies weighing information, analysis results and monetary values. Direct access to OliveTrack is not assumed in the current scope.

Project Team. Analyses requirements, proposes the solution, implements and validates the agreed scope, and maintains the project documentation.

Course Faculty and Evaluators. Establish academic expectations and assess the project deliverables, management process and demonstrated results.

Business validation and academic acceptance are related but distinct. Client agreement does not replace the course's assessment requirements.

2.2 User Personas

The profiles below describe working assumptions for requirements discussion. They represent roles and usage contexts; they are not evidence of a completed user study.

2.2.1 Farm Owner

Role and context. Responsible for the business's properties, workers and harvest decisions. May prepare administrative information with connectivity and consult or record operational information in the field.

Main goals.

- Organise properties, parcels and campaigns.
- Prepare work without repeatedly entering the same information.
- Understand who attended and what work was actually performed.
- Access the correct worker documents.
- Follow deliveries and later results.
- Review campaign totals and investigate their supporting records.

Main difficulties. Information arrives at different times, manual records can be fragmented, and field connectivity is unreliable.

Design implications. Use clear business terminology, group entry with individual adjustments, explicit save states and understandable summaries with access to detail.

2.2.2 Field Manager

Status. Conditional persona, pending confirmation of the need for a separate application role.

Role and context. May coordinate workers and field operations on behalf of the owner. Uses the application within an explicitly authorised operational scope.

Main goals.

- Consult prepared schedules.
- Record attendance and actual work.
- Record transport and delivery information.
- Access relevant worker insurance during field operations.

Access limitations. Work-entry permission does not automatically include correction rights, contract access, monetary information or configuration rights.

Design implications. Prioritise essential field tasks, prepared offline information and clear permission boundaries.

2.2.3 Worker as a Participant

Role and context. A person whose work and documents are managed. May review and sign a document in person on the owner's or authorised manager's device.

Main needs.

- Be correctly identified in records.
- Review the document intended for signing.
- Review, redraw or cancel before confirming a signature.
- Avoid unintended access to other workers' information.

No worker self-service application or management account is assumed.

2.3 System Actors

Farm Owner is the formal actor name; Owner is its shortened form in the use cases, requirements and business rules.

Farm Owner. Primary application actor. Performs authorised management, operational and consultation tasks within the owner's business.

Field Manager. Conditional application actor with delegated permissions.

Worker. Participating actor in the signing interaction. Does not manage attendance or business information through an account.

The cooperative is currently an information source, rather than a direct application actor. Likewise, identifying a worker as a driver does not create another application role.

The owner and manager are user roles. A worker record is a business record. These concepts must not be treated as interchangeable identities.

Visitor. A Visitor is a person who is not authenticated. Visitors may register as Owners, verify their email, log in or initiate account recovery. They cannot access business information.

Authenticated account holders. An Owner holds an application account associated with the business created through registration.

If the Field Manager role is approved, a manager receives business access through an Owner invitation and acts within the assigned scope and permissions.

Worker records are operational business records. Creating a worker record does not create an account. Workers review and sign documents through an authorised account holder's device without logging in themselves.

2.4 Access Permissions

Permissions are evaluated within the authorised business context and, where applicable, the relevant properties, campaigns, workers and documents.

The same person may have independent worker records in different businesses. Those records are not automatically merged or exposed to another owner.

The following matrix summarises the proposed access principles.

Consultation, creation, modification and correction permissions are distinct. A permitted interface action must also be authorised by the server.

Offline access is limited to previously prepared authorised information. Changes to permissions must be rechecked when the device reconnects. The detailed access-expiry and revocation policy remains to be defined.

The complete permission matrix must be reviewed if the Field Manager role is confirmed.

Account and access administration. Self-registration establishes ownership of a newly created business workspace. Email verification is required before active access is granted.

Only the Owner may administer manager invitations, responsibility scopes and permissions under the current role model.

Authentication, business membership and operation-specific permissions are separate checks. A valid account does not grant access to another Owner's business.

Manager permissions distinguish consultation, modification, correction, document operations and financial access. Insurance consultation is mandatory within an assigned worker-responsibility scope.

Financial information remains restricted to authorised online access.

The Field Manager role and its responsibility-scope granularity remain subject to stakeholder confirmation.

Capability	Farm Owner	Field Manager, if included
Manage properties, parcels and campaigns	Within the business scope.	Explicit permission required.
Manage worker profiles	Within the business scope.	Relevant consultation or editing permission required.
Plan and record work	Within the business scope.	Explicit operational permission required.
Correct confirmed work	Permitted under correction and campaign rules.	Separate correction permission required.
Consult worker insurance	Within the business scope.	Required for workers within the authorised operational scope.
Consult contracts and other documents	Within the business scope.	Specific document-type permission required.
Upload or update documents	Within the business scope.	Separate modification permission required.
Initiate worker signing	Within the business scope.	Explicit signing-initiation permission required.
Consult or edit monetary values	Within the business scope.	Specific monetary-data permission required.
Consult campaign indicators	Within the business scope.	Limited to authorised data and indicators.

3 Scope and Feature Overview

3.1 Included Scope

The proposed scope focuses on managing olive harvest operations within an owner's business.

It includes the following functional areas.

Properties and Parcels. Create and manage properties with optional internal parcels. Retain historical operational associations when descriptive information changes.

Harvest Campaigns. Organise campaigns covering whole properties or selected parcels. Manage planned and actual dates, closure, reopening and archival.

Worker Records. Maintain business-specific worker profiles, contact information and machinery capability. Support inactivation without automatically deleting historical records.

Work Planning. Prepare daily or multi-day plans, apply group values with individual adjustments, and reuse previous plans or actual work as proposals for future dates.

Attendance and Actual Work. Confirm daily attendance and actual assignments. Record task-specific periods, areas and optional breaks. Support authorised corrections.

Worker Documents. Upload, consult and update contracts, insurance and other relevant documents. Apply category-specific version rules and prepare insurance for offline access.

Worker Signing. Support the proposed in-person drawn-signature interaction, including prepared offline signing. Operational requirements remain subject to validation before deployment.

Transport and Deliveries. Record departure and delivery, registered drivers, vehicles, destinations and load provenance.

Weighing and Analysis. Manually record available ticket values, adjustments and analysis results. Preserve unknown information without inventing calculations.

Reported Monetary Values. Record the cooperative's current provisional or final amount for a ticket. Replace superseded values without counting them twice.

Campaign Indicators. Consult current work, production and permitted monetary indicators. Support property summaries, parcel detail and PDF export.

The use cases provide the detailed proposed behaviour. This overview groups capabilities without replacing their rules or acceptance criteria.

Account access. The proposed scope includes free Owner registration, email verification, login, logout and password recovery.

Manager invitations and permission administration are conditional on confirmation of the Field Manager role.

Initial authentication and access administration require internet connectivity. Supported field operations may continue offline after prior authentication and preparation of the required data.

3.2 Excluded Scope

The current proposal excludes:

- Payroll execution, wage calculation, worker remuneration rates and payment processing.
- Invoicing, full accounting and commercial olive oil sales.
- Application-calculated price estimates for deliveries.
- Worker self-service accounts for attendance or business management.
- Direct cooperative or laboratory integrations.
- Automatic extraction of values from ticket images or documents.
- Independent transport-trip management beyond the delivery-associated transport record.
- Management of agricultural activities unrelated to the agreed olive-harvest scope.
- Automatic allocation of mixed-load weights or monetary amounts without approved rules.

The following are optional enhancements, rather than committed first-version capabilities:

- Map-based property and parcel boundaries, visual area selection and estimated map areas.
- Cross-campaign or cross-year comparisons.
- Detailed Excel exports.
- Dedicated document-printing workflows.

Document generation through UC-20 remains a candidate capability whose concrete scope must be confirmed.

3.3 MVP Proposal Pending Approval

The Minimum Viable Product has not yet been formally prioritised and approved.

The current proposed baseline is an end-to-end harvest workflow in which the owner can:

1. Prepare properties, campaign coverage, workers and necessary documents.
2. Plan work and consult prepared schedules.
3. Confirm attendance and record actual periods and tasks in the field, including without connectivity.
4. Record transport and delivery, then complete ticket, analysis and reported monetary information as it becomes available.
5. Consult relevant insurance, use the agreed signing interaction and inspect campaign results.

This baseline is a proposal for prioritisation, not a declaration that every drafted feature is equally essential or already feasible within the available resources.

Before estimates are finalised, the team and client must identify which capabilities are mandatory for academic demonstration and which are required for operational use.

In particular, operational signing, offline document protection and synchronisation recovery require explicit validation and cannot be considered complete solely because their interface is demonstrated.

Optional map features and unconfirmed document-generation requirements must not silently expand the baseline.

3.4 Feature Overview

Area	Main capability	Use cases
Account access	Owner registration, email verification, login, logout and recovery.	UC-01–03
Manager access	Invitations, permissions and revocation, conditional on confirmation of the role.	UC-04
Land and campaign organisation	Properties, optional parcels and campaign lifecycle.	UC-05–06
Workforce	Worker profiles and machinery capability.	UC-07
Planning	Daily and multi-day plans, reuse and schedule consultation.	UC-08–09
Actual work	Attendance, task periods, breaks and corrections.	UC-10–11
Documents and signing	Document access, uploads, versions and worker signatures.	UC-12–14
Harvest logistics	Transport-associated deliveries, weighing tickets and analyses.	UC-15–17
Reported values	Current cooperative-reported ticket amounts.	UC-18
Campaign review	Indicators, supporting detail and PDF summaries.	UC-19
Document generation	Template-based generation, pending scope confirmation.	UC-20

Administrative configuration normally requires connectivity. Essential field entry and consultation use prepared authorised information and explicit synchronisation states.

Offline capability is defined per operation. It does not imply that the entire application or every historical record is available without a connection.

3.5 Assumptions and Dependencies

3.5.1 Business Assumptions

The current requirements review assumes that:

- Worker profiles and necessary documentation are prepared before fieldwork.
- A worker may perform different tasks and work in several areas during the same day.
- Campaigns may cover several properties or selected parcels.
- Source information may arrive after the physical delivery or campaign closure.
- One transport record is associated with each delivery in the first version.

- Each ticket and analysis belongs to one delivery. The reverse relationships remain subject to confirmation.
- Monetary values are supplied by the cooperative and associated with weighing tickets.

These assumptions must be revised when validated examples contradict them.

3.5.2 Business Dependencies

The team depends on client or intermediary input to confirm:

- Weighing-ticket terminology, deduction rules and source-value meanings.
- Analysis reference weights, result selection and yield calculations.
- Document types, templates and intended signing requirements.
- The Field Manager role and delegated permissions.
- First-version priorities and operational acceptance expectations.

Real examples should be provided with unnecessary personal information removed.

3.5.3 Technical Dependencies

The solution depends on:

- Devices and browsers capable of supporting the selected application approach.
- Successful preparation of authorised reference data before offline work.
- Sufficient local storage for pending operations and prepared documents.
- A central service for validation, persistence and synchronisation.
- Defined recovery behaviour for rejected operations, concurrent changes and interrupted transfers.
- Consistent deletion propagation for document versions removed under the retention rules.

The existing architecture proposes a PWA, local browser storage, a backend API and central relational persistence. The technical design must be reviewed against the consolidated requirements before being treated as final.

3.5.4 Project Dependencies

The academic schedule, team availability, hosting resources and required course deliverables constrain the implementation plan.

Use-case approval, task ownership, effort estimates and acceptance evidence must be tracked explicitly. The existence of a documented use case does not establish that its implementation has been completed or validated.

Identity and email delivery. Registration, invitations and password recovery depend on an email delivery mechanism and an identity-management implementation.

The initial proposal creates one independent business workspace for each self-registered Owner. Multiple business memberships and ownership transfer require confirmation.

Offline access depends on a previously authenticated session, protected local storage and a bounded offline authorisation period.

Session expiry, token lifetimes, password policy, authentication request limits and security-event retention must be specified in the non-functional requirements.

4 Use Cases

This section describes the interactions required to access the application and perform its business operations.

Account holders are distinguished from worker records. Owners authenticate to manage their business. Field Manager access remains conditional on confirmation of that role. Workers do not require application accounts.

UC-04 and UC-20 remain subject to the scope decisions identified in their descriptions.

4.1 Scope and Review Status

The use cases describe the proposed behaviour of OliveTrack, based on the current requirements review.

They remain subject to team review and client validation. Open business decisions are explicitly identified.

The Field Manager is a conditional role. Its inclusion has not yet been confirmed. Whenever included, it operates only within the permissions granted by the Farm Owner.

4.2 Shared Definitions

Business context. Each business maintains its own authorised properties, workers, documents and operational data. Records belonging to different businesses are not automatically shared or merged.

Property and parcel. A property may have optional named parcels. Operational information may identify the whole property or a particular parcel. Unallocated property-level information is not automatically distributed among parcels.

Campaign. A harvest campaign may cover several properties, including whole properties or selected parcels.

Working date. A work period belongs to the date on which it starts, even when it ends after midnight. Full start and end timestamps are used for duration and overlap validation.

Daily attendance. Attendance is recorded once per worker and working date within the business. Actual work assignments identify the relevant campaign and work area.

Confirmation and synchronisation. User confirmation is distinct from server acceptance. A locally confirmed operation may still be pending synchronisation or require resolution.

Current values. Worker-profile changes, work corrections, ticket corrections, analysis-entry corrections and monetary corrections do not require a permanent field-change history.

Distinct records. Document versions, individual signatures and separate source analyses are business records, not merely field-change history. Their preservation follows their specific rules.

Offline preparation. Offline operations use previously prepared, authorised reference information. A two-week download window does not define the permitted duration of an offline session.

Authorisation. Authentication, business membership and responsibility scope are checked separately from permission for the operation. The server checks current authority before accepting protected operations. An access denial must not disclose protected information. The Field Manager participates only if that role is approved and the Owner has granted the relevant scope and permission.

Cancellation and validation. Cancelling before confirmation discards the unsaved proposal and preserves accepted records. Invalid input is identified and recoverable information is retained for correction. Case-specific withdrawal, deletion and post-confirmation recovery follow their own explicitly described flows.

Saving and retry. A failed or uncertain transmission does not establish server acceptance or rejection. The interface reports the known storage state, preserves recoverable input and checks or resumes the same operation. A retry must not duplicate its business effect.

Offline submission. Only explicitly supported operations use prepared authorised data. Local confirmation remains pending until server acceptance. Reconnection revalidates current permissions and business rules. Rejected or conflicting proposals remain distinguishable from accepted records and require the applicable recovery workflow.

Financial information. Consultation, registration and correction of monetary amounts and monetary adjustments require online access and financial permission. Offline datasets, summaries and reports exclude them. Mixed-content attachment handling remains pending under Q-AT-01.

Task and capability. Machinery-operation capability describes a worker profile. Machinery operation and general work describe actual or planned tasks. An unknown actual task remains unclassified; a valid period may still contribute to the counted-hours total.

Provenance and collection. Provenance identifies a source property or parcel. Collection origin distinguishes olives collected from trees or from the ground. Harvesting method identifies how the olives were harvested. These attributes and olive variety are distinct.

Completion and correction. The single definition in Section 6.3.8 governs initial completion, changes to confirmed work and additional periods in closed campaigns.

4.3 Use Case Catalogue

UC-01 Register as an Owner.

UC-02 Log In and Log Out.

UC-03 Recover Account Access.

UC-04 Manage Manager Access and Permissions — conditional on confirmation of the Field Manager role.

UC-05 Manage Properties and Parcels.

UC-06 Manage Harvest Campaigns.

UC-07 Manage Worker Records.

UC-08 Plan Work.

UC-09 Consult Work Schedules.

UC-10 Record Attendance and Actual Work Periods.

UC-11 Correct Work Records.

UC-12 Consult Worker Documents.

UC-13 Upload and Update Worker Documents.

UC-14 Review and Sign a Document.

UC-15 Record Transport and Olive Deliveries.

UC-16 Record Weighing Data and Adjustments.

UC-17 Record Analysis Results.

UC-18 Manage Cooperative-Reported Ticket Values.

UC-19 Consult Campaign Indicators.

UC-20 Generate Worker Documents — scope awaiting confirmation.

4.4 UC-01 — Register as an Owner

Purpose
Allow a new Owner to create an account and establish an independent business workspace.

Primary Actor
Visitor.

Trigger
The visitor selects the registration option.

Preconditions
An internet connection is available. The visitor is not authenticated in the current session.

Required Information
Owner name, email address, password and business name.

Main Success Scenario

1. The visitor opens the registration form.
2. The system requests the required information.
3. The visitor enters the information and submits the registration request.
4. The system validates the submitted information and checks whether the email can be registered.
5. The system creates a pending registration and sends an email verification link.
6. The visitor follows the verification link.
7. The system validates the verification request.
8. The system activates the account and creates its business workspace.

9. The system assigns the Owner role for that workspace to the new account.
10. The system confirms activation and provides access to the login process.

Alternative and Exception Flows A1. **Invalid information:** The system identifies the fields requiring correction. Registration is not completed.

A2. **Email unavailable for registration:** The system does not create a duplicate account and provides a route to login or account recovery without unnecessarily disclosing account information.

A3. **Verification link expired or invalid:** The system does not activate the account and allows a new verification message to be requested.

A4. **Email delivery failure:** Registration remains pending. The visitor can retry delivery without creating a second account.

A5. **Repeated activation request:** An already processed verification request does not create another account or business workspace.

A6. **Connection failure:** The system reports the incomplete operation and supports a retry without duplicate creation.

Postconditions on Success
A verified account exists. A business workspace exists. The account is its Owner. No operational properties, campaigns or workers are created automatically.

Postconditions on Failure
No active business access is granted. A pending registration may remain available for verification or retry.

Connectivity
Registration and email verification require an internet connection.

Related Business Rules
BR-001, BR-002, BR-003, BR-004, BR-005, BR-006, BR-010.

Related Use Cases
UC-02 — Log In and Log Out. UC-03 — Recover Account Access.

Open Decisions
The initial registration creates one independent business workspace. Multiple memberships and lifecycle changes remain under Q-AA-03 and Q-AA-07.

4.5 UC-02 — Log In and Log Out

Purpose
Authenticate an account holder, establish access to an authorised business workspace and allow the user to terminate the session.

Primary Actors
Owner. Field Manager, if this role is included in the approved scope.

Trigger
The user requests login or logout.

Preconditions
For login, an activated account exists and an internet connection is available. For logout, a session exists on the device.

Main Success Scenario — Login

1. The user opens the login screen.
2. The user enters an email address and password.
3. The system validates the credentials and account status.
4. The system determines the user's current business access and permissions.
5. The system establishes an authenticated session.
6. The system opens the authorised workspace.
7. The system makes offline preparation available for the operations supported by the application.

Main Success Scenario — Logout

1. The user selects logout.
2. The system identifies any local changes that have not yet been accepted by the server.
3. The system informs the user of pending changes and provides a synchronisation option when online.
4. The user confirms logout.
5. The system terminates the session on the device and blocks access to the workspace.
6. The system preserves pending changes in protected local storage for recovery through the originating account, subject to renewed authorisation.

Alternative and Exception Flows

A1. **Invalid credentials:** Access is refused using a response that does not unnecessarily reveal which credential was incorrect.

A2. **Unverified account:** Workspace access is refused and the user is offered a route to complete verification.

A3. **No authorised business access:** Authentication does not grant access to a business from which the account has been removed.

A4. **No connection during login:** A new online authentication cannot be completed. Previously authorised offline use follows the established offline-session policy.

A5. **Expired or invalid session:** The system requests renewed authentication before protected online operations continue.

A6. **Logout while offline:** The local session is terminated and workspace access is blocked. Any server-side invalidation requiring connectivity is completed when communication becomes possible.

A7. **Pending synchronisation failure:** The user may still log out. Pending changes are not silently discarded or exposed to another account.

Postconditions on Login

The user can access only the authorised business information and operations.

Postconditions on Logout

The workspace is inaccessible through the terminated local session. Pending data, if any, remains protected and is not transferred to another account.

Connectivity

Login and online permission validation require internet. Supported offline work requires previous authentication and locally available authorised data. Logout is available online and offline.

Related Business Rules

BR-006, BR-009, BR-010, BR-011, BR-012, BR-013.

Related Use Cases
UC-01 — Register as an Owner. UC-03 — Recover Account Access. UC-04 — Manage Manager Access and Permissions.

Open Decisions
Offline authorisation, session expiry and simultaneous-device limits remain under Q-AA-04 and Q-AA-05.

4.6 UC-03 — Recover Account Access

Purpose
Allow an account holder who has forgotten the password to recover access without changing business ownership or permissions.

Primary Actor
Account holder.

Trigger
The account holder selects the password recovery option.

Preconditions
An internet connection is available. The account holder can access the registered email inbox.

Main Success Scenario

1. The account holder opens the recovery form.
2. The account holder enters the account email.
3. The system displays a generic acknowledgement.
4. For an eligible account, the system sends a password reset link to the registered email.
5. The account holder follows the link.
6. The system validates the reset request.
7. The account holder enters and confirms a new password.
8. The system validates and saves the new password.
9. The system consumes the reset authorisation and invalidates existing online sessions.
10. The system confirms the change and directs the account holder to login.

Alternative and Exception Flows

A1. **Unknown or ineligible email:** The system displays the same generic acknowledgement without disclosing whether an account exists.

A2. **Invalid, expired or previously used link:** The password is not changed. A new recovery request may be submitted.

A3. **Invalid password or confirmation mismatch:** The system requests correction before completing the reset.

A4. **Email delivery failure:** The existing credentials remain unchanged. The account holder may submit another request.

A5. **Connection failure:** The system reports that completion could not be confirmed and requires validation before reporting a successful reset.

A6. **No access to the registered inbox:** Automatic recovery cannot be completed. An alternative identity-verification process is outside the currently defined workflow.

Postconditions on Success
The new password is active. The reset authorisation cannot be reused. Existing online sessions are invalidated. Business associations and permissions remain unchanged.

Postconditions on Failure
No password change is confirmed. Existing account permissions are unchanged.

Connectivity
Recovery requires an internet connection. Disconnected devices detect session invalidation when they reconnect or reach the offline authorisation limit.

Related Business Rules
BR-014.

Related Use Cases
UC-02 — Log In and Log Out.

Open Decisions
Reset lifetimes and request limits remain under Q-AA-05. Assisted recovery and account-lifecycle scope remain under Q-AA-07.

4.7 UC-04 — Manage Manager Access and Permissions

Status
Proposed. Inclusion depends on confirmation of the Field Manager role by the project team and business stakeholders.

Purpose
Allow an Owner to invite a Field Manager, define the authorised scope and permissions, and later modify or revoke that access.

Primary Actor
Owner.

Supporting Actor
Invited Field Manager.

Trigger
The Owner opens access management.

Preconditions
The Owner is authenticated. An internet connection is available. The Owner controls the selected business workspace.

Main Success Scenario — Invite a Manager

1. The Owner selects the invitation option.
2. The Owner enters the manager's email address.
3. The Owner selects the responsibility scope and permitted operations.
4. The system checks the permission configuration.
5. The system creates an invitation and sends an acceptance link.
6. The recipient follows the invitation link.
7. An existing account holder authenticates; a new recipient creates and verifies an account.
8. The system confirms that the authenticated email matches the invitation.

9. The recipient explicitly accepts the invitation.
10. The system activates the manager's access to the Owner's business.

Main Success Scenario — Change Permissions
1. The Owner selects an existing manager.
2. The system presents the current scope and permissions.
3. The Owner changes the configuration.
4. The system validates and saves the changes.
5. Subsequent online requests use the updated authorisation.

Main Success Scenario — Revoke Access
1. The Owner selects an existing manager.
2. The Owner requests revocation.
3. The system requests confirmation.
4. The Owner confirms.
5. The system revokes the manager's business access.
6. Subsequent online requests to that business are refused.

Alternative and Exception Flows
A1. **Invitation already pending:** The system allows the Owner to review, resend or cancel the existing invitation without creating duplicate memberships.

A2. **Manager already has access:** The system directs the Owner to the existing permission configuration.

A3. **Invalid, expired or cancelled invitation:** Access is not granted. The Owner may issue a new invitation.

A4. **Different authenticated email:** The invitation cannot be accepted by that account.

A5. **Incomplete mandatory permissions:** The system identifies the incompatible configuration. A manager responsible for workers must retain access to their insurance documents within that scope.

A6. **Invitation cancelled before acceptance:** The invitation no longer grants access.

A7. **Manager device offline during revocation:** Central access is revoked immediately. The disconnected device enforces the change upon reconnection or expiry of offline authorisation.

A8. **Pending changes after revocation:** Synchronisation revalidates permissions. Unauthorised changes are not automatically accepted or silently discarded.

Postconditions on Success

The invitation, permission change or revocation is saved. Access is limited to the Owner's business and the authorised responsibility scope.

Postconditions on Revocation

The account remains distinct from its business membership. Existing operational records are preserved. Revocation does not delete worker records or historical business information.

Connectivity

Invitations, acceptance and permission changes require an internet connection.

Related Business Rules

BR-005, BR-006, BR-007, BR-008, BR-009, BR-011, BR-012.

Related Use Cases

UC-02 — Log In and Log Out. UC-12 — Consult Worker Documents. UC-18 — Manage Cooperative-Reported Ticket Values.

Open Decisions

Role inclusion and responsibility scope remain under Q-AA-01 and Q-AA-02. An invitation recipient who already belongs to another business requires the membership policy pending under Q-AA-03; acceptance must not silently establish an unapproved second membership.

4.8 UC-05 — Manage Properties and Parcels

Purpose

Allow an authorised user to create, consult, update, archive, reactivate and, where permitted, delete properties and their optional parcels.

The current agricultural scope concerns olive harvesting.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors

Primary actor: Farm Owner.

Conditional actor: Field Manager, when explicitly granted property-management permissions by the owner.

Permission to record operational work does not automatically grant permission to manage properties.

Trigger

The user needs to register a property, organise its optional parcels, consult its details or manage its information and availability.

Preconditions

- The user has permission for the requested operation.
- Creation, modification, archival, reactivation and deletion require an internet connection.
- For operations on an existing record, that record exists within the authorised scope.
- To create a parcel, its property exists.
- Offline consultation requires previously downloaded authorised information.

Main Flow

1. The user opens property management.
2. The user chooses to create a property or selects an existing property.
3. The system presents the existing information or an entry form.

4. The user enters or updates the property name and textual location.
5. The user may enter the declared area in hectares and one or more olive varieties.
6. The system identifies the owner associated with the property.
7. If internal divisions are needed, the user creates or updates named parcels. Parcel area and olive varieties are optional.
8. The system validates the information and access permissions.
9. The user reviews and confirms the changes.
10. The system saves the records and confirms the result.

For consultation only, the flow ends after the requested information is displayed.

Alternative Flows and Exceptions

A1 — No internal divisions are needed. The user saves the property without parcels. The absence of parcels does not prevent use of the property in operational records.

A2 — Optional information is unknown. The user leaves area or variety information unfilled and may add it later. An unknown area is not treated as zero.

A3 — The user archives a record. The system requests confirmation and archives the selected property or parcel. Historical records remain accessible within the user's permissions. The archived record is unavailable for new operational associations.

A4 — The user reactivates a record. The system validates the request and restores its availability, subject to its parent property's state.

A5 — The user requests deletion. The system checks for associated records. If none exist, the user may confirm deletion. Otherwise, deletion is refused and archival is offered.

A6 — Information is invalid or incomplete. The system identifies the affected fields and preserves unaffected input for correction.

A7 — Access is denied. The shared authorisation conditions apply.

A8 — Connectivity is unavailable. The system permits consultation of previously downloaded authorised records and identifies their synchronisation status. Management operations are unavailable until connectivity is restored.

A9 — Saving fails. The system reports that the change has not been confirmed by the server and preserves recoverable input.

A10 — The user cancels. The shared cancellation conditions apply.

Postconditions

On successful creation or update: Property and parcel information is saved with the correct ownership and parent associations.

On archival: Historical information is preserved, and new operational use is blocked for the archived scope.

On reactivation: The record becomes available again, subject to the property's availability.

On deletion: Only the eligible record without associated records is removed.

On consultation, cancellation or rejection: Stored information remains unchanged.

Related Business Rules

Property and parcel structure, declared areas, archival, deletion and connected management follow BR-015, BR-016, BR-017, BR-018, BR-019, BR-028.

Case-specific details:

- A property requires a name and textual location.
- A parcel requires a name and its parent property.
- Several olive varieties may be recorded.
- The owner is associated with the record rather than repeatedly entered as free text.
- Record identity remains stable when descriptive information is corrected.
- Maps and boundary drawing are optional enhancements.
- Creating or changing parcels must not silently change existing campaign coverage or reassign historical operational records.

Acceptance Criteria

1. A property can be created with a name and textual location, without area, varieties, parcels or a map.
2. A parcel can be created with a name and an existing authorised parent property.
3. Area and varieties can be added later.
4. An unfilled area is distinguishable from a recorded numerical value.
5. Updating names preserves existing operational associations.
6. A manager without specific permission cannot modify property or parcel records.
7. Archiving preserves historical records and blocks new operational associations.
8. An archived record can be reactivated when the parent context permits it.
9. Deletion is refused when associated records exist.
10. Permitted deletion requires explicit user confirmation.
11. Management operations cannot be completed without connectivity.
12. Prepared authorised information can be consulted offline.
13. Failed saves are not presented as server-confirmed changes.
14. A future map estimate does not replace the owner's declared area.

Related Use Cases

- UC-06 — Manage Harvest Campaigns.
- UC-08 — Plan Work.
- UC-09 — Consult Work Schedules.
- UC-10 — Record Attendance and Actual Work Periods.
- UC-15 — Record Transport and Olive Deliveries.
- UC-19 — Consult Campaign Indicators.

Review Status

Optional map and area-estimation scope remains under Q-MVP-01. Coverage changes with existing records follow Q-PC-03.

4.9 UC-06 — Manage Harvest Campaigns

Purpose

Allow an authorised user to create, consult, update, close, reopen, archive, unarchive and, where permitted, delete harvest campaigns.

A campaign may cover one or more properties, including whole properties or selected parcels.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors

Primary actor: Farm Owner.

Conditional actor: Field Manager, with the relevant permissions granted by the owner.

Permissions apply to the requested operation. Operational access does not automatically grant all campaign-management permissions.

Trigger

The user needs to prepare a campaign, consult its information, change its scope or manage its lifecycle.

Preconditions

- The user has permission for the requested operation.
- Campaign-management operations require an internet connection.
- Selected properties and parcels exist within the authorised scope.
- For operations on an existing campaign, that campaign exists.
- Offline consultation requires previously downloaded authorised information.

Main Flow

1. The user opens campaign management.
2. The user chooses to create a campaign or selects an existing one.
3. The system presents its information or an entry form.
4. The user enters or updates the campaign name and planned start date.
5. The user selects one or more properties.
6. For each property, the user chooses the whole property or specific parcels.
7. The system validates the selected areas, dates and permissions.
8. The system checks for conflicting campaign coverage.
9. The user reviews and confirms the proposed information.
10. The system saves the campaign and confirms the result.

For consultation only, the flow ends after the authorised campaign information, dates and covered areas are displayed.

Alternative Flows and Exceptions

A1 — Parcels do not yet exist. The user may create parcels through UC-05, provided they have the required permission. The parcels remain associated with the property for future use. The user then returns to campaign selection.

A2 — Invalid or incomplete information. The system identifies the affected fields and preserves unaffected input for correction.

A3 — Conflicting campaign coverage. The system rejects overlapping campaign coverage of the same area and period. Campaigns covering different parcels of the same property may coexist.

A4 — Areas are added. The system checks permissions and campaign conflicts before accepting the additional coverage.

A5 — Areas are removed. Direct removal is permitted only when no operational records are associated with that area within the campaign. Otherwise, an explicit correction process is required.

A6 — The user closes the campaign. The user provides or confirms the actual end date. The system validates the closure against existing activity and the agreed date rules. On confirmation, the campaign is closed.

A7 — External information arrives after closure. An authorised user may add or complete ticket information, analysis results and cooperative-reported values for existing deliveries and tickets. This does not require reopening the campaign.

A8 — Earlier work is completed. Completion follows Section 6.3.8. Adding periods to a date with confirmed work in that closed campaign requires authorised reopening. Confirmed values are not replaced through initial entry.

A9 — Confirmed work requires correction. The campaign must be reopened by an authorised user before confirmed work records are corrected. The correction follows UC-11.

A10 — Work occurred after the end date. The campaign must be reopened and its dates reviewed before that work is accepted.

A11 — The user reopens the campaign. The system checks reopening permission and conflicting campaign coverage. Existing records are preserved. Previous closure information must not be mistaken for the campaign's current lifecycle status.

A12 — The user archives or unarchives. The system changes visibility in the usual lists without changing the closure status. Archiving does not close a campaign. Unarchiving does not reopen one. Archiving must not bypass campaign-overlap validation.

A13 — The user requests deletion. The system checks for operational records. If none exist, deletion may be confirmed. Otherwise, deletion is refused and archival is offered.

A14 — Connectivity is unavailable. Management operations cannot be completed. Previously downloaded authorised campaign information remains available for consultation and supported field operations.

A15 — Access is denied. The shared authorisation conditions apply.

A16 — Saving fails. The system preserves recoverable input and does not claim successful storage without server confirmation.

A17 — The user cancels. The shared cancellation conditions apply.

Postconditions

On successful creation or update: The campaign is saved with its name, planned start date and selected coverage.

On closure: The campaign has an actual end date. Later completion of relevant external information remains possible without reopening. Initial completion and additional periods follow Section 6.3.8. Work-record corrections require authorised reopening.

On reopening: The campaign permits further work within its revised valid scope and dates.

On archival or unarchival: Visibility changes without deleting history or changing the closure status.

On deletion: The eligible campaign is removed without deleting operational history.

On consultation, cancellation or rejection: Stored information remains unchanged.

Related Business Rules

Campaign coverage, dates, lifecycle, late information and archival follow BR-018, BR-020, BR-021, BR-022, BR-023, BR-024, BR-025, BR-026, BR-027, BR-028.

Case-specific details:

- A campaign requires a name, a planned start date and at least one covered area.
- Parcels created during campaign setup belong to the property, not exclusively to that campaign.
- Creating or changing parcel divisions must not bypass existing campaign coverage.
- Map-based area selection is an optional future enhancement.

Acceptance Criteria

1. A campaign can cover one or several authorised properties.
2. For each property, the user can select the whole property or specific parcels.
3. Duplicate whole-property and parcel selection is prevented.
4. A campaign cannot be accepted without a name, planned start date and covered area.
5. Revising the planned start date does not overwrite the actual start date.
6. Campaigns on different parcels may coexist during the same period.
7. Conflicting coverage of the same area and period is rejected.
8. An area with operational records cannot be directly removed.
9. Closing records the actual end date without requiring all analysis results or cooperative values to be available.
10. Completion follows Section 6.3.8. Adding periods to a date with confirmed work in a closed campaign requires reopening and cannot overwrite accepted information.
11. A closed campaign prevents correction of confirmed work until authorised reopening.
12. External delivery information can be completed after closure without reopening.
13. Archiving a campaign does not release its covered areas for conflicting campaign activity.
14. Work after the end date requires authorised reopening and date review.
15. Archiving preserves history and does not close an open campaign.
16. Unarchiving does not reopen a closed campaign.
17. Deletion is refused when operational records exist.
18. Managers can perform only explicitly permitted operations.
19. Offline consultation is limited to prepared authorised information.
20. Management changes require server confirmation.

Related Use Cases

- UC-05 — Manage Properties and Parcels.
- UC-08 — Plan Work.
- UC-10 — Record Attendance and Actual Work Periods.
- UC-11 — Correct Work Records.
- UC-15 — Record Transport and Olive Deliveries.
- UC-16 — Record Weighing Data and Adjustments.
- UC-17 — Record Analysis Results.
- UC-18 — Manage Cooperative-Reported Ticket Values.
- UC-19 — Consult Campaign Indicators.

Review Status

Campaign overlap, actual-start recording and coverage correction remain under Q-PC-01 to Q-PC-03. Completion and correction follow Section 6.3.8 and confirmed decision D-WF-01.

4.10 UC-07 — Manage Worker Records

Purpose

Allow an authorised user to create, consult, update, inactivate, reactivate and, where permitted, delete worker records within a business.

Worker records support operational management and document workflows. They do not constitute worker application accounts.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors

Primary actor: Farm Owner.

Conditional actor: Field Manager, if this role is included and the owner grants the relevant permissions.

Permission to access worker information does not automatically grant access to every associated document.

Trigger

The user needs to register a worker, consult or update their information, or manage their availability for future work.

Preconditions

- The user has permission for the requested operation.
- Management operations require an internet connection.
- For operations on an existing worker, the record exists within the authorised business context.
- Offline consultation requires previously downloaded authorised information.

Main Flow

1. The user opens worker management.
2. The user searches for an existing worker or chooses to create a record.
3. The system displays the authorised information or an entry form.
4. The user enters or updates the worker's full name and telephone number.
5. The user indicates whether the worker can operate machinery and may provide an optional description.

6. The system validates required fields and checks for matching names or telephone numbers within the same business.
7. The user reviews and confirms the information.
8. The system saves the current values and updates the last-modified date.
9. The system confirms the result.

For consultation only, the flow ends after the authorised worker information is displayed.

Alternative Flows and Exceptions

A1 — No worker matches the search. The system displays an explicit no-results message and permits authorised creation of a new record.

A2 — A matching name or telephone is found. The system warns of a possible duplicate within the same business. The user may open the existing record or confirm that a separate record is needed. Records are not automatically merged.

A3 — Invalid or incomplete information. The system identifies the affected fields and preserves unaffected input for correction.

A4 — The user inactivates a worker. The system requests confirmation and removes the worker from the usual selections for new work. Historical work and documents are preserved.

A5 — The user reactivates a worker. The system restores the worker's availability for selection without creating another identity.

A6 — The user requests deletion. The system checks for associated records. Deletion may be confirmed only if none exist. Otherwise, deletion is refused and inactivation is offered.

A7 — Access is denied. The shared authorisation conditions apply.

A8 — Retrieval or saving fails. The system explains the failure and preserves recoverable input. A failed search is not presented as proof that the worker does not exist.

A9 — A creation request is repeated. Retrying the same submission does not unintentionally create duplicate records.

A10 — The user cancels. The shared cancellation conditions apply.

A11 — Connectivity is unavailable. The system permits consultation of previously downloaded authorised information. Creating, editing, inactivating, reactivating or deleting records requires reconnection.

Postconditions

On successful creation or update: The current worker information is saved within the correct business context, with its last-modified date.

On inactivation or reactivation: Selection availability changes without deleting historical records.

On deletion: Only an eligible record without associations is removed.

On consultation, cancellation or rejection: Previously saved information remains unchanged.

In all cases: The operation does not automatically create an application account, schedule work or confirm attendance.

Related Business Rules

Business-specific worker identity, capability, inactivation and profile corrections follow BR-005, BR-029, BR-030, BR-031, BR-032, BR-033, BR-065.

Case-specific details:

- Full name and telephone number are mandatory.
- Each worker record receives an automatic internal identifier.
- A description of that capability is optional.
- Not being selected on a particular day does not make a worker inactive.
- Worker registration and necessary documentation are prepared before fieldwork.
- Offline worker creation is not required.

Acceptance Criteria

1. A worker cannot be accepted without the required name and telephone.
2. A new record receives an internal identifier.
3. Machinery-operation capability can be recorded with an optional description.
4. A matching name or telephone within the business produces a warning.
5. The user may confirm a separate record without automatically merging identities.
6. Records from another business are not exposed by duplicate checking.
7. Updating profile information preserves links to existing operational records.
8. Changing machinery capability does not change previously recorded tasks or hours.
9. Inactivation preserves history and removes the worker from usual new-work selections.
10. Reactivation restores availability using the existing identity.
11. Deletion is refused when associated records exist.
12. Profile updates refresh the last-modified date.
13. Management operations require connectivity and server-side permission checks.
14. Prepared authorised information remains consultable offline.
15. Cancellation preserves saved information.
16. Repeated creation submissions do not unintentionally create duplicate records.

Related Use Cases

- UC-08 — Plan Work.
- UC-10 — Record Attendance and Actual Work Periods.
- UC-11 — Correct Work Records.
- UC-12 — Consult Worker Documents.

- UC-13 — Upload and Update Worker Documents.
- UC-20 — Generate Worker Documents.

Review Status

Future plans identify inactive workers for review under BR-032. Document eligibility remains under Q-DOC-01; automatic compliance checking is not assumed.

4.11 UC-08 — Plan Work

Purpose

Allow an authorised user to plan workers, work areas, optional periods and tasks for one or more working days.

Planning may start from a new entry, an existing plan or previously recorded actual work. It never establishes attendance or actual hours for the target dates.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors

Primary actor: Farm Owner.

Conditional actor: Field Manager, if this role is included and the owner grants the relevant planning permissions.

Trigger

The user needs to prepare or adjust upcoming work, including the current day, or reuse information to avoid repetitive planning.

Preconditions

- The user has permission to plan work within the selected scope.
- The campaign, work areas and worker records exist.
- Selected areas are included in the campaign coverage.
- The requested dates comply with the campaign lifecycle rules.
- Offline planning requires the necessary authorised reference data to be available on the device.

Main Flow

1. The user selects the campaign and one or more target dates.
2. The user selects a property or an included parcel.
3. The system displays existing plans for the selected context.

4. The user selects the workers expected to work.
5. The user may enter planned periods, task types and optional breaks. Common values can be applied to a group and adjusted individually.
6. The system checks required information, worker availability, area eligibility and known scheduling conflicts.
7. Where a worker is planned for several areas on the same day, the system requires periods for the relevant assignments.
8. The system presents the proposed daily plans and any unresolved issues.
9. The user reviews and confirms.
10. The system saves independent daily plans and displays their local and server status.

Alternative Flows and Exceptions

A1 — A minimal plan is sufficient. For a worker assigned to one area on a working day, the user may save the campaign, date, area and worker without specifying hours or task type.

A2 — Individual details differ. The user adjusts one worker's periods or tasks without unintentionally changing other workers' entries.

A3 — Several areas apply. The user creates the relevant assignments. Periods are required for all assignments of that worker to different areas on the same working day.

A4 — Several target dates are selected. The system applies the reviewed proposal to each selected date. Each date receives an independent plan, validated against its own context.

A5 — An existing plan is copied. The user selects a source plan and the target dates. Workers, areas, periods and tasks are proposed for review. Source plans remain unchanged.

A6 — Actual work is used as a basis. The user selects previously recorded work. Its workers, areas, periods and tasks are proposed as planning information for the target dates. Confirmed attendance and actual work records are not copied to those dates.

A7 — Target dates already contain plans. The system identifies the existing plans. Reuse does not silently overwrite or duplicate them. Changes require explicit review.

A8 — A worker is inactive. The system identifies the inactive worker and requires an eligible selection or authorised reactivation. Existing future plans are flagged rather than silently edited.

A9 — Periods overlap. The system blocks conflicting periods, considering the worker's relevant plans across properties, parcels and campaigns.

A10 — Work crosses midnight. The user records the start and end with sufficient date information. The period belongs to the working date on which it starts. Validation considers the full interval, including its continuation into the next day.

A11 — A worker is removed or replaced. Before actual work has been recorded for the affected daily area context, the user may confirm removal or replacement. Other assignments remain unchanged.

A12 — Actual work already exists. The corresponding plan remains the reference for what was expected. Differences are recorded as actual work or handled through the work-correction process.

A13 — The device is offline. The user may create or adjust plans using prepared authorised data. The system identifies the changes as pending synchronisation.

A14 — Server validation rejects a change. The system identifies the rejected change and the required action. It does not label the change as synchronised.

A15 — Saving fails. The system preserves recoverable input and accurately reports what was saved locally or acknowledged by the server.

A16 — The user cancels. The shared cancellation conditions apply.

A17 — Planned breaks are entered. The user identifies break start and end times and their intended counting treatment. The system prevents duplicate deductions in any displayed planned-duration calculation. These values remain subject to actual confirmation.

Postconditions

On success: Daily plans identify the selected campaign, working dates, areas and workers, with any supplied periods and tasks.

On reuse: Target plans are independent of their source. Editing a target day does not modify the source or other target days.

On pending synchronisation: Local plans remain distinguishable from server-accepted plans.

On cancellation or rejection: Previously accepted information remains unchanged.

In all cases: Planning does not confirm attendance, create actual hours or modify source actual work.

Related Business Rules

Planning reuse, required detail, worker status, overnight dates and synchronisation follow BR-031, BR-032, BR-034, BR-035, BR-036, BR-041, BR-044.

Case-specific details:

- A daily plan requires a campaign, working date, work area and selected workers.
- The area must be within campaign coverage.
- Planned periods for the same worker must not overlap across areas or campaigns.
- Planned breaks are optional. Each recorded break identifies its start and end and whether it is intended to count towards working time.
- A planned break is a proposal. Its actual occurrence and treatment require confirmation through UC-10.
- Planned intervals and confirmed actual intervals are validated separately. The same planned and actual interval does not constitute a duplicate work entry.
- End dates and times must establish a valid positive-duration interval.

Acceptance Criteria

1. A minimal single-area daily plan can be saved without hours or tasks.

2. Multiple-area planning for the same worker and day requires the relevant periods.
3. Overlapping defined periods are rejected.
4. Overnight overlap validation uses complete start and end timestamps, including periods beginning on an adjacent date.
5. An overnight period is associated with its start date.
6. Group values apply only to the selected workers.
7. Individual adjustments preserve other workers' entries.
8. Multi-day planning produces independent daily plans.
9. Copying preserves the source plan and requires review of the target context.
10. Copying a plan or reusing actual work produces reviewed planning information only, without confirmed attendance, periods or breaks.
11. Existing target plans are not silently overwritten or duplicated.
12. Inactive workers in future plans are identified without automatic removal.
13. Plan changes do not rewrite existing actual work records.
14. Supported offline plans remain available locally with an explicit pending status.
15. Server-rejected changes are not presented as synchronised.
16. Cancelling preserves saved plans.
17. A plan may include optional breaks without requiring them for every worker.
18. Planned breaks are not automatically treated as confirmed actual breaks.

Related Use Cases

- UC-05 — Manage Properties and Parcels.
- UC-06 — Manage Harvest Campaigns.
- UC-07 — Manage Worker Records.
- UC-09 — Consult Work Schedules.
- UC-10 — Record Attendance and Actual Work Periods.
- UC-11 — Correct Work Records.

Review Status

Existing-target-plan interaction and pending-change recovery remain under Q-WF-04. Offline access limits follow Q-AA-04; timestamp configuration follows Q-WF-02.

4.12 UC-09 — Consult Work Schedules

Purpose

Allow an authorised user to consult planned work by day, by worker or by week, within the selected harvest campaign.

Consultation does not modify plans or establish actual attendance.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors

Primary actor: Farm Owner.

Conditional actor: Field Manager, if this role is included and the owner grants the relevant consultation permissions.

Trigger

The user needs to know who is expected to work, where, when and in which planned role.

Preconditions

- The selected campaign exists.
- The user is authorised to consult the requested information.
- Offline consultation requires the relevant plans and reference information to be available on the device.

An existing plan is not a precondition: a valid query may return no planned work.

Main Flow

1. The user opens schedule consultation and selects a campaign.
2. The user chooses a daily, worker-based or weekly view.
3. The user selects the relevant date, week or worker consultation period.
4. The user may filter by property, parcel, worker and planned task type.
5. The system retrieves the matching authorised plans.
6. The system displays the workers, working dates, properties or parcels, planned periods, task types and any recorded planned breaks.
7. Missing hours or task types are identified as not yet defined.
8. The system identifies relevant synchronisation and freshness information.
9. The user may change the view or filters and repeat the consultation.

Alternative Flows and Exceptions

A1 — No matching plan exists. After a successful query, the system states that no planned work matches the selection. This does not establish worker absence.

A2 — A plan has incomplete details. The system displays the known information. Missing hours or task types are shown as not yet defined, not as zero hours or an assumed task.

A3 — Several areas are included. The system can display all authorised areas of the selected campaign together. Each assignment retains its property and optional parcel identification.

A4 — A period crosses midnight. The system displays the period under its start date and identifies the end date and time.

A5 — A planned worker is inactive. The system identifies the inactive status. The saved assignment is not silently removed. Any change follows UC-08.

A6 — Local changes are pending. The system identifies locally saved plans or changes awaiting synchronisation. They are not presented as server-confirmed.

A7 — The device is offline. The system displays prepared authorised plans for the current and following week, together with the last successful synchronisation information.

A8 — Requested information is not local. The system explains that the information is unavailable without connectivity. It does not claim that no plan exists.

A9 — Retrieval fails. The system explains the failure and offers a retry where appropriate. A failed query is distinguishable from a successful query with no results.

A10 — Access is restricted. The system excludes unauthorised information and identifies the scope being displayed without exposing protected details.

Postconditions

On success: Matching authorised plans are displayed, or a successful query confirms no matching plans.

On offline or incomplete consultation: The system identifies the available scope and relevant limitations.

In all cases: Consultation does not modify plans, confirm attendance or create actual hours.

Related Business Rules

Planning status, incomplete detail, working dates and freshness follow BR-034, BR-035, BR-041, BR-044.

Case-specific details:

- Daily, worker-based and weekly views represent the same underlying plans.
- Filters include campaign, property, parcel, worker and planned task type.
- Property-level assignments remain distinguishable from parcel assignments.
- Unspecified hours and tasks are displayed as not yet defined.
- Downloaded information remains limited to the authorised scope.
- The current and following weeks define the standard offline preparation window. Older or additional information is available offline only when explicitly prepared.

Acceptance Criteria

1. The user can consult plans by day, by worker and by week.

2. Available filters return the matching authorised assignments.
3. The campaign can be consulted across its authorised areas without requiring a single-property filter.
4. The same assignment has consistent information in each view.
5. Missing hours and task types are explicitly identified.
6. Overnight periods appear under their start date with a clear end date.
7. Inactive workers remain identifiable in their existing plans.
8. Prepared plans for the current and following week are consultable offline.
9. Offline consultation identifies the last successful synchronisation.
10. Pending local changes are distinguishable from server-confirmed information.
11. A successful empty query is distinguishable from information unavailable on the device; missing local data is not proof of no planned work.
12. Retrieval failures are distinguishable from successful queries with no results.
13. Consultation does not alter plans or actual work records.
14. Planned breaks are visible where they have been recorded.

Related Use Cases

- UC-05 — Manage Properties and Parcels.
- UC-06 — Manage Harvest Campaigns.
- UC-07 — Manage Worker Records.
- UC-08 — Plan Work.
- UC-10 — Record Attendance and Actual Work Periods.

Review Status

Offline preparation and freshness follow the shared conditions. Session duration remains under Q-AA-04 and Q-AA-05; the two-week preparation window does not define session validity.

4.13 UC-10 — Record Attendance and Actual Work Periods

Purpose

Allow an authorised user to confirm daily attendance and record actual work periods, tasks, locations and optional breaks.

Planned information may support entry, but becomes actual work only after explicit confirmation.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors

Primary actor: Farm Owner.

Conditional actor: Field Manager, if this role is included and the owner grants the relevant permissions.

Workers are not required to use an application account to confirm their own attendance.

Trigger

The user needs to confirm attendance or record work performed on the current or a previous working day.

Preconditions

- The user has permission for the requested operation.
- Worker records already exist.
- The relevant campaign and work areas exist within the authorised scope.
- Work dates comply with the campaign lifecycle rules.
- Offline entry requires the necessary authorised reference data to be available.

Main Flow

1. The user selects the campaign and working date.
2. The system displays planned workers, available proposed periods and any existing attendance and work records.
3. Workers without an attendance decision are shown as awaiting confirmation.
4. The user marks workers as present or absent. Daily attendance is shared within the business for that worker and working date.
5. For present workers, the user identifies the actual campaign and work-area assignments. These may remain incomplete as to hours or tasks.
6. The user reviews proposed periods or enters actual periods, tasks and areas. Common values may be applied to a group with individual adjustments.
7. The user reviews or records actual breaks, including their start, end and whether they count towards working time. Planned breaks are not accepted automatically.
8. The system validates the entries and calculates available working-time totals.
9. The user reviews and explicitly confirms the submitted information.
10. The system saves the records and displays their local storage and synchronisation status.

Alternative Flows and Exceptions

A1 — Attendance is known before hours. The user saves presence or absence and completes work details later. Missing hours remain incomplete, not a confirmed zero total.

A2 — An unplanned worker attends. The user selects an existing authorised worker record and records attendance. A planned assignment is not required.

A3 — No plan exists. The user selects existing workers and enters the actual information directly.

A4 — Actual work differs from the plan. The user adjusts workers, periods, tasks or areas before confirmation. The original plan remains unchanged. A valid period with an unknown task contributes to total counted hours as unclassified, separately from named task totals.

A5 — A worker changes task or area. The user records separate periods for the relevant campaign and area. Known tasks are identified; an unknown task remains unclassified. Valid unclassified hours contribute to total counted hours.

A6 — Work periods exclude a break. The interval between those periods does not contribute to working time. The same excluded interval is not deducted again.

A7 — A break occurs within a period. The user records its start and end. It is deducted only when explicitly marked as not counted towards working time.

A8 — A worker is absent. The system does not convert planned periods into actual work. Contradictory actual work entries require resolution.

A9 — Work crosses midnight. The period retains its start-date association. Its end date and time establish the full interval for calculations and overlap validation.

A10 — Work is entered retrospectively. The user selects the actual working date. The system preserves the distinction between work date and entry date.

A11 — The campaign is closed. Completion follows Section 6.3.8. Adding periods to a date with confirmed work in that campaign requires authorised reopening. Corrections of confirmed work follow UC-11 after reopening.

Work after the actual end date requires reopening and date review.

A12 — Confirmed information needs correction. The user follows UC-11. Existing values are not silently replaced through another initial submission.

A13 — Invalid or overlapping periods. The system identifies the problem and preserves unaffected input. Validation considers the worker's relevant periods across areas and campaigns.

A14 — The device is offline. The system saves supported entries locally. Their synchronisation status remains visible.

A15 — Saving or synchronisation fails. The system preserves recoverable information and reports the actual storage state. Retrying does not unintentionally duplicate attendance or work periods.

A16 — The user cancels. The shared cancellation conditions apply.

Postconditions

On success: The submitted attendance and available actual work details are saved in the correct context.

On attendance-only confirmation: Presence is recorded while hours and tasks remain explicitly incomplete where unknown.

On offline save: User-confirmed entries are stored locally and remain pending server acknowledgement.

On cancellation or rejected changes: Previously saved information remains unchanged.

In all cases: Actual work entry does not rewrite the plan.

Related Business Rules

Attendance, explicit actual-work confirmation, periods, breaks and synchronisation follow BR-025, BR-037, BR-038, BR-039, BR-040, BR-041, BR-044.

Case-specific details:

- Attendance states are awaiting confirmation, present and absent.
- No confirmation does not mean absence.
- A late arrival is reflected in actual times, not a separate attendance state.
- Work date and entry date remain distinct.
- This use case calculates working hours, not wages or payment amounts.
- A present worker may have an actual assignment whose periods are still incomplete.

Acceptance Criteria

1. Planned workers without an attendance decision are shown as awaiting confirmation.
2. An existing unplanned worker can be added to daily attendance.
3. Attendance can be saved before actual periods are complete.
4. One worker's attendance is not duplicated because they work in several areas.
5. Planned periods become actual only after explicit confirmation.
6. Group confirmation supports individual adjustments without changing the source plan.
7. Changes of task or area remain distinguishable in actual periods.
8. A period from 08:00 to 18:00 without a deducted break contributes 10 hours.
9. Periods from 08:00 to 12:00 and 13:00 to 18:00 contribute 9 hours.
10. A period from 08:00 to 18:00 with a non-counted break from 12:00 to 13:00 contributes 9 hours.
11. A break marked as counted does not reduce hours. Another worker's breaks never reduce this worker's periods.
12. Valid hours with an unknown task contribute once to total counted hours and are displayed as unclassified.
13. The same interval is not deducted twice.
14. Overlapping actual periods are rejected.
15. An overnight period is associated with its start date and calculated using its full interval.
16. Retrospective entry preserves the actual work date.

17. Closed-campaign entry respects valid campaign dates and Section 6.3.8, including reopening before adding periods to a date with confirmed work.
18. Supported entries can be saved offline with an explicit pending synchronisation status.
19. Retrying does not unintentionally duplicate saved records.
20. A present worker with incomplete hours can still have an explicitly identified actual campaign and work area.
21. Changing the actual area does not rewrite the planned assignment.
22. Daily attendance remains consistent across that worker's actual work contexts.
23. A break affecting several consecutive task periods is deducted only once across the affected durations.

Related Use Cases

- UC-05 — Manage Properties and Parcels.
- UC-06 — Manage Harvest Campaigns.
- UC-07 — Manage Worker Records.
- UC-08 — Plan Work.
- UC-09 — Consult Work Schedules.
- UC-11 — Correct Work Records.
- UC-19 — Consult Campaign Indicators.

Review Status

Precision, timestamps and break validation remain under Q-WF-01 to Q-WF-03. Completion and reopening follow D-WF-01 and Section 6.3.8.

4.14 UC-11 — Correct Work Records

Purpose

Allow an authorised user to correct saved attendance and actual work information, including assignments without hours, periods, tasks, work areas and breaks.

Corrections update the affected totals without rewriting the original work plan.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors

Primary actor: Farm Owner.

Conditional actor: Field Manager, if included and granted explicit correction permissions by the owner.

Permission to record work does not automatically grant permission to correct confirmed records.

Trigger

The user identifies an error in saved attendance or actual work.

Preconditions

- The target record exists.
- The user has correction permission for the affected records and contexts.
- Campaigns whose work records are being changed are open.
- Offline correction requires the relevant authorised data to be available on the device.

Main Flow

1. The user selects the worker, working date and relevant context.
2. The system displays current attendance, periods, tasks, areas, breaks and totals.
3. The user proposes the correction.
4. The system validates permissions, campaign status, dates, associations, time intervals and break treatment.
5. The system checks the worker's relevant periods for overlaps.
6. The system presents the corrected information and affected totals for review.
7. The user confirms.
8. The system saves the current values and updates the affected local totals.
9. The system displays whether the change is pending synchronisation or accepted by the server.

Alternative Flows and Exceptions

A1 — A campaign is closed. Correction cannot proceed until an authorised user reopens the campaign through UC-06. No additional correction deadline applies.

A2 — Presence is changed to absence. The system identifies all actual campaign-area assignments, including those without hours, and all periods and breaks associated with that worker and working date within the business.

The user must have the required permissions, and affected campaigns must satisfy the reopening rules.

The system requests confirmation before removing the affected work and changing attendance. It must not leave the worker absent while contradictory confirmed assignments, periods or breaks remain.

A3 — Record associations are incorrect. The user may correct the worker, working date, campaign, property or parcel.

The system validates the origin and destination contexts, campaign status and time overlaps. It also checks daily-attendance consistency.

Only the selected work records are reassigned. Other periods belonging to the original worker remain unchanged.

Affected origin and destination totals are updated after acceptance.

A4 — Periods or breaks are invalid. The system identifies the problem and preserves unaffected input. Corrections must not create overlapping actual periods or duplicate break deductions.

A5 — The device is offline. The system saves the correction locally and identifies it as pending. Local totals reflect the pending change. Server permissions, campaign status and consistency are checked on reconnection.

A6 — The server record has changed. The system presents the current server values and the pending correction to an authorised user for manual resolution. No conflicting update is silently substituted.

A7 — The user resolves a conflict. The user may retain the server values or review and confirm a revised correction. Any submitted correction is validated again.

A8 — The server rejects a pending change. The system identifies the rejected operation and required action. It does not present the local correction as server-accepted.

A9 — Saving or transmission fails. The system preserves recoverable information and reports the known storage state. Retrying does not unintentionally duplicate or repeatedly apply the change.

A10 — The user cancels. The shared cancellation conditions apply.

Postconditions

On local acceptance: The corrected values and affected local totals are saved with their synchronisation status.

On server acceptance: Central records and dependent totals reflect the accepted correction. Other clients receive the change through synchronisation.

On cancellation or rejection: Authoritative accepted records remain unchanged. Any unresolved local proposal is clearly identified.

In all cases: Original plans and unrelated records are not modified.

Related Business Rules

Correction permissions and reopening, attendance consistency, counted time and conflict resolution follow BR-025, BR-037, BR-039, BR-040, BR-041, BR-042, BR-043, BR-044.

Case-specific details:

- Worker, date and context associations may be corrected directly when valid.
- Corrections do not rewrite plans or calculate wages.
- Moving one work period must not silently remove or duplicate daily attendance.
- Existing work on the same date must be considered before changing the worker's attendance status.

- Corrections involving several campaigns must satisfy permissions and reopening rules for every affected campaign.

Acceptance Criteria

1. Unauthorised corrections are rejected, including direct server requests.
2. A closed campaign prevents correction until authorised reopening.
3. Old records can be corrected without an additional age-based deadline.
4. Presence-to-absence changes identify actual assignments without hours as well as periods and breaks, and require confirmation before removing contradictory actual work.
5. Cancelling that confirmation preserves the existing attendance, assignments, periods and breaks.
6. Valid worker, date and area corrections update origin and destination totals.
7. Invalid associations and overlapping actual periods are rejected.
8. Corrected breaks are not deducted twice.
9. Plans and unrelated workers' records remain unchanged.
10. Local totals update after local acceptance and identify pending synchronisation.
11. Central totals reflect server-accepted corrections.
12. Concurrent changes require an authorised choice rather than silent replacement.
13. Conflict resolution is validated before server acceptance.
14. Rejected local changes are not displayed as server-confirmed.
15. The application does not maintain a permanent history of previous field values.

Related Use Cases

- UC-06 — Manage Harvest Campaigns.
- UC-07 — Manage Worker Records.
- UC-08 — Plan Work.
- UC-10 — Record Attendance and Actual Work Periods.
- UC-19 — Consult Campaign Indicators.

Review Status

Detailed recovery interactions remain under Q-WF-04. Manual conflict resolution and rejection handling follow the correction rules; D-WF-01 records the confirmed reopening clarification.

4.15 UC-12 — Consult Worker Documents

Purpose

Allow an authorised user to find, open and download worker documents, including uploaded files.

Uploaded documents are included in the current scope.

Application-generated documents may appear in the same consultation workflow if generation is confirmed through UC-20.

Insurance documents must support prepared offline consultation for field operations.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors

Primary actor: Farm Owner.

Conditional actor: Field Manager, if this role is included.

The owner may consult documents within their business.

A Field Manager must have access to insurance documents for workers within their authorised operational scope. Contracts and other documents require additional owner-granted permissions.

Trigger

The user needs to inspect or download a worker document, including insurance information needed in the field.

Preconditions

- The worker record exists.
- The user is authorised to consult the requested document type and worker scope.
- Offline access requires the relevant authorised files and metadata to have been prepared on the device.

An existing document is not a precondition: the document list may be empty.

Main Flow

1. The user selects a worker and opens the document list.
2. The system displays authorised documents, organised as contracts, insurance or other documents.
3. The system identifies each document by its title, type, relevant date or version and signature status where applicable.
4. Insurance entries display the recorded expiry date and validity status.
5. The user selects an available version.
6. The system verifies access and opens the PDF, JPG or PNG file.

7. The user may download the authorised document.

Alternative Flows and Exceptions

A1 — No documents are available.
The system states that no documents are available within the authorised scope.

A2 — A previous version is requested.
For contracts, all retained versions are consultable.

For insurance and other documents, only the current and immediately previous versions are retained and consultable.

Older insurance and other-document versions are deleted under UC-13, including versions that were signed.

A3 — Insurance validity is incomplete.
If the expiry date is missing, the system displays validity as unconfirmed. It does not infer validity from upload date or version order.

A4 — Insurance is expired.
The system identifies the expired status while allowing authorised consultation. Expiry does not hide the document.

A5 — The device is offline.
The system opens prepared authorised insurance files and displays the last successful synchronisation information. The current and previous versions are available where they exist and have been successfully prepared.

A6 — A required file is not local.
The system explains that the document is unavailable offline. It does not claim that no document exists.

A7 — Additional workers need preparation.
While connected, the authorised user may prepare insurance documents for additional existing workers who may be called to work.

A8 — Access is denied.
The shared authorisation conditions apply.

A9 — Retrieval or download fails.
The system explains the failure and offers a retry where appropriate. Failure is distinguishable from an empty document list.

A10 — A shared document is opened.
The document may be associated with several workers without duplicate file storage.

Within the selected worker's view, application-captured signature information is limited to that worker.

A11 — A document was signed outside the application.
The interface identifies the external-signature declaration separately from application signing. It does not claim that the system verified the external signature.

A12 — A retained version is withdrawn.
The interface identifies it as withdrawn and does not present it as available for operational use. Its retained signature information is not changed to unsigned.

A13 — A deleted version remains on an offline device.
The device applies the authorised deletion when it reconnects and synchronises. The interface must not promise knowledge of server changes that have not yet been received.

Postconditions

On success: The selected authorised document version is displayed or downloaded.

On offline consultation: The available local version and its freshness are identifiable.

On failure: The system explains the limitation without exposing protected information.

In all cases: Consultation does not change document contents, version identity or signature status.

Related Business Rules

Document scope, retention, insurance validity, worker-specific signatures and offline preparation follow BR-045, BR-046, BR-047, BR-048, BR-050, BR-051, BR-052, BR-053, BR-054, BR-055, BR-058, BR-059, BR-060, BR-061.

Case-specific details:

- Uploaded documents are included in the current scope.

  Application-generated documents may appear in the same consultation workflow if generation is confirmed through UC-20.
- Supported preview formats are PDF, JPG and PNG.
- Opening and downloading are included. Printing is a future option to assess.
- Cached consultation identifies the last successful synchronisation.

Acceptance Criteria

1. The list includes only documents associated with the selected worker and authorised business scope.
2. The owner can consult documents within their business permissions.
3. An authorised Field Manager can consult relevant worker insurance.
4. Contracts and other documents are unavailable to a manager without the required additional permission.
5. Contract versions remain individually identifiable and consultable.
6. Insurance and other documents expose the current and immediately previous versions.
7. Displayed signature status corresponds to the selected version.
8. PDF, JPG and PNG files can be opened and downloaded when available and authorised.
9. Insurance expiry dates produce the corresponding displayed status.
10. A missing expiry date does not produce a valid status.
11. Prepared insurance files can be opened without connectivity.

12. Additional workers' insurance can be prepared before fieldwork.
13. Offline-unavailable files are distinguishable from missing documents.
14. Retrieval failures are distinguishable from successful queries with no results.
15. Direct unauthorised file access is rejected.
16. Consultation does not modify documents or mark them as signed.

Related Use Cases

- UC-07 — Manage Worker Records.
- UC-08 — Plan Work.
- UC-09 — Consult Work Schedules.
- UC-13 — Upload and Update Worker Documents.
- UC-14 — Review and Sign a Document.
- UC-20 — Generate Worker Documents.

Review Status

Offline protection and source-file sharing remain under Q-DOC-03. Pending signatures on removed versions follow Q-DOC-04. Printing remains optional under Q-MVP-01.

4.16 UC-13 — Upload and Update Worker Documents

Purpose

Allow an authorised user to upload documents and create new versions associated with one or more workers.

Document generation and signing are handled through separate use cases.

This use case covers existing files supplied for upload. Document generation remains a separate candidate capability under UC-20.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors

Primary actor: Farm Owner.

Conditional actor: Field Manager, if included and granted upload or update permissions for the relevant document type.

Insurance consultation permission does not automatically permit document modification.

Trigger

The user receives a document or an updated version to store.

Preconditions

- An internet connection is available.
- The user has permission for the document type and affected workers.
- Associated worker records exist within the same authorised business.
- For a version update, the target document exists.
- A supported file is available.

Main Flow

1. The user opens document management.
2. The user chooses a new document or a new version of an existing document.
3. The user selects or reviews the associated workers.
4. The user selects the document category: contract, insurance or other.
5. The user provides or confirms the document title and selects the file.
6. For insurance, the user enters the expiry date where known.
7. The user may add observations and declare whether the uploaded version was signed outside the application.
8. The system validates permissions, required fields, file format and size.
9. The system presents the submission and relevant version-retention consequences.
10. The user confirms.
11. The system stores the accepted version, records its upload date and version identity, and makes it current.
12. The system applies the retention policy and confirms the outcome.

Alternative Flows and Exceptions

A1 — An insurance expiry date is unknown. The system allows the document to be saved with validity identified as unconfirmed. It does not infer validity from upload date.

A2 — A document covers several workers. The user associates the same document with the relevant authorised workers. The file is not duplicated for each worker.

A3 — A version was signed externally. The user explicitly identifies that status for the uploaded version. It is distinguishable from an application signing event and is not presented as externally verified by the system.

A4 — An existing signed document is updated. The new version becomes current without inheriting an earlier signature. Earlier versions retain their own status while they remain stored.

A5 — Retention removes an older version. For insurance and other documents, the system retains the current and immediately previous versions only. Older versions are deleted, including versions marked as signed. This policy applies to the shared version, not separately to each worker association.

A6 — An incorrect upload may be deleted. An authorised user may confirm deletion when the target has no signature and no dependent records.

A7 — An incorrect upload is signed. The user may withdraw it from use, preserving its file and signature association while it remains retained. The rolling retention policy for insurance and other documents remains applicable.

A8 — Validation fails. The system identifies unsupported formats, files exceeding the size limit, missing required information or invalid associations. No version is accepted until corrected.

A9 — Upload or storage fails. The system reports the failure. The previously current version and its retained history remain unchanged. Failed uploads do not trigger retention deletion.

A10 — A submission is repeated. Retrying the same upload operation does not unintentionally publish duplicate versions or trigger extra deletion.

A11 — The user cancels. The shared cancellation conditions apply.

A12 — Connectivity is unavailable. Upload and update operations cannot complete. Prepared offline insurance consultation remains available through UC-12.

Postconditions

On success: The accepted version becomes current and is available to authorised consultation through the associated worker records.

On insurance or other-document update: Only the current and immediately previous versions remain stored.

On contract update: All accepted versions remain stored, subject to separately authorised document actions.

On failure or cancellation: Previously accepted versions are unchanged and no retention deletion is performed.

In all cases: A new version does not inherit signatures from another version.

Related Business Rules

Document metadata, supported files, shared versions, signature declarations, retention and withdrawal follow BR-046, BR-047, BR-048, BR-049, BR-050, BR-051, BR-052, BR-053, BR-054, BR-055, BR-058, BR-061.

Case-specific details:

- A new accepted version becomes current without an additional approval stage.
- Signature information must distinguish: no registered signature, an external-signature declaration, and an application-captured signature.
- External signature status requires an explicit declaration for the uploaded version.
- A current version may be unsigned, expired, validity-unconfirmed or withdrawn. The interface must not conflate those states.

Acceptance Criteria

1. Required document fields and worker associations are validated.
2. PDF, JPG and PNG files within the agreed limit can be accepted.
3. Unsupported or oversized files are rejected without replacing the current version.
4. Insurance without an expiry date is shown with validity unconfirmed.
5. A new accepted version becomes current.
6. A new version does not inherit an earlier version's signature.
7. External signature declarations remain distinguishable from application signing.
8. A shared document is accessible through its authorised worker associations without duplicate file storage.
9. Updating a shared document exposes the same current version to its authorised associated worker records.
10. All contract versions remain available after a normal version update.
11. After a third accepted insurance or other-document version, only the two most recent versions remain stored.
12. The retention rule also applies to signed older versions.
13. Cancellation and failed uploads do not delete retained versions.
14. Retrying an upload does not create unintended duplicate versions.
15. Withdrawal identifies a document as unavailable for operational use without presenting it as unsigned.
16. A manager's consultation permission alone does not permit modification.
17. Upload and update cannot complete offline.

Related Use Cases

- UC-07 — Manage Worker Records.
- UC-12 — Consult Worker Documents.
- UC-14 — Review and Sign a Document.
- UC-20 — Generate Worker Documents.

Review Status

The proposed file limit remains under Q-DOC-06. Withdrawal recovery and offline cleanup follow Q-DOC-02 and Q-DOC-03. Generation scope remains pending under Q-DOC-05.

4.17 UC-14 — Review and Sign a Document

Purpose

Allow a worker to review a specific document version and confirm a drawn signature in person on an authorised user's device.

The signing result belongs to that worker and the reviewed version.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors

Initiating actor: Farm Owner.

Conditional initiating actor: Field Manager, if included and granted the relevant permissions.

Signing actor: Worker.

The worker does not require an application management account.

This process collects the worker's signature only. Any required owner signature is handled separately.

Trigger

An authorised user initiates signing of a document associated with a worker.

Preconditions

- The worker and document exist.
- The initiating user has permission to access the document and initiate signing.
- The selected version is available and eligible for signing.
- The worker is physically present.
- The worker has not already completed this signing process for the selected version.
- Offline signing requires the authorised document and relevant data to be available on the device.

Main Flow

1. The initiating user selects the worker and document version.
2. The system clearly identifies the intended worker and document.
3. The initiating user confirms that the selected worker is present.
4. The system presents the document for the worker to review.
5. The worker chooses to proceed.
6. The system displays a signature drawing area.
7. The worker draws a signature using a finger or compatible pen.

8. The worker reviews the drawing and its association with the document.
9. The worker explicitly confirms the signing action.
10. The system saves the signature result with the worker, version, initiating user and recorded date and time.
11. The system displays whether the result is pending synchronisation or accepted by the server.

Alternative Flows and Exceptions

A1 — The worker redraws the signature.
Before confirmation, the worker may clear the drawing and try again.

A2 — The worker cancels.
No completed signature is created by the cancelled interaction. The document is not marked as signed for that worker by this interaction.

A3 — No signature has been drawn.
The system requests a drawing before accepting confirmation.

A4 — The document cannot be reviewed.
Signing cannot proceed. The system explains the access or retrieval limitation.

A5 — A signing result already exists.
If the worker has already confirmed a signature for the selected version, the system displays its actual state.

A pending or uncertain result is checked or resumed rather than replaced by another signing interaction.

An accepted signature is not overwritten.

A6 — The document is shared.
Each worker completes an individual review and signing interaction. One worker's signature does not establish another worker's signature.

A7 — The device is offline.
The system saves the confirmed result locally as pending synchronisation. It does not claim server acceptance.

A8 — A newer version exists.
The signature remains linked to the version actually reviewed. It is not transferred to the newer version.

A9 — An older version is synchronised.
If the original version remains available and eligible, the server may accept the signature for that version. The system identifies that a newer version exists.

A10 — The original version is withdrawn or deleted.
The pending result requires explicit resolution.

A withdrawn version does not automatically accept a pending signature.

A version deleted under the retention policy is not recreated merely because an offline signing operation reconnects.

No newer version is marked as signed by that pending operation.

A11 — The submission outcome is uncertain.
The system checks the existing operation before initiating another signing event. It does not automatically ask the worker to sign again.

A12 — Saving or synchronisation fails.
The system reports the actual known state and preserves recoverable pending information. A failed or uncertain submission is not presented as server-confirmed.

Postconditions

On local confirmation: The signature result and identifying information are saved on the device. The result is pending until server acceptance where connectivity is unavailable.

On server acceptance: The signature is associated with the intended worker and reviewed version and is available for authorised consultation.

On cancellation: No completed signature is created by the cancelled interaction.

On unresolved submission: The pending or rejected state is explicit. No other document version is marked as signed.

In all cases: Participation does not grant additional application permissions.

Related Business Rules

Worker-and-version signing, explicit review and confirmation, individual outcomes and retained evidence follow BR-051, BR-052, BR-054, BR-055, BR-056, BR-057, BR-058, BR-059, BR-060, BR-061, BR-062.

Acceptance Criteria

1. Only an authorised user can initiate signing.
2. The worker and document version are identified before confirmation.
3. The worker can review the document, clear the drawing and cancel.
4. An empty drawing cannot be confirmed.
5. Cancellation does not create a completed signing result.
6. Confirmation records the worker, version, initiating user and date and time.
7. A confirmed drawing cannot be replaced through another signing submission.
8. A shared document supports separate worker signatures.
9. One worker's view does not display other workers' application-captured signatures.
10. Prepared documents can be signed offline.
11. Pending local results are distinguishable from server-accepted results.
12. Retrying the same submission does not create duplicate signing events.
13. A newer version does not inherit an older version's signature.
14. A pending signature for an unavailable or ineligible version requires resolution.
15. No owner signature is required by this application workflow.

Related Use Cases

- UC-07 — Manage Worker Records.
- UC-12 — Consult Worker Documents.
- UC-13 — Upload and Update Worker Documents.
- UC-20 — Generate Worker Documents.

Review Status

Operational validation of the signing procedure remains under Q-DOC-01. Removed-version resolution and offline protection remain under Q-DOC-04 and Q-DOC-03; the case-specific pending-signature flows remain applicable.

4.18 UC-15 — Record Transport and Olive Deliveries

Purpose

Allow an authorised user to record and ordinarily correct the departure and delivery details of an olive load, including provenance and transport information.

A delivery may be recorded in two stages or entered retrospectively after arrival.

Weighing tickets, analysis results and monetary values are handled separately.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors

Primary actor: Farm Owner.

Conditional actor: Field Manager, if included and granted the relevant permissions.

The driver is identified through an existing worker record and does not need an application account.

Trigger

An olive load leaves a property, or the user needs to record a delivery that has already occurred.

Preconditions

- The user is authorised to record the transport or confirm the delivery.
- The campaign and relevant properties or parcels exist within the authorised scope.
- The driver has an existing worker record within the business.
- Selected areas are included in the campaign coverage.
- Offline entry requires the relevant authorised reference data to be available.

Main Flow

1. The user starts a delivery record and selects the campaign.
2. The user identifies the property or properties of origin and may select specific parcels.
3. The user may identify collection origin and olive variety for each recorded part of the load.
4. The user selects the driver from existing worker records.
5. The user identifies the vehicle.
6. The user selects a previously used destination or enters a new destination name.
7. The user records the actual departure date and time.
8. The system validates required information, associations and permissions.
9. The user reviews and confirms.
10. The system saves the delivery as in transport and displays its synchronisation status.
11. After arrival, an authorised user opens the record and confirms delivery, providing the actual delivery date and time.
12. The system validates and saves the delivered status.

Alternative Flows and Exceptions

A1 — The delivery already occurred. The user enters departure and delivery information in the same operation. After validation and confirmation, the record is saved as delivered.

A2 — The load has several provenances. The user records the relevant properties and optional parcels separately. Missing quantities are not invented or automatically allocated.

A3 — Collection origin or variety is unknown. The user may leave the information unfilled. The system does not assume tree-collected or ground-collected olives.

A4 — The destination is new. The user enters its name directly. A separate cooperative-management process is not required.

A5 — The driver has no worker record. The user cannot select an external driver by free-text substitution. A worker record must first be created through UC-07 while connected.

A6 — Weighing information is unavailable. Transport and delivery can be recorded without ticket weights. Missing weights are not treated as zero. Each ticket will belong to this delivery under UC-16. Whether one delivery may have several tickets remains an open business question.

A7 — Analysis or monetary values are unavailable. The delivery remains recorded. Results and reported amounts may be added later through their respective use cases.

A8 — Information is invalid or incomplete. The system identifies the affected fields and preserves unaffected input. Delivery time must not precede departure time.

A9 — The device is offline. Departure and delivery confirmation can be saved locally using prepared data. A new destination name may be entered. Changes remain pending synchronisation.

A10 — Server validation rejects a change. The system identifies the rejected operation and required action. It does not present the record as server-accepted.

A11 — Saving or transmission fails. The system preserves recoverable information and reports the actual storage state. Retrying does not unintentionally create duplicate deliveries.

A12 — The user cancels. The shared cancellation conditions apply.

A13 — Saved transport or delivery details require correction. An authorised user may review and correct existing details within the same campaign and operational state. The system validates permissions, chronology and affected associations before accepting the correction. Reversing confirmation, cancellation or moving the delivery to another campaign requires a separately approved exceptional procedure under Q-DA-06.

Postconditions

After departure entry: The delivery has its required transport and provenance information and is identified as in transport.

After delivery confirmation: The record identifies the actual delivery date and time and is marked as delivered.

On offline save: The physical delivery status is distinguishable from its pending synchronisation status.

On cancellation or rejected submission: Previously accepted information remains unchanged.

In all cases: Unknown weights, analysis results and monetary values remain unknown, rather than being replaced with zero.

Related Business Rules

Transport and delivery stages, source areas, driver identity, chronology and source-record separation follow BR-005, BR-063, BR-064, BR-065, BR-066, BR-067, BR-077, BR-078, BR-079.

Case-specific details:

- The destination may reuse a previous name or introduce a new one.
- Delivery requires explicit confirmation by the owner or authorised manager.
- Actual departure and delivery timestamps remain distinct from entry timestamps.
- Any driver's working time is recorded through UC-10 using the agreed task and time-accounting rules.
- Ticket weights and monetary values are obtained through their own source records, not duplicated as independent delivery totals.

Acceptance Criteria

1. A departure record cannot be accepted without its required information.
2. The driver is selected from an authorised existing worker record.
3. A new destination name can be entered without a separate management workflow.
4. One delivery can identify several source properties or parcels.
5. Unknown collection origin and variety can remain unfilled.

6. Unknown quantities are not automatically allocated among source areas.
7. Departure can be saved before arrival and ticket information are available.
8. An authorised user can later confirm the actual delivery.
9. A completed delivery can also be entered in one operation.
10. Delivery time cannot precede departure time.
11. Both stages can be saved offline with the necessary prepared data.
12. New destination names can be recorded during offline entry.
13. In-transport and delivered states remain distinguishable from pending or completed synchronisation.
14. Missing weights or results are not displayed as zero.
15. Repeated submission does not unintentionally duplicate the delivery.
16. Unauthorised operations are rejected by the server.
17. Cancelling preserves saved information.
18. Authorised ordinary corrections validate permissions, chronology and associations within the same campaign and state. They do not authorise reversal, cancellation or campaign reassignment.

Related Use Cases

- UC-05 — Manage Properties and Parcels.
- UC-06 — Manage Harvest Campaigns.
- UC-07 — Manage Worker Records.
- UC-16 — Record Weighing Data and Adjustments.
- UC-17 — Record Analysis Results.
- UC-18 — Manage Cooperative-Reported Ticket Values.
- UC-19 — Consult Campaign Indicators.

Review Status

Exceptional reversal, cancellation and reassignment remain under Q-DA-06. Registered drivers and the transport baseline remain under Q-DA-07 and Q-DA-10. Ordinary correction is covered by A13.

4.19 UC-16 — Record Weighing Data and Adjustments

Purpose

Allow an authorised user to record weighing ticket information and reported adjustments for an olive delivery.

Values are entered manually and retain their source meaning. Incomplete information may be saved.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors

Primary actor: Farm Owner.

Conditional actor: Field Manager, if included and granted the relevant permissions.

The cooperative supplies ticket information. Direct integration and automatic ticket reading are outside the current scope.

Trigger

The user receives a weighing ticket, additional ticket information or identifies an error in previously entered values.

Preconditions

- The delivery exists.
- The user has permission for the requested entry or correction.
- Source information is available for manual entry.
- Offline entry requires the relevant authorised delivery information on the device and is limited to non-financial data. Monetary information follows FR-119. Mixed-content attachments require the policy pending under Q-AT-01.

Main Flow

1. The user selects the delivery and opens its weighing information.
2. The user creates a ticket record or opens an existing one.
3. The user enters the available ticket number or reference, cooperative and weighing date and time.
4. The user enters the known weights, identifying their meaning and units.
5. The user records known operational weight adjustments as reported by the cooperative. Monetary adjustment information requires online access and the financial permissions of UC-18.
6. Where known, the user identifies whether an adjustment is already reflected in a reported weight. Monetary effects are handled online as financial information and are not copied into offline datasets.
7. The user may attach an image or PDF of the ticket.
8. The system validates the available information without inventing missing values.
9. The system presents source values, adjustments and missing information for review.
10. The user confirms the submission.

11. The system saves the current values and displays the synchronisation and attachment-upload status.

Alternative Flows and Exceptions

A1 — Ticket information is incomplete. The user saves the known information. Missing fields remain explicitly identified. Unknown weights are not replaced with zero.

A2 — A reported weight is ambiguous. The system permits the source information to be retained as incomplete. The value is not used in calculations requiring an unconfirmed interpretation.

A3 — An adjustment is already applied. The system retains that indication and does not deduct the adjustment again.

A4 — Adjustment treatment is unknown. The information remains unresolved. The system does not assume that the adjustment must be applied or has already been applied.

A5 — Calculation rules are unconfirmed. The system stores reported values without generating an unsupported final weight or combined deduction.

A6 — The delivery contains a mixed load. Ticket totals remain associated with the delivery. They are not automatically divided among its properties or parcels.

A7 — A saved value requires correction. An authorised user reviews and confirms the replacement value. The system stores the current information without a permanent correction history and updates affected results.

A8 — A possible duplicate is detected. The system presents the available ticket and delivery information for review. A ticket number alone is not assumed to be globally unique.

A9 — The device is offline. Only non-financial structured ticket information is saved locally as pending synchronisation. Monetary fields and adjustments are excluded. Upload deferral does not authorise offline disclosure of monetary content in an attachment. Mixed-content file handling remains pending under Q-AT-01.

A10 — Attachment upload fails. The system identifies the attachment as unavailable or pending. It does not imply that the source file has been stored on the server. The structured record retains its actual storage status.

A11 — Validation or access fails. The system identifies the affected submission and preserves recoverable input without exposing protected information.

A12 — Saving or synchronisation fails. The system reports the known storage state. Retrying the same operation does not unintentionally duplicate tickets or adjustments.

A13 — The user cancels. The shared cancellation conditions apply.

Postconditions

On success: Known ticket values and adjustments are saved against the selected delivery, with their meaning and units.

On incomplete entry: Unresolved information remains visible and is not treated as confirmed calculation input.

On correction: Current values replace the incorrect information, and affected totals are updated according to the approved calculation rules.

On offline save: Local changes and pending attachment uploads remain distinguishable from server acceptance.

On cancellation or rejected submission: Previously accepted information remains unchanged.

Related Business Rules

Ticket identity, weight interpretation, adjustment eligibility, online financial boundaries and correction follow BR-026, BR-068, BR-069, BR-070, BR-076, BR-077, BR-078, BR-079.

Case-specific details:

- Ticket references are interpreted with their cooperative and source context.
- Source attachments do not automatically populate structured fields.
- Decimal values are supported. Precision and rounding rules require validation against cooperative examples.
- Local save, server acceptance and attachment upload are separate states.
- Reported ticket amounts are managed through UC-18 and associated with this ticket.
- A recorded adjustment does not automatically modify the cooperative-reported monetary amount.

Acceptance Criteria

1. A saved ticket is associated with its intended authorised delivery.
2. Ticket identification includes the available source context.
3. Known weights retain their meaning and units.
4. An incomplete ticket can be saved.
5. Missing weights are not displayed as zero.
6. Unknown adjustment treatment remains explicitly unresolved.
7. Already-applied deductions are not reapplied automatically.
8. Unconfirmed formulas do not produce an apparently definitive final weight.
9. Mixed-load ticket totals are not automatically distributed among areas.
10. An image or PDF may be attached without automatic value extraction.
11. Offline structured entry remains identifiable as pending synchronisation.
12. Pending or failed attachment uploads are distinguishable from stored files.
13. Authorised corrections update the current values and affected results.
14. No permanent correction history is presented for ticket fields.
15. Repeated submission does not unintentionally duplicate records.
16. Unauthorised changes are rejected by the server.
17. Cancelling preserves saved information.

Related Use Cases

- UC-06 — Manage Harvest Campaigns.
- UC-15 — Record Transport and Olive Deliveries.
- UC-17 — Record Analysis Results.
- UC-18 — Manage Cooperative-Reported Ticket Values.
- UC-19 — Consult Campaign Indicators.

Open Business Questions

See Q-DA-01 and Q-DA-02 for ticket multiplicity and weight/adjustment interpretation, and Q-AT-01 for mixed-content files.

Review Status

Ticket multiplicity, weight interpretation and completeness remain under Q-DA-01, Q-DA-02 and Q-DA-08. Mixed-content attachments and source-file recovery remain under Q-AT-01 and Q-AT-02.

4.20 UC-17 — Record Analysis Results

Purpose

Allow an authorised user to record an olive-oil yield analysis associated with one delivery.

The system preserves the distinction between pending results, reported values and any application-calculated indicators.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors

Primary actor: Farm Owner.

Conditional actor: Field Manager, if included and granted the relevant permissions.

The cooperative or laboratory supplies the analysis information. Direct integration is not assumed.

Trigger

The user receives analysis information, needs to complete a pending result or identifies an entry error.

Preconditions

- The delivery exists.
- The user has permission for the requested operation.

- Offline entry requires the relevant authorised delivery information to be available on the device.

Complete analysis information is not required: a pending or incomplete entry may be saved.

Main Flow

1. The user selects a delivery and opens its analysis information.
2. The user creates an analysis entry or opens an existing one.
3. The user records the available analysis reference, reporting entity and analysis date.
4. The user enters the reported yield percentage where available.
5. The user records the reference weight and its meaning where known.
6. The user may attach an image or PDF of the source result.
7. The system validates the available information using the approved rules.
8. The system presents the reported values and identifies missing information.
9. The user reviews and confirms.
10. The system saves the current information and displays its synchronisation and attachment-upload status.

Alternative Flows and Exceptions

A1 — The result has not arrived. The entry remains pending. The system does not assign a zero yield percentage.

A2 — The result is known but reference information is incomplete. The system saves the reported result and identifies the missing information. Calculations requiring that information remain unavailable or explicitly incomplete.

A3 — An entry error is corrected. An authorised user reviews and confirms the corrected current values. The system updates affected indicators according to the approved rules, without maintaining detailed correction history.

A4 — A different analysis is received. The system distinguishes a new source analysis from a correction to an existing entry. Whether the new analysis replaces or supplements an earlier result follows the confirmed business policy. No automatic selection or double counting is assumed.

A5 — The campaign is closed. Analysis information for an existing delivery may be added or completed without reopening the campaign.

A6 — The device is offline. The system saves the available information locally as pending synchronisation. An attachment may remain pending upload until connectivity returns.

A7 — An attachment upload fails. The system identifies the attachment as pending or unavailable. Structured analysis information retains its actual storage status.

A8 — Information is invalid. The system identifies the affected values and preserves unaffected input. Validation uses approved value ranges and precision rules.

A9 — Access or server validation fails. The system rejects the affected operation and explains the required action without exposing protected information.

A10 — Saving or transmission fails. The system preserves recoverable information and reports the actual storage state. Retrying the same submission does not unintentionally duplicate analysis records.

A11 — The user cancels. The shared cancellation conditions apply.

Postconditions

On success: The analysis information is saved against the selected delivery, with its available source references.

On incomplete entry: Pending results and missing reference information remain explicitly identified.

On correction: Current values reflect the accepted correction. Dependent indicators use the updated information according to approved calculation rules.

On offline save: Local information and pending attachments remain distinguishable from server-accepted data.

On cancellation or rejection: Previously accepted information remains unchanged.

In all cases: Recording an analysis does not change the delivery date or overwrite official weighing values.

Related Business Rules

Analysis identity, reference weights, missing results, late entry and aggregation eligibility follow BR-026, BR-071, BR-072, BR-074, BR-076, BR-078, BR-079, BR-085, BR-086.

Case-specific details:

- Other analysis parameters require an identified business need.
- Analysis reference, reporting entity, analysis date and reference weight are recorded where available.
- Analysis date, delivery date and entry date have distinct meanings.
- Partial indicators identify their known coverage.
- Theoretical oil quantities remain distinguishable from oil actually received.
- Source attachments do not automatically populate structured fields.
- Monetary valuation is not calculated from yield within the current scope.

Acceptance Criteria

1. A saved analysis belongs to the selected authorised delivery.
2. The same analysis is not associated with several deliveries.
3. An analysis can remain pending without a yield percentage.
4. A pending result is distinguishable from an explicitly reported zero.

5. A known yield can be saved with missing reference information clearly identified.
6. Missing reference weights are not replaced with assumed values.
7. Unsupported calculations are not presented as definitive results.
8. Analysis entry does not alter the original delivery date or ticket weights.
9. An image or PDF may be attached without automatic extraction.
10. Authorised entry corrections update the current values.
11. A different source analysis is not silently merged with an earlier result.
12. Aggregates exclude unresolved result selections and prevent duplicate contributions.
13. Results for existing deliveries can be added after campaign closure.
14. Supported entry works offline with an explicit pending status.
15. Attachment-upload status is distinguishable from structured-data status.
16. Repeated submission does not unintentionally create duplicate analyses.
17. Unauthorised modifications are rejected.
18. Cancelling preserves saved information.

Related Use Cases

- UC-06 — Manage Harvest Campaigns.
- UC-15 — Record Transport and Olive Deliveries.
- UC-16 — Record Weighing Data and Adjustments.
- UC-19 — Consult Campaign Indicators.

Open Business Questions

See Q-DA-03 and Q-DA-04 for analysis multiplicity, reference weights, selection, precision and candidate yield calculations.

Review Status

Analysis selection, reference weights and completeness remain under Q-DA-03, Q-DA-04 and Q-DA-08. Attachment handling follows Q-AT-01 and Q-AT-02.

4.21 UC-18 — Manage Cooperative-Reported Ticket Values

Purpose

Allow an authorised user to record, consult and correct the current monetary value reported by the cooperative for a weighing ticket.

The ticket identifies the associated delivery. The application does not calculate price estimates or process payments.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors

Primary actor: Farm Owner.

Conditional actor: Field Manager, if included and granted the relevant monetary-data permissions.

Operational work permissions do not automatically grant access to monetary information.

The cooperative supplies the reported amount. Direct integration is not assumed.

Trigger

The user receives a provisional or final value, needs to consult it, or identifies an entry error.

Preconditions

- The weighing ticket exists and is associated with a delivery.
- The user has permission for the requested operation.
- Consultation, registration and correction of monetary information require an internet connection.
- For entry of an amount, source information is available.

A known amount is not required for consultation: the ticket may still be awaiting its value.

Main Flow

1. The user selects a delivery and its weighing ticket, or opens the ticket directly.
2. The system displays the current monetary information or identifies the value as awaiting receipt.
3. The user chooses to consult, record or correct the reported value.
4. For entry or correction, the user provides the amount in euros, reporting date and provisional or final status.
5. The system identifies the cooperative from the ticket context.
6. The user may add a source reference, observations and an image or PDF.
7. The system validates permissions, required information and ticket association.
8. The system presents the proposed current value for review.
9. The user confirms.
10. The system saves the current amount and status, replacing any previous value.
11. The system updates affected monetary indicators and confirms the result.

For consultation only, the flow ends after the available information is displayed.

Alternative Flows and Exceptions

A1 — No amount has been reported. The system identifies the ticket as awaiting its monetary value. It does not create a zero amount.

A2 — A final value replaces a provisional value. The system replaces the current amount and changes its status to final. The provisional amount is not retained as a separate historical value.

A3 — An entered value is incorrect. An authorised user reviews and confirms the corrected amount and status. Only the current information is retained.

A4 — Source details are incomplete. Optional references or observations may be added later. A reported amount must not be invented to complete an otherwise pending entry.

A5 — The ticket covers a mixed load. The amount remains associated with the ticket. It is not automatically allocated across properties or parcels.

A6 — The campaign is closed. The user may record or update the reported value of an existing ticket without reopening the campaign.

A7 — An attachment fails to upload. The system identifies the attachment failure separately from the amount's save status. It does not claim that the file was successfully stored.

A8 — Information is invalid or access is denied. The system rejects the affected operation and preserves recoverable input without exposing protected information.

A9 — Connectivity is unavailable. Monetary information cannot be consulted, registered or corrected offline. Prepared operational summaries and offline PDF reports exclude monetary information. Previously accepted monetary values remain unchanged.

A10 — Saving fails. The system reports the failure. The previously accepted amount and its status remain unchanged.

A11 — Submission is repeated. Retrying the same operation does not create additional monetary contributions for the same ticket.

A12 — The user cancels. The shared cancellation conditions apply.

Postconditions

On successful entry or correction: The ticket has one current reported amount, its provisional or final status and the required source information.

On replacement: The accepted amount replaces the previous value. Affected totals use the current value only.

On consultation: Current information is displayed, or the absence of a reported value is explicit.

On cancellation or rejected changes: Previously accepted information remains unchanged.

In all cases: The operation does not create a payment, alter ticket weights or modify analysis results.

Related Business Rules

Ticket valuation, single current amount, replacement, online permission and source allocation follow BR-073, BR-074, BR-075, BR-076, BR-077, BR-078, BR-079, BR-092.

Case-specific details:

- Currency is euros, with two decimal places.
- Reporting date and application entry date have different meanings.
- Source attachments are optional and do not automatically populate fields.
- If several tickets per delivery are approved, delivery-level monetary totals require a confirmed eligibility rule.
- Monetary source attachments are not worker documents. They do not automatically inherit the two-version retention policy of UC-13.

Acceptance Criteria

1. A saved amount is associated with the selected authorised ticket.
2. Consultation through different delivery or ticket screens refers to the same value and does not duplicate its contribution.
3. Unknown amounts are shown as awaiting receipt, not zero.
4. An explicitly reported zero remains distinguishable from missing data.
5. Amounts use euros and two decimal places.
6. The current value identifies its provisional or final status.
7. A final amount replaces the provisional amount and removes its superseded contribution from current totals.
8. Corrections retain only the current values.
9. Totals use the accepted replacement without adding the previous amount.
10. Reported deductions are not automatically applied again.
11. Mixed-load amounts are not automatically distributed across areas.
12. An image or PDF may be attached.
13. Attachment failure is distinguishable from failure to save the amount.
14. Existing-ticket values can be completed after campaign closure without reopening.
15. Unauthorised access or modification is rejected by the server.
16. Consultation, registration and correction require connectivity.
17. Repeated submission does not create duplicate monetary contributions.
18. Cancelling preserves accepted information.
19. No payment transaction is created.
20. No monetary content is included in the prepared offline operational summary.

Related Use Cases

- UC-06 — Manage Harvest Campaigns.
- UC-15 — Record Transport and Olive Deliveries.
- UC-16 — Record Weighing Data and Adjustments.
- UC-19 — Consult Campaign Indicators.

Review Status

Ticket eligibility and mixed-source allocation remain under Q-DA-01 and Q-DA-05. Reclassification of a final amount remains under Q-DA-09. Source-attachment handling follows Q-AT-01 and Q-AT-02.

4.22 UC-19 — Consult Campaign Indicators

Purpose

Allow an authorised user to understand the progress and results of a harvest campaign through operational, production and permitted monetary indicators.

The same consultation supports ongoing work and the final campaign review. Closing a campaign does not freeze its indicators.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors

Primary actor: Farm Owner.

Conditional actor: Field Manager, if included and granted the relevant consultation permissions.

Monetary indicators require separate monetary-data permission.

Trigger

The user wants to inspect campaign progress, review final results, investigate an indicator or export a summary.

Preconditions

- The campaign exists.
- The user has permission to consult the requested scope.
- Offline consultation requires a previously prepared operational summary and relevant authorised local information.

Operational activity is not a precondition: a campaign may have no recorded work or deliveries yet.

Main Flow

1. The user selects one harvest campaign.
2. The system identifies the campaign and the authorised consultation scope.
3. The user selects the campaign overview, a property summary or parcel detail.
4. The user may apply date, worker and task filters where relevant.
5. The system retrieves or calculates the eligible indicators using approved rules.
6. The system presents totals, tables and useful simple charts.
7. Each indicator identifies its meaning, unit, scope and relevant completeness.
8. The user may open an indicator to inspect its contributing records.
9. The user may export the selected summary as a PDF.

Indicator Groups

Work indicators:

- Distinct workers with confirmed daily presence and an explicitly identified actual assignment within the selected campaign.
- Distinct start dates with confirmed actual work periods.
- Total confirmed actual working hours.
- Hours by worker.
- Hours by task type: machinery operation and general work.
- Actual campaign assignments whose hours or task information remain incomplete.

Production indicators:

- Confirmed deliveries.
- Known weight totals, separated by weight meaning.
- Available yield results and approved yield aggregates.
- Pending analyses and relevant result coverage.

Monetary indicators:

- Current cooperative-reported ticket amounts.
- Separate provisional and final totals.
- Tickets awaiting monetary values.

Monetary consultation requires connectivity in the current version.

Alternative Flows and Exceptions

A1 — No matching activity exists. The system explains that no records match. Zero is used only where it is a valid result for the defined indicator. An undefined average is not shown as zero.

A2 — Data is incomplete. The system identifies pending information. Meaningful partial results show their known coverage and exclusions.

A3 — The campaign is closed. The user may consult its current results. Late analysis results, ticket information and reported values update the relevant indicators when accepted.

A4 — An accepted correction exists. Indicators use the current accepted values under the applicable correction rules. Previous values are not added as additional contributions.

A5 — Loads have mixed provenance. Known totals may contribute to the campaign summary. Property or parcel allocation is shown only when supported by known quantities or approved rules. Unallocated information remains identifiable.

A6 — Yield calculation is unsupported. The system displays available source results but does not invent reference weights or aggregation formulas.

A7 — Access covers only part of the campaign. Results are limited to the authorised scope. A restricted view is not labelled as the unrestricted campaign total.

A8 — The device is offline. The system displays the prepared operational summary and its last successful update. Monetary indicators are unavailable offline.

A9 — Local changes are pending. The operational summary combines the prepared server baseline with relevant local changes.

Creates are included once. Updates replace their corresponding baseline values. Locally accepted removals or cancellations remove the affected local contributions.

Affected indicators are identified as provisional until server acceptance. The interface does not imply knowledge of unsynchronised changes from other devices.

A10 — A pending change is rejected. The system distinguishes the unresolved proposal from server-accepted information.

The accepted summary is restored or recalculated from accepted values. Any still-visible local proposal is separately identified and is not presented as part of the accepted total.

A11 — Requested detail is unavailable offline. The system explains that connectivity is required for that detail. It does not present unavailable records as an empty result.

A12 — Retrieval or calculation fails. The system identifies the affected indicator and offers a retry where appropriate. Failure is distinguishable from zero or a successful query with no records.

A13 — A PDF is requested. The system exports the selected authorised summary with its campaign, filters, generation date and time, completeness and synchronisation limitations. The export represents that moment, rather than a permanently frozen campaign. An offline PDF contains operational information only. It excludes monetary indicators and identifies pending local contributions and the last successful synchronisation.

Postconditions

On success: The user can inspect the current indicators for the selected authorised scope and access available supporting records.

On offline consultation: The displayed operational summary identifies its synchronisation date and pending local changes.

On export: A PDF reflects the selected summary and its relevant limitations at generation time.

In all cases: Consultation does not modify source records, create payments or close the campaign.

Related Business Rules

Actual-work measures, source eligibility, allocation, freshness, filters and financial/report access follow BR-029, BR-037, BR-040, BR-041, BR-080, BR-081, BR-082, BR-083, BR-084, BR-085, BR-086, BR-087, BR-088, BR-089, BR-090, BR-091, BR-092.

Case-specific details:

- Property summaries are primary, with parcel detail where available.
- Property-level records without a parcel remain identifiable.
- Cross-campaign comparison and detailed Excel export are future enhancements.

Acceptance Criteria

1. The selected campaign and authorised scope are clearly identified.
2. Property summaries and available parcel detail can be consulted.
3. Applicable filters return the relevant contributing records.
4. A worker present on several days counts once in the campaign worker total.
5. Working-day counts do not increase once for each worker on the same date.
6. Hours distinguish machinery operation from general work.
7. Hours follow confirmed actual periods and the agreed break rules.
8. Incomplete hours remain visible as incomplete information.
9. Overnight work uses its start date.
10. Mixed loads do not duplicate weights or amounts across source areas.
11. Unsupported yield calculations remain unavailable rather than invented.
12. Monetary totals use current ticket values and distinguish provisional from final amounts.
13. Unknown values are distinguishable from reported zeros.
14. Available indicator detail matches the contributing authorised records.
15. Offline operational summaries identify their last successful update.
16. Pending local changes affect provisional totals without duplicate counting.
17. Offline monetary indicators are not exposed.
18. Retrieval failures are distinguishable from valid empty results.
19. PDF exports identify the campaign, filters, generation time and relevant limitations.
20. Consultation does not modify source data.

Related Use Cases

- UC-05 — Manage Properties and Parcels.
- UC-06 — Manage Harvest Campaigns.
- UC-07 — Manage Worker Records.
- UC-08 — Plan Work.
- UC-10 — Record Attendance and Actual Work Periods.
- UC-11 — Correct Work Records.
- UC-15 — Record Transport and Olive Deliveries.
- UC-16 — Record Weighing Data and Adjustments.
- UC-17 — Record Analysis Results.
- UC-18 — Manage Cooperative-Reported Ticket Values.

Review Status

Source eligibility and allocation follow Q-DA-01 to Q-DA-05. Indicator date bases and offline/PDF preparation remain under Q-IN-01 and Q-IN-02.

4.23 UC-20 — Generate Worker Documents

Status

Scope awaiting confirmation with the team and client.

The use case identifier is reserved. The detailed generation workflow is not yet an approved requirement.

Candidate Purpose

Generate selected worker documents from agreed templates and reviewed information, reducing repeated manual entry.

This capability is distinct from uploading existing files through UC-13 and signing a version through UC-14.

Candidate Actors

Farm Owner.

Field Manager only if the role is included and the relevant generation permissions are granted.

Questions for Confirmation

The complete generation-scope questions are recorded under Q-DOC-05. UC-20 remains reserved pending that confirmation.

Consistency Constraints

If generation is included:

- The user must review the document before accepting it.
- Generation must not automatically mark the document as signed.
- Editing generation inputs must not silently modify source worker records.
- The accepted output must identify its document category and version.
- Access and version retention must align with UC-12 and UC-13.
- Signing must follow UC-14 for the specific reviewed version.

Related Use Cases

- UC-07 — Manage Worker Records.
- UC-12 — Consult Worker Documents.
- UC-13 — Upload and Update Worker Documents.
- UC-14 — Review and Sign a Document.

Acceptance Status

Detailed acceptance criteria and implementation estimates remain pending scope confirmation.

5 Functional Requirements

This section defines the system behaviours derived from the use cases and reviewed business decisions.

Each requirement has a unique identifier and acceptance criteria. The mapping identifies its originating use cases.

Conditional requirements and unresolved business decisions are explicitly identified. Catalogue inclusion does not by itself establish MVP priority or stakeholder approval. Acceptance details tied to a pending decision remain provisional; the decision register is maintained in Chapter 10.

The integrated catalogue covers FR-001 to FR-139, including account access, properties and campaigns, workforce, documents and signing, deliveries, weighing, analysis, reported values and campaign indicators.

5.1 Requirements Catalogue

5.1.1 Registration, Authentication and Access Management

Requirements FR-001 to FR-012 and FR-019 cover account access and access revalidation independently of whether the Field Manager role is included.

Requirements FR-013 to FR-018 and FR-020 are conditional on approval of the Field Manager role. The manager-specific application of FR-019 is conditional on that role; its general behaviour also applies to Owner sessions and pending operations.

Catalogue inclusion does not by itself establish MVP priority or stakeholder approval.

ID	Requirement	Acceptance Criteria
FR-001	The system shall allow a visitor to self-register as an Owner.	The registration requests the person's name, email address, password and business name. No invitation from another user is required.
FR-002	The system shall validate registration information and prevent duplicate accounts for the same email identity.	Invalid information is rejected. Email comparison follows a consistent normalisation policy across registration, login, recovery and invitations.
FR-003	The system shall verify the registered email before granting active business access.	A pending registration can request a replacement verification message. Invalid or expired verification links do not activate access.
FR-004	The system shall activate the verified Owner account and create its independent business workspace.	The account receives ownership of that workspace. Repeated activation requests do not create duplicate accounts or businesses. No operational records are created automatically.

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ID	Requirement	Acceptance Criteria
FR-005	The system shall allow activated account holders to log in using email and password.	Successful authentication establishes a session and loads the currently authorised business access. Unverified accounts do not receive active workspace access.
FR-006	The system shall allow the user to log out online or offline.	Logout blocks access through the local session. Pending operational changes are identified and preserved in protected storage rather than silently discarded.
FR-007	The system shall validate the account session and business permissions before accepting protected online operations.	Authentication alone does not grant access to every business. Expired sessions require renewed authentication.
FR-008	The system shall support authorised offline continuation after a previous successful online authentication.	Only supported operations and locally available data are accessible. Initial login and renewal of offline authorisation require connectivity.
FR-009	The system shall allow an account holder to request password recovery by email.	The acknowledgement does not disclose whether the submitted email belongs to an eligible account. Eligible accounts receive a recovery message.
FR-010	The system shall allow a new password to be established through a valid recovery authorisation.	Invalid, expired or used authorisations are rejected. The new password must satisfy the configured password policy.
FR-011	The system shall invalidate existing online sessions after a successful password reset.	The consumed reset authorisation cannot be reused. Business ownership and permissions remain unchanged. Disconnected sessions follow the offline authorisation policy.
FR-012	The system shall isolate local workspace data and pending changes between accounts.	An account logging in on a shared device cannot access another account's cached business information or submit its pending changes.
FR-013	The system shall allow an Owner to invite a Field Manager to the Owner's business by email.	The invitation identifies the business, intended recipient, responsibility scope and initial permissions.
FR-014	The system shall require explicit acceptance of a valid manager invitation.	The authenticated and verified email matches the invitation. Creating a manager account through an invitation does not automatically create an Owner business workspace.
FR-015	The system shall allow an Owner to inspect, resend and cancel pending manager invitations.	Cancelled or expired invitations cannot grant access. Resending does not create duplicate memberships.

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ID	Requirement	Acceptance Criteria
FR-016	The system shall allow the Owner to define and modify a manager's responsibility scope and permissions.	Consultation, operational changes, corrections, document operations and financial access can be distinguished. Saved changes govern subsequent online requests.
FR-017	The system shall enforce insurance-document consultation for managers responsible for the relevant workers.	The configuration cannot remove required insurance access while retaining that responsibility. Other document access remains separately controlled.
FR-018	The system shall allow the Owner to revoke a manager's business access.	Revocation requires confirmation and blocks subsequent online requests without deleting the account or business records.
FR-019	The system shall revalidate permissions when a device reconnects or submits pending changes.	Revoked or reduced access is enforced. Unauthorised proposals are not silently accepted, and the user is informed of unresolved pending changes.
FR-020	The system shall prevent a Field Manager from changing their own authorisation or granting business access to other accounts.	Invitation, permission administration and access revocation are restricted to the Owner under the current role model.

5.1.2 Properties and Parcels

ID	Functional Requirement	Acceptance Criterion
FR-021	The system shall allow an authorised user to create a property with a name and textual location.	A property can be saved with both fields. Missing either field prevents acceptance and produces an explanatory message.
FR-022	The system shall allow the property's declared area in hectares and olive varieties to be recorded and updated optionally.	The property can be saved without these values. They can be added later. An unknown area is not displayed as zero.
FR-023	The system shall allow an authorised user to create named parcels within a property, with optional area and variety information.	A parcel can be created with a name and parent property, without requiring area, varieties or map information.
FR-024	The system shall allow authorised consultation and updating of properties and parcels while preserving existing record associations.	Changing descriptive information does not reassign operational records or change the identity of the property or parcel.
FR-025	The system shall allow authorised archival and reactivation of properties and parcels.	Archival preserves history and blocks new operational associations. Reactivation restores availability, subject to the parent property's availability.

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ID	Functional Requirement	Acceptance Criterion
FR-026	The system shall allow authorised deletion of an eligible property or parcel after explicit confirmation.	Deletion succeeds only without associated records. Otherwise, it is refused and archival is offered. Cancelling confirmation preserves the record.
FR-027	The system shall allow offline consultation of previously downloaded authorised property and parcel information.	Prepared information remains consultable without connectivity. Information unavailable locally is not presented as proof that no record exists.

5.1.3 Harvest Campaigns

ID	Functional Requirement	Acceptance Criterion
FR-028	The system shall allow an authorised user to create a campaign with a name, planned start date and selected coverage.	Creation requires all three elements. An actual end date is not required when preparing a campaign.
FR-029	The system shall allow campaign coverage to include whole properties or selected parcels from several properties.	Both selection modes are supported. A whole property and its parcels cannot be selected as duplicate coverage within the same campaign.
FR-030	The system shall allow parcel creation during campaign preparation through the property-management workflow.	A user with property-management permission can create a parcel and return to campaign selection. The parcel remains associated with its property for future use.
FR-031	The system shall allow authorised consultation and updating of campaign details and coverage.	Valid areas can be added. Direct removal of an area with operational records in that campaign is blocked. Existing records are not silently reassigned.
FR-032	The system shall maintain distinct planned-start, actual-start and actual-end information for a campaign.	Changing the planned start does not overwrite the actual start. The interface distinguishes planned and actual dates. The actual-start recording mechanism remains pending under Q-PC-02. The dependent acceptance detail remains pending under Q-PC-02.
FR-033	The system shall allow an authorised user to close a campaign by confirming its actual end date.	Closure records the end date and preserves existing records. Later external information for existing deliveries and tickets can still be completed.
FR-034	The system shall allow authorised reopening of a campaign after validation of its scope and dates.	Reopening preserves existing records and checks relevant campaign conflicts. It does not silently discard prior activity.
FR-035	The system shall allow authorised archival, unarchival and deletion of eligible campaigns.	Archival changes list visibility without changing open or closed status. Deletion requires no operational records and explicit confirmation.
FR-036	The system shall detect and prevent campaign overlap in the same area and period.	For determinable periods, conflicting coverage is rejected. Simultaneous campaigns on different parcels of the same property are permitted. Open-ended period handling remains pending under Q-PC-01. The dependent acceptance detail remains pending under Q-PC-01.

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ID	Functional Requirement	Acceptance Criterion
FR-037	The system shall allow offline consultation of previously downloaded authorised campaigns.	Prepared campaigns and coverage are consultable without connectivity, with synchronisation information. Management operations remain unavailable offline.

5.1.4 Common Acceptance Conditions

The following conditions apply to the management operations in this group:

- The server shall reject operations without the relevant permission.
- A Field Manager, if included, shall receive only the permissions granted by the Farm Owner.
- Creation, modification, archival, reactivation, unarchival, closure, reopening and deletion require connectivity.
- A failed save shall not be presented as a successful server-confirmed operation.
- Cancelling a proposed change shall preserve previously saved information.
- Destructive actions require explicit confirmation.

The requirements are proposed specification statements. They do not establish that the corresponding behaviour has been implemented or verified.

5.1.5 Worker Records

ID	Functional Requirement	Acceptance Criterion
FR-038	The system shall allow an authorised user to create a worker record with a full name, telephone number and automatic internal identifier.	Name and telephone are required. Creation assigns an identifier without creating an application user account.
FR-039	The system shall record whether a worker can operate machinery, with an optional capability description.	Capability and description can be recorded. Changing them does not modify previously recorded tasks.
FR-040	The system shall allow authorised consultation and updating of worker records, preserving associations and recording the last-modified date.	Updates preserve links to plans, work and documents. Current values and the last-modified date are retained without detailed field-change history.
FR-041	The system shall warn of matching worker names or telephone numbers within the business, while allowing confirmation of a distinct record.	A match generates a warning. The user may confirm a separate person. Other businesses' records are not exposed or automatically merged.
FR-042	The system shall allow authorised worker inactivation and reactivation without deleting operational history.	Inactive workers leave the usual new-work selections. Existing future plans are flagged, not silently changed. Reactivation reuses the same identity.
FR-043	The system shall allow worker deletion only without associated records and after explicit confirmation.	Associated records prevent deletion. Cancelling confirmation preserves the worker.
FR-044	The system shall allow offline consultation of prepared authorised worker information.	Downloaded information remains consultable offline. Creation, editing, inactivation, reactivation and deletion require connectivity.

5.1.6 Work Planning and Schedule Consultation

ID	Functional Requirement	Acceptance Criterion
FR-045	The system shall allow planning for one or several dates by selecting a campaign, eligible work areas and workers.	Each selected date receives an independent plan with the intended campaign, areas and workers.
FR-046	The system shall support optional planned hours and tasks, requiring periods when a worker is assigned to several areas on the same working day.	Single-area planning can remain without hours or tasks. Multiple-area assignments for the same worker and day require their periods.
FR-047	The system shall allow common planned periods, tasks and optional breaks to be applied to selected workers with individual adjustments.	Group values affect selected workers only. Individual changes preserve other entries. Planned breaks remain subject to actual confirmation.
FR-048	The system shall allow an existing plan or actual work to be reused as a reviewed planning proposal for new dates.	Reuse preserves source records and does not create confirmed attendance or actual work for target dates. Existing target plans are not silently overwritten.
FR-049	The system shall validate planned intervals and prevent known overlaps for the same worker across areas and campaigns.	Overlapping defined periods are rejected. Missing hours are not represented as successfully checked conflict-free periods.
FR-050	The system shall represent overnight planned and actual periods using complete start and end information and the start-date association.	A period from 22:00 to 02:00 the next day has four hours before deductions and belongs to its start date. Validation considers adjacent dates.
FR-051	The system shall allow plan changes, including worker removal or replacement, before actual work exists in the affected context.	Permitted changes require review. Once actual work exists for the affected daily area context, the corresponding plan is preserved as the original expectation.
FR-052	The system shall provide daily, worker-based and weekly schedule consultation with the agreed filters.	Views represent the same underlying plans. Filters include campaign, property, parcel, worker and task type.
FR-053	The system shall prepare authorised schedules for the current and following week for offline consultation.	Prepared schedules identify their last synchronisation and pending local changes. Unavailable information is distinguishable from a successful query with no plans.
FR-054	The system shall allow offline plan creation and adjustment using prepared authorised data, with server validation on reconnection.	Changes remain pending until accepted. Rejected operations are identified rather than marked as synchronised.

5.1.7 Attendance and Actual Work

ID	Functional Requirement	Acceptance Criterion
FR-055	The system shall record daily attendance as awaiting confirmation, present or absent, without duplication by area.	One worker has one attendance decision per working date within the business. No confirmation does not imply absence.
FR-056	The system shall allow attendance and actual campaign-area assignments to be saved before hours and tasks are complete. Existing unplanned workers may be included.	A worker with an existing record can be added. Missing hours remain incomplete, not a confirmed zero total. Actual campaign participation is explicit. A valid period with an unknown task contributes to total hours as unclassified.
FR-057	The system shall require explicit confirmation before planned information becomes actual work, individually or in a reviewed group.	The user can adjust proposed periods, tasks, areas and breaks. Confirmation does not change the source plan.
FR-058	The system shall record distinct actual periods for changes of task or work area.	Different known tasks and locations remain identifiable. Valid hours with an unknown task remain unclassified and contribute once to total hours, separately from named task totals.
FR-059	The system shall record optional breaks with start, end and counting treatment, and calculate each worker's hours without duplicate deductions.	Only that worker's non-counted breaks reduce their periods, and only for the overlapping duration. Gaps already excluded by separate periods are not deducted again.
FR-060	The system shall validate actual intervals and prevent overlapping actual work for the same worker across contexts.	Invalid and overlapping periods are rejected, including overnight conflicts. Matching planned and actual intervals are not treated as two actual work entries.
FR-061	The system shall support offline attendance and actual-work entry with explicit local and server states.	Entries can be confirmed and saved locally. They remain pending until server acceptance. Retries do not unintentionally duplicate records.
FR-062	The system shall support retrospective work entry while distinguishing work date from entry date.	Entry follows the completion and correction definition in Section 6.3.8. Adding periods to a date with confirmed work in a closed campaign requires reopening. Confirmed values are not replaced through initial entry.

5.1.8 Work Corrections

ID	Functional Requirement	Acceptance Criterion
FR-063	The system shall allow authorised correction of attendance, actual assignments, periods, tasks, breaks and worker, date, campaign and area associations.	Origin and destination contexts, permissions, dates and overlaps are validated. Assignments without periods are included. Closed campaigns require reopening before confirmed work is corrected.
FR-064	The system shall require confirmation before changing presence to absence and removing contradictory actual work.	The review includes confirmed actual assignments without hours, as well as periods and breaks for that worker and working date. Cancellation preserves them. Acceptance cannot leave absence with contradictory confirmed assignments, periods or breaks.
FR-065	The system shall support offline corrections and authorised manual resolution of concurrent changes.	The user compares current server values with the pending correction. No silent replacement occurs. The comparison does not become a permanent correction history.
FR-066	The system shall update affected totals after corrections while distinguishing pending local and server-accepted results.	Local totals reflect locally accepted changes. Rejected proposals are not presented as part of the accepted central total. Origin and destination totals are updated when relevant.

5.1.9 Common Acceptance Conditions

The following conditions apply to this group:

- Permissions are enforced by the server, including during synchronisation.
- Worker-profile management requires connectivity.
- Prepared planning, attendance and work-correction operations support the agreed offline behaviour.
- A locally confirmed operation is not automatically server-accepted.
- Cancellation preserves previously accepted information.
- Missing information is not silently converted into zero or a confirmed business decision.
- Repeated submission must not unintentionally create duplicate records.
- The application records working hours and task types without calculating wages.

5.1.10 Document Consultation

ID	Functional Requirement	Acceptance Criterion
FR-067	The system shall list authorised documents associated with a selected worker, organised by document category.	Contracts, insurance and other documents are distinguishable. The list identifies title, version information and relevant signature information.
FR-068	The system shall allow consultation of retained document versions according to their category.	All retained contract versions are consultable. Insurance and other documents expose only the current and immediately previous versions.
FR-069	The system shall allow authorised opening and downloading of PDF, JPG and PNG documents.	Supported available files can be opened and downloaded. Retrieval failure is distinguishable from an empty document list.
FR-070	The system shall display insurance expiry information and its corresponding validity status.	Recorded expiry information produces valid, expired or validity-unconfirmed status. Current-version status does not imply validity.
FR-071	The system shall prepare authorised insurance documents for offline consultation for workers planned in the current and following week, with additional worker preparation where requested.	Current and previous retained insurance versions can be prepared and opened offline. The interface identifies preparation failures and the last successful synchronisation.
FR-072	The system shall enforce document consultation permissions by business, worker scope and document type.	An authorised Field Manager, if included, can consult relevant worker insurance. Contracts and other documents require additional permission. Direct file requests are also checked.

5.1.11 Document Upload and Version Management

ID	Functional Requirement	Acceptance Criterion
FR-073	The system shall allow authorised upload of a document with its category, title, file and associated workers.	Required fields are validated. Observations are optional. Upload date and version identity are recorded automatically.
FR-074	The system shall validate uploaded files against the supported formats and proposed 10 MB per-file limit.	Unsupported or oversized files are rejected without replacing an accepted version. PDF, JPG and PNG files within the limit can be accepted. The dependent acceptance detail remains pending under Q-DOC-06.
FR-075	The system shall allow insurance upload before its expiry date is known.	The file can be saved with validity unconfirmed. No date or valid status is invented. The date can be added later.
FR-076	The system shall allow one document to be associated with several authorised workers within the same business.	The same document is accessible through the relevant worker records without separate file copies. Shared version updates affect the same document.
FR-077	The system shall make a newly accepted document version current without inheriting previous signatures.	No additional approval stage is required. The new version has its own signature information. Failed or cancelled uploads do not replace the current version.
FR-078	The system shall allow an uploaded version to be identified as signed outside the application.	The declaration is distinguishable from application-captured signing. It does not fabricate a verified signing event or signatures for all associated workers.
FR-079	The system shall apply category-specific version retention after successful version acceptance.	Contracts retain all versions. Insurance and other documents retain only the current and previous versions. Older versions are deleted even when signed.
FR-080	The system shall allow authorised deletion of an incorrect unsigned upload without dependent records.	Eligibility is checked and confirmation required. Signed or dependent records are not deleted through this ordinary deletion action.
FR-081	The system shall allow authorised withdrawal of an incorrect signed document version.	Withdrawal identifies the version as unavailable for operational use without changing it to unsigned. Category-specific retention still applies.
FR-082	The system shall require connectivity and the relevant modification permissions for document upload and version management.	Consultation permission alone does not permit modification. Management operations do not complete offline and are checked by the server.

5.1.12 Worker Signing

ID	Functional Requirement	Acceptance Criterion
FR-083	The system shall allow an authorised user to initiate in-person signing for an identified worker and document version.	The worker and version are displayed. The initiating user confirms the intended worker's presence. No worker management account is required.
FR-084	The system shall allow the worker to review the document, draw a signature, clear and redraw it or cancel.	The document is available before confirmation. An empty drawing is not accepted. Cancellation creates no completed signature.
FR-085	The system shall require explicit worker confirmation of the drawing and reviewed document version.	A drawing alone does not complete signing. After confirmation, it cannot be silently replaced or transferred to another version.
FR-086	The system shall associate each signing result with its worker, document version, initiating user and recorded date and time.	The accepted result retains those associations. Document generation or upload does not create this signing event automatically.
FR-087	The system shall prevent repeated completed signing for the same worker and version and check uncertain submission outcomes.	Existing accepted results are displayed. Pending or uncertain operations are checked or resumed rather than replaced by another signing interaction.
FR-088	The system shall maintain individual signing results for workers associated with a shared document.	One signature does not mark other workers as signed. A worker's view presents their own application-captured signature only.
FR-089	The system shall support explicit preparation and offline signing of authorised documents.	A prepared version can be reviewed and signed without connectivity. The confirmed result is saved locally with pending synchronisation status.
FR-090	The system shall synchronise a signature against the version actually reviewed and handle changed version eligibility.	A newer version never inherits the signature. A withdrawn or deleted original version causes explicit resolution rather than automatic reassignment or recreation.
FR-091	The system shall display signing outcomes without conflating local confirmation, server acceptance and unresolved results.	Pending, accepted and resolution-required results are distinguishable. Failed or uncertain submissions are not presented as server-confirmed.
FR-092	The system shall support safe recovery and retry of pending signing operations.	Recoverable pending information is preserved. Retrying the same operation does not create duplicate signing events.
FR-093	The system shall propagate document-version deletion to prepared offline copies when devices synchronise.	Deleted versions and their related local files are removed according to the synchronisation policy. Immediate removal from disconnected devices is not assumed.

5.1.13 Common Acceptance Conditions

- Document access remains limited to the authorised business and worker scope.
- Consultation, modification and signing-initiation permissions are distinct.
- Version currency, withdrawal, insurance validity and signature information are separate attributes.
- Cancelled or failed uploads do not trigger replacement or retention deletion.
- The signing workflow collects the worker's signature only. An owner's signature is not required by this application workflow.
- Operational use of the selected signing method remains subject to validation for the intended document types.
- Document generation remains pending scope confirmation under UC-20. It is not specified as an approved capability in this requirement group.

5.1.14 Transport and Olive Deliveries

The following requirements cover the departure, transport and delivery of olives. Weighing data, analysis results and monetary values may be recorded later and are managed separately.

ID	Requirement	Acceptance Criteria
FR-094	The system shall allow the Owner or an authorised Field Manager to register and ordinarily correct an olive transport and its associated delivery.	Registration requires a campaign, at least one source area, driver, vehicle, destination and departure date and time. Corrections validate permissions, chronology and associations within the same campaign and operational state. State reversal, cancellation and campaign reassignment require a separately approved exceptional procedure.
FR-095	The system shall allow a delivery to contain olives from multiple properties or parcels belonging to the selected campaign.	Each source is recorded without duplicating the same area through both its property and its parcels. Collection origin and olive variety may be recorded when known.
FR-096	The system shall allow the user to select a registered worker as the driver and identify the vehicle used.	A worker record is required for the driver under the current proposal. This selection does not automatically create attendance or work records.
FR-097	The system shall allow the destination to be selected from previously used destinations or entered as a new destination.	A destination is mandatory. A new destination can be entered without requiring a separate cooperative management module.

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ID	Requirement	Acceptance Criteria
FR-098	The system shall support registration in two stages: departure and delivery confirmation.	The departure can be saved before delivery occurs. Confirmation requires a delivery date and time that is not earlier than departure.
FR-099	The system shall allow the complete transport and delivery record to be entered retrospectively.	The user may enter departure and delivery information together after delivery, subject to the same validations.
FR-100	The system shall allow transport and delivery information to be recorded without weighing data, analysis results or monetary values.	Missing downstream information does not prevent departure registration or delivery confirmation.
FR-101	The system shall support offline registration of departures and delivery confirmations using previously available reference data.	Local records are marked as pending synchronisation. Server acceptance follows validation of permissions, campaign context and associated records.

5.1.15 Weighing Data and Adjustments

ID	Requirement	Acceptance Criteria
FR-102	The system shall allow the Owner or an authorised Field Manager to associate a weighing ticket with a delivery.	Each ticket belongs to exactly one delivery. The number of tickets permitted for one delivery remains subject to business confirmation. The dependent acceptance detail remains pending under Q-DA-01.
FR-103	The system shall allow available weighing information to be entered manually.	The record supports the ticket reference, issuing cooperative, weighing date and time, and available gross, tare, net or final weight values, with their meaning and unit identified.
FR-104	The system shall allow incomplete weighing records to be saved.	Unavailable values remain unspecified rather than being replaced with zero or inferred from unrelated fields.
FR-105	The system shall allow reported operational adjustments or deductions to be recorded with their known interpretation.	The value, unit or type and whether it is already reflected in a reported weight are identified where known. Unknown treatment remains unresolved. Monetary adjustment information follows FR-119, including its online-only access rule. The dependent acceptance detail remains pending under Q-DA-02.

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ID	Requirement	Acceptance Criteria
FR-106	The system shall allow a PDF or image of a weighing ticket to be attached to its record.	An attachment supplements the manually entered information. Automatic extraction from the attachment is not required.
FR-107	The system shall allow authorised corrections to weighing and adjustment information.	Corrections replace the current values, trigger recalculation of affected indicators and do not create a permanent history of previous field values.
FR-108	The system shall support offline entry of non-financial weighing information.	Operational values can be saved locally for available deliveries. Monetary fields and adjustments are excluded. Eligible attachments may remain pending upload; mixed-content attachment handling remains pending under Q-AT-01.

5.1.16 Analysis Results

ID	Requirement	Acceptance Criteria
FR-109	The system shall allow the Owner or an authorised Field Manager to associate an analysis result with a delivery.	Each analysis belongs to exactly one delivery. The number of analyses permitted for one delivery remains subject to business confirmation. The dependent acceptance detail remains pending under Q-DA-03.
FR-110	The system shall allow available analysis information to be entered manually.	The record supports the reported oil yield percentage, issuing entity, analysis reference, analysis date and reference weight when supplied. The entry timestamp is recorded separately.
FR-111	The system shall allow incomplete analysis records to be saved.	Missing results remain pending or unspecified. A missing oil yield percentage is not interpreted as zero.
FR-112	The system shall allow a PDF or image containing the analysis results to be attached to the record.	The attachment can be consulted alongside the entered results. Automatic extraction is not required.
FR-113	The system shall allow authorised corrections to analysis results.	Corrections replace current values and update affected indicators. Recording a genuinely new analysis is distinguished from correcting an existing result.
FR-114	The system shall support offline entry of analysis results.	Results can be recorded locally for available deliveries. Pending attachments and data are synchronised when connectivity is restored.

5.1.17 Cooperative-Reported Ticket Values

ID	Requirement	Acceptance Criteria
FR-115	The system shall allow the Owner or a Field Manager with specific financial permissions to record the amount reported by the cooperative for a ticket.	The amount is associated with a ticket and, through that ticket, with its delivery. It is not entered as an independent delivery-level amount.
FR-116	The system shall record the amount, currency, reporting cooperative, reporting date and provisional or final classification.	Amounts use EUR and two decimal places under the current scope. A reference, observations and a supporting PDF or image may also be supplied.
FR-117	The system shall maintain a single current reported amount for each ticket.	A final amount replaces the provisional amount. Corrections replace the current value without retaining a permanent history of previous amounts.
FR-118	The system shall distinguish a ticket with no reported amount from a ticket with an explicitly reported zero amount.	Missing monetary information remains pending and is not automatically converted to zero.
FR-119	The system shall require an internet connection and financial permissions to consult, register or correct monetary information.	Amounts and monetary adjustments are excluded from offline datasets, operational summaries and reports. A ticket's operational classification does not authorise offline exposure of monetary content. Mixed-content attachment handling remains pending under Q-AT-01.
FR-120	The system shall allow weighing information, analysis results and reported amounts to be completed after campaign closure.	The information must relate to existing deliveries or tickets. These updates do not require campaign reopening.
FR-121	The system shall update the eligible source data used by indicators after accepting changes to weighing, analysis or reported amounts.	The updated source contribution replaces the previous one without duplication. Eligibility, missing information and unresolved aggregation dependencies remain identifiable. Presentation of refreshed results follows FR-137.
FR-122	The system shall enforce business ownership and operation-specific permissions across delivery-related records.	A user can only access or modify records within the authorised business context. Financial permissions are checked separately from operational permissions.

5.1.18 Campaign Indicators

The following requirements cover the consultation of campaign indicators during and after operational activity. Indicators must distinguish actual work from planning, available information from missing information, and accepted records from local changes awaiting synchronisation.

ID	Requirement	Acceptance Criteria
FR-123	The system shall allow the Owner or an authorised Field Manager to consult indicators for a selected campaign.	Indicators are available during the campaign and after closure, within the user's authorised business and campaign scope.
FR-124	The system shall provide a campaign overview with breakdowns by property and, where applicable, by parcel.	The user can consult campaign totals and inspect more detailed information without counting the same underlying record more than once.
FR-125	The system shall present the number of distinct workers who actually participated in the campaign.	A worker is counted once when daily presence is confirmed and an actual work assignment exists for that campaign. Planned assignments alone do not establish participation. Missing work hours are identified separately.
FR-126	The system shall present actual working days and worked hours.	Working days use the start date of actual work periods. Worked hours follow the approved interval and break rules. Planned hours are excluded from actual-work totals.
FR-127	The system shall allow counted working hours to be inspected by worker and actual task.	Only the relevant worker's breaks reduce their overlapping periods. Machinery operation and general work follow the recorded task. Valid hours with an unknown task contribute to total hours and appear separately as unclassified.
FR-128	The system shall present the number of confirmed deliveries and available weighing indicators.	Only confirmed deliveries contribute to the delivery count. Weight indicators identify the weight concept and unit, exclude duplicate contributions and disclose missing data.
FR-129	The system shall present an aggregated oil yield indicator only when approved calculation rules and eligible data exist.	The indicator uses the approved reference weights and analysis selection rules. Partial coverage is disclosed. An unavailable result is not displayed as zero. The dependent acceptance detail remains pending under Q-DA-04.
FR-130	The system shall present cooperative-reported monetary totals to users with financial permissions.	Current final and provisional amounts are shown separately. Tickets without a reported amount are identified. Replaced amounts are not included in totals.
FR-131	The system shall identify information that cannot be allocated to an individual property or parcel.	Mixed-source values without an approved allocation remain visible at the appropriate campaign level and are labelled as unallocated in detailed breakdowns.

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ID	Requirement	Acceptance Criteria
FR-132	The system shall allow indicators to be filtered by date range and, where applicable, property, parcel, worker and task.	Each filter affects only relevant indicators. The interface identifies the scope of each result and does not imply that a worker or task filter applies to unrelated delivery or financial information. The dependent acceptance detail remains pending under Q-IN-01.
FR-133	The system shall allow the user to inspect records supporting an indicator.	Available source records can be opened subject to permissions. The displayed records correspond to the indicator's scope and applicable filters.
FR-134	The system shall disclose the completeness and freshness of the information used by the indicators.	The interface identifies missing data, unresolved calculation rules, partial access, pending local changes and the last successful synchronisation where relevant.
FR-135	The system shall support offline consultation of operational campaign indicators using locally available data.	Offline indicators identify their available coverage and last successful synchronisation. They do not claim to include changes made on other devices after that synchronisation.
FR-136	The system shall reflect pending local operational changes in supported offline indicators without duplicating records.	A local creation contributes once, an update replaces the cached contribution and a removal suppresses it. Affected results are identified as provisional.
FR-137	The system shall present refreshed indicator results after accepted source updates under FR-121, including for closed campaigns.	Consultation reflects current eligible source records and permitted corrections. Campaign closure does not freeze results; previously exported PDFs remain unchanged snapshots.
FR-138	The system shall allow an authorised user to export a campaign indicator report as PDF.	The report identifies the campaign, applicable scope and filters, generation date and time, and relevant completeness or synchronisation limitations.
FR-139	The system shall enforce indicator and export permissions consistently with access to the underlying information.	Financial information requires online access and financial permissions. Restricted values are excluded from displays, supporting records and exported reports.

5.2 Mapping to Use Cases

5.2.1 Account and Access Requirements

Requirement	Use Case	Purpose
FR-001	UC-01	Self-register as an Owner.
FR-002	UC-01	Validate registration and email identity.
FR-003	UC-01	Verify the registered email.
FR-004	UC-01	Activate ownership and create the business workspace.
FR-005	UC-02	Authenticate an activated account.
FR-006	UC-02	Log out while protecting pending changes.
FR-007	UC-02	Validate sessions and business authorisation.
FR-008	UC-02	Continue eligible work offline.
FR-009	UC-03	Request password recovery.
FR-010	UC-03	Establish a new password securely.
FR-011	UC-03	Invalidate sessions after password reset.
FR-012	UC-02	Isolate cached data between accounts.
FR-013	UC-04	Invite a manager to a business.
FR-014	UC-04	Accept an invitation through the intended account.
FR-015	UC-04	Manage pending invitations.
FR-016	UC-04	Configure manager scope and permissions.
FR-017	UC-04	Preserve required insurance-document access.
FR-018	UC-04	Revoke business access.
FR-019	UC-02, UC-04	Revalidate access and pending changes.
FR-020	UC-04	Restrict access administration to the Owner.

5.2.2 Properties and Campaigns

Requirement	Use Case	Behaviour
FR-021	UC-05	Create a property.
FR-022	UC-05	Record optional area and varieties.
FR-023	UC-05	Create optional parcels.
FR-024	UC-05	Consult and update property information.
FR-025	UC-05	Archive and reactivate properties or parcels.

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Requirement	Use Case	Behaviour
FR-026	UC-05	Delete eligible records.
FR-027	UC-05	Consult prepared property information offline.
FR-028	UC-06	Create a campaign.
FR-029	UC-06	Select whole properties or parcels.
FR-030	UC-05, UC-06	Create a reusable parcel during campaign preparation.
FR-031	UC-06	Consult and update campaign details and coverage.
FR-032	UC-06	Distinguish planned and actual campaign dates.
FR-033	UC-06	Close a campaign.
FR-034	UC-06	Reopen a campaign.
FR-035	UC-06	Archive, unarchive or delete a campaign.
FR-036	UC-06	Prevent conflicting campaign coverage.
FR-037	UC-06	Consult prepared campaign information offline.

The mapping identifies the primary use cases. Other use cases may consume the resulting records, such as selecting campaign areas during planning or associating a delivery with its provenance.

5.2.3 Workforce and Work Management

Requirement	Use Case	Behaviour
FR-038	UC-07	Create a worker record.
FR-039	UC-07	Record machinery capability.
FR-040	UC-07	Consult and update current worker information.
FR-041	UC-07	Review possible duplicates.
FR-042	UC-07	Inactivate and reactivate workers.
FR-043	UC-07	Delete eligible worker records.
FR-044	UC-07	Consult prepared worker information offline.
FR-045	UC-08	Create independent daily plans.
FR-046	UC-08	Handle optional and required planning details.
FR-047	UC-08	Apply group values and individual exceptions.
FR-048	UC-08	Reuse plans or actual work as proposals.
FR-049	UC-08	Validate planned intervals and overlaps.

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Requirement	Use Case	Behaviour
FR-050	UC-08, UC-10	Represent overnight work periods.
FR-051	UC-08	Update eligible plans.
FR-052	UC-09	Consult schedules through views and filters.
FR-053	UC-09	Prepare two-week offline consultation.
FR-054	UC-08	Create and adjust plans offline.
FR-055	UC-10	Confirm daily attendance.
FR-056	UC-10	Save actual context with incomplete hours.
FR-057	UC-10	Explicitly confirm proposed work.
FR-058	UC-10	Distinguish actual tasks and work areas.
FR-059	UC-10	Record breaks and calculate counted hours.
FR-060	UC-10	Prevent actual-work overlaps.
FR-061	UC-10	Save actual work offline.
FR-062	UC-10	Enter work retrospectively.
FR-063	UC-11	Correct work and its associations.
FR-064	UC-11	Resolve all contradictory assignments, periods and breaks before absence.
FR-065	UC-11	Resolve offline correction conflicts.
FR-066	UC-11	Update affected totals.

5.2.4 Documents and Worker Signing

Requirement	Use Case	Behaviour
FR-067	UC-12	List documents by worker and category.
FR-068	UC-12	Consult retained versions.
FR-069	UC-12	Open and download supported files.
FR-070	UC-12, UC-13	Display recorded insurance validity.
FR-071	UC-12	Prepare insurance consultation offline.
FR-072	UC-12	Enforce document consultation permissions.
FR-073	UC-13	Upload files and document metadata.
FR-074	UC-13	Validate file format and size.
FR-075	UC-13	Save insurance with unknown expiry.

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Requirement	Use Case	Behaviour
FR-076	UC-12, UC-13	Share one document across workers.
FR-077	UC-13	Accept a new current version.
FR-078	UC-13	Declare external signature information.
FR-079	UC-12, UC-13	Apply version-retention rules.
FR-080	UC-13	Delete eligible incorrect uploads.
FR-081	UC-12, UC-13	Withdraw signed versions from use.
FR-082	UC-13	Require connected authorised management.
FR-083	UC-14	Initiate in-person worker signing.
FR-084	UC-14	Review, draw, redraw or cancel.
FR-085	UC-14	Explicitly confirm the reviewed result.
FR-086	UC-14	Record signing associations and evidence.
FR-087	UC-14	Prevent repeated or uncertain duplicate signing.
FR-088	UC-12, UC-14	Present individual shared-document signatures.
FR-089	UC-14	Prepare and perform signing offline.
FR-090	UC-14	Validate the original version on synchronisation.
FR-091	UC-14	Display the actual signing outcome.
FR-092	UC-14	Recover and retry pending submissions.
FR-093	UC-12, UC-13, UC-14	Propagate version deletion to prepared devices.

5.2.5 Delivery-Related Requirements

Requirement	Use Case	Purpose
FR-094	UC-15	Register and ordinarily correct transport and delivery details.
FR-095	UC-15	Identify campaign properties and parcels supplying the load.
FR-096	UC-15	Identify the registered driver and vehicle.
FR-097	UC-15	Select or enter the destination.
FR-098	UC-15	Register departure and subsequently confirm delivery.
FR-099	UC-15	Register a completed transport retrospectively.

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Requirement	Use Case	Purpose
FR-100	UC-15	Separate delivery registration from downstream information.
FR-101	UC-15	Record operational delivery information offline.
FR-102	UC-16	Associate a weighing ticket with a delivery.
FR-103	UC-16	Record reported weighing information.
FR-104	UC-16	Preserve incomplete weighing information explicitly.
FR-105	UC-16	Record operational adjustments and separate financial information.
FR-106	UC-16	Attach supporting weighing documents.
FR-107	UC-16	Correct current weighing information.
FR-108	UC-16	Record non-financial weighing information offline.
FR-109	UC-17	Associate analysis results with a delivery.
FR-110	UC-17	Record reported analysis information.
FR-111	UC-17	Preserve incomplete analysis information explicitly.
FR-112	UC-17	Attach supporting analysis documents.
FR-113	UC-17	Correct current analysis information.
FR-114	UC-17	Record analysis results offline.
FR-115	UC-18	Associate cooperative-reported amounts with tickets.
FR-116	UC-18	Record financial information and supporting evidence.
FR-117	UC-18	Maintain the current provisional or final amount.
FR-118	UC-18	Distinguish missing amounts from reported zero amounts.
FR-119	UC-18	Restrict financial operations to authorised online access.
FR-120	UC-16, UC-17, UC-18	Complete information for existing records after campaign closure.
FR-121	UC-16, UC-17, UC-18, UC-19	Update eligible source contributions used by indicators.
FR-122	UC-15, UC-16, UC-17, UC-18	Enforce business ownership and operation-specific permissions.

5.2.6 Campaign Indicator Requirements

Requirement	Use Case	Purpose
FR-123	UC-19	Consult indicators for an authorised campaign.
FR-124	UC-19	Inspect campaign, property and parcel summaries.
FR-125	UC-19	Count distinct actual campaign participants.
FR-126	UC-19	Consult actual working days and worked hours.
FR-127	UC-19	Inspect counted hours by worker and task, including unclassified hours.
FR-128	UC-19	Consult confirmed deliveries and eligible weight indicators.
FR-129	UC-19	Consult oil yield indicators under approved calculation rules.
FR-130	UC-19	Consult current provisional and final reported amounts.
FR-131	UC-19	Identify values without property or parcel allocation.
FR-132	UC-19	Apply filters with an explicit scope.
FR-133	UC-19	Inspect records supporting an indicator.
FR-134	UC-19	Identify completeness, access and freshness limitations.
FR-135	UC-19	Consult available operational indicators offline.
FR-136	UC-19	Reflect pending local changes without double counting.
FR-137	UC-19	Present refreshed indicators after accepted source updates.
FR-138	UC-19	Export a scoped campaign indicator report as PDF.
FR-139	UC-19	Enforce access restrictions in indicators and exports.

6 Business Rules and Record States

This section defines the constraints, record states, correction principles and calculation rules governing system behaviour.

Rules are separated from functional requirements so that the same condition can be referenced by several use cases without being redefined.

The current content covers account registration and access, properties, parcels, harvest campaigns, worker records, planning, attendance, actual work, corrections, documents and worker signing, transport and deliveries, weighing, analysis, reported values and campaign indicators. The integrated business rules cover BR-001 to BR-092. Unresolved business and calculation decisions remain explicitly identified in their respective domains.

6.1 Business Rules

6.1.1 Account Registration and Business Access

ID	Business Rule
BR-001	Owners may register freely. The initial registration requires a name, email, password and business name. An existing Owner invitation is not required.
BR-002	An email identity identifies at most one account. A consistent email normalisation policy applies throughout the account lifecycle.
BR-003	A self-registered account does not receive active business access until its email has been verified. Repeated verification must not create duplicate business workspaces.
BR-004	Under the initial proposal, verified Owner registration creates one independent business workspace. Supporting several business memberships or ownership transfer requires separate confirmation.
BR-005	An account, a business membership and a worker record are distinct concepts. Workers do not require application accounts. An invited manager account does not automatically create an Owner workspace.
BR-006	Authentication establishes account identity. Business membership, responsibility scope and permissions determine which data and operations are accessible. Business data must remain isolated.
BR-007	Only the Owner may invite managers, administer their permissions and revoke their business access under the current role model. Managers cannot grant themselves additional permissions.

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ID	Business Rule
BR-008	Manager access requires explicit acceptance by the account with the verified email identified in the invitation. Pending, cancelled or expired invitations do not grant operational access.
BR-009	Manager permissions distinguish consultation from modification, correction, document operations and financial access. A manager responsible for workers must be able to consult their insurance documents within the authorised scope.
BR-010	Initial login, account activation, password recovery and access administration require internet connectivity. Offline activity requires previous authentication and valid offline authorisation.
BR-011	Revocation blocks subsequent online access immediately. A disconnected device enforces the change when it reconnects or its offline authorisation expires. Immediate remote removal from a disconnected device is not assumed.
BR-012	Pending offline changes are revalidated on submission. Revocation or permission reduction may prevent their acceptance. Such proposals must not be silently accepted or discarded.
BR-013	Logout blocks local workspace access and preserves pending changes in protected storage. Cached information and pending operations must not become accessible to another account using the same device.
BR-014	Password reset authorisations are time-limited and single-use. A successful reset invalidates existing online sessions without changing business ownership or permissions. Recovery responses must not disclose account existence.

6.1.2 Properties and Campaigns

ID	Business Rule
BR-015	A property may exist without parcels. Each parcel belongs to one property.
BR-016	Declared area is optional and is the reference area when supplied by the owner. A future map-derived estimate must remain separately identified and must not automatically replace the declared area.

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ID	Business Rule
BR-017	Property-level information without a parcel association is not automatically distributed among parcels.
BR-018	Properties and parcels with associated records are archived rather than deleted. Deletion requires the absence of associated records. Campaign deletion requires the absence of operational records.
BR-019	An archived property is unavailable for new operational associations, including through its parcels. Archival preserves existing historical information. A parcel cannot be operationally available while its parent property is archived.
BR-020	A campaign covers at least one whole property or one parcel within the authorised business. It may cover several properties and selected parcels.
BR-021	Selecting a whole property and its parcels must not duplicate campaign coverage.
BR-022	Campaigns may coexist on different parcels, but must not overlap in the same area and period. Whole-property coverage overlaps any of that property's parcels.
BR-023	An area with operational records in a campaign cannot be directly removed from that campaign. Any exceptional reassignment requires an explicit correction process.
BR-024	The planned start date is mandatory. Actual start is recorded separately. Actual end is recorded when closing. Changing a planned date does not overwrite an actual date.
BR-025	Correcting confirmed work in a closed campaign requires authorised reopening. Adding periods to a date already containing confirmed work in that campaign also requires reopening. Work after its actual end requires reopening and date review. Completion follows Section 6.3.8.
BR-026	Ticket information, analysis results and reported monetary values for existing deliveries and tickets may be completed after campaign closure without reopening.
BR-027	Campaign archival affects list visibility, not open or closed status. Archiving does not release covered areas from campaign-overlap validation. Unarchiving does not reopen a campaign.

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ID	Business Rule
BR-028	Property, parcel and campaign management require connectivity and the relevant permission. Supported field operations follow their own offline rules. Prepared authorised information may be consulted offline.

6.1.3 Workers, Planning and Actual Work

ID	Business Rule
BR-029	Each business maintains independent worker records. Within the business, a worker identity is reused across properties and campaigns. Records in different businesses are not automatically merged.
BR-030	Matching names or telephone numbers generate a possible-duplicate warning within the business. They do not automatically prevent creation or cause identity merging.
BR-031	Machinery-operation capability belongs to the worker profile. The task planned or actually performed belongs to the relevant work period. Changing capability does not relabel past work.
BR-032	Inactivation preserves historical records and does not silently remove a worker from existing future plans. Those plans identify the inactive worker for review.
BR-033	Worker profiles retain current information and the last-modified date without detailed field-change history.
BR-034	Plans created for several dates are independent. Reusing planned or actual work produces a reviewed proposal, not confirmed attendance or actual hours for the target dates.
BR-035	Hours and tasks are optional for a worker planned in one area on a working day. Several areas on that day require explicit periods for the relevant assignments.
BR-036	Once actual work exists in the affected daily area context, the corresponding plan remains the reference for expected work. It is not rewritten to match the outcome.
BR-037	Attendance is daily per worker within the business. Actual participation in each campaign and work area is identified separately.

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ID	Business Rule
BR-038	Confirming presence does not automatically confirm proposed hours, tasks or breaks. Actual work requires explicit confirmation.
BR-039	Changes of actual task or work area require distinguishable periods. The same worker's actual time must not be counted twice across contexts.
BR-040	Only a worker's own non-counted breaks reduce their working hours, by intersection with that worker's actual periods. Time already excluded by separate periods is not deducted again.
BR-041	Overnight periods belong to their start date. Duration and conflict validation use complete start and end timestamps.
BR-042	Work corrections follow the campaign reopening rule in BR-025. No additional age-based correction deadline applies.
BR-043	Work corrections retain current values only. Temporary comparison for conflict resolution does not establish a permanent correction history.
BR-044	Pending local values are distinguishable from server-accepted values. Rejected proposals must not remain represented as accepted central contributions.

6.1.4 Documents and Signatures

ID	Business Rule
BR-045	Document categories are contract, insurance and other. Generation remains conditional on confirmation of UC-20.
BR-046	One document may be associated with one or more workers in the same business. Shared access does not require separate copies of the source file.
BR-047	Document consultation, modification and signing initiation require distinct permissions. An authorised Field Manager must be able to consult relevant worker insurance. Other document access requires the relevant owner-granted permission.
BR-048	Supported upload formats are PDF, JPG and PNG. The proposed per-file limit is 10 MB.

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ID	Business Rule
BR-049	Associated workers, category, title and file are mandatory for upload. Observations are optional. Upload date and version identity are recorded automatically.
BR-050	Insurance expiry is recorded where known. Missing expiry information produces validity-unconfirmed status. Current-version status alone does not establish insurance validity.
BR-051	An accepted new version becomes current without inheriting previous signatures. Current-version status, withdrawal, expiry and signature information remain distinct.
BR-052	Contracts retain all accepted versions. Insurance and other documents retain the current and immediately previous versions only. Older versions are deleted even when signed, after successful replacement acceptance.
BR-053	An incorrect unsigned version without dependencies may be deleted after authorisation and confirmation. An incorrect signed version is withdrawn rather than ordinarily deleted. The rolling retention rule in BR-052 remains an explicit exception.
BR-054	External-signature declarations are distinct from application-captured signatures. They do not fabricate verified application events or signatures for every associated worker.
BR-055	Application signing is associated with a specific worker and document version. Worker participation does not create a management account.
BR-056	Signing occurs in person on an authorised owner's or manager's device. The initiating user confirms the intended person. The worker reviews the document and drawing before explicitly confirming.
BR-057	The drawing may be cleared or repeated before confirmation. After confirmation, the same worker's signature for that version is not replaced through another signing interaction.
BR-058	Shared documents maintain separate application signing results per worker. The worker-specific view does not display other workers' application-captured signatures.
BR-059	Offline insurance consultation is prepared for workers planned in the current and following week, with additional workers prepared where requested. Other documents may be explicitly prepared for the agreed offline signing workflow. This does not imply automatic offline download of all contracts.

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ID	Business Rule
BR-060	A pending signature remains associated with the reviewed version. A newer version does not inherit it. A withdrawn or deleted original version requires explicit resolution.
BR-061	Version deletion applies across shared worker associations and related stored files. Prepared devices apply deletion on synchronisation. Immediate deletion from a disconnected device is not assumed.
BR-062	A signing result records the worker, document version, initiating user and recorded date and time. Local confirmation remains distinct from server acceptance. The workflow does not collect the owner's signature.

6.1.5 Transport, Delivery and Cooperative Records

ID	Business Rule
BR-063	Under the current proposal, each delivery has one associated transport. Departure registration and delivery confirmation are stages of the same operational record. Independent trip management is outside the current baseline.
BR-064	A delivery belongs to one campaign and may contain olives from several of that campaign's properties or parcels. The same source area must not be counted twice through overlapping property and parcel selections.
BR-065	The driver must be a registered worker under the current proposal. Selecting a driver does not establish attendance, actual work periods or remuneration. Allowing external drivers remains subject to confirmation.
BR-066	Driver, vehicle, destination and departure date and time are mandatory. Collection origin and olive variety may remain unspecified when unknown. Delivery confirmation requires a delivery date and time not earlier than departure.
BR-067	A departure may be recorded before delivery or both stages may be recorded retrospectively. Weighing tickets, analysis results and monetary values are not prerequisites for delivery confirmation.
BR-068	Each weighing ticket belongs to exactly one delivery. Whether a delivery can have multiple tickets, and how those tickets relate to each other, remains unresolved. No additive aggregation may be assumed.

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ID	Business Rule
BR-069	Each weight must retain its reported meaning and unit. Gross weight, tare, net weight and final or payable weight are not interchangeable. Missing values must not be represented as zero.
BR-070	Operational adjustments retain their known type, unit and application status before automatic use. Already reflected adjustments are not applied again. Monetary adjustments are financial information and follow BR-078. Handling of mixed-content source attachments remains pending under Q-AT-01.
BR-071	Each analysis belongs to exactly one delivery. Whether several analyses may exist for one delivery, and which results are eligible for aggregation, remains unresolved. Correcting a result is distinct from recording a new analysis.
BR-072	Oil yield is recorded as a percentage when reported. Any reference weight used for aggregation must have a known, approved meaning. The system must not silently assume that gross, net or payable weight is the appropriate basis.
BR-073	A cooperative-reported monetary amount belongs to one ticket, which has at most one current amount classified as provisional or final. Current-value replacement and retention follow BR-075.
BR-074	The application records monetary amounts reported by the cooperative. It does not independently estimate ticket prices, reapply reported deductions or infer payment from an amount. Payment processing and payroll are outside this functionality.
BR-075	A provisional amount is replaced by the final amount when the latter becomes available. Only the current amount is retained; previous monetary values are not maintained as a permanent correction history.
BR-076	After campaign closure, authorised users may complete or correct weighing information, analysis results and reported amounts for existing deliveries or tickets without reopening the campaign. This permission does not authorise new operational work or a new delivery outside the campaign period.
BR-077	When a load contains olives from several properties or parcels, its weight, analysis results and monetary amount must not be assigned in full to every source area. Allocation requires reported quantities or an explicitly approved allocation rule.

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ID	Business Rule
BR-078	Non-financial transport, weighing and analysis information may be captured offline using available authorised data. Consultation, registration and correction of monetary amounts and adjustments require online access and financial permissions, including in indicators and reports. Operational ticket access does not exempt monetary content from this restriction.
BR-079	Authorised corrections retain current values rather than a permanent field-change history. Local changes remain provisional until accepted by the server, and conflicts must not silently overwrite accepted information.

6.1.6 Campaign Indicators and Reporting

ID	Business Rule
BR-080	Indicators describe one selected campaign at a time. Cross-campaign comparisons are outside the current baseline.
BR-081	Actual-work indicators are derived from confirmed attendance and actual work information. Plans remain a separate reference and must not be counted as work performed.
BR-082	A campaign participant is a distinct worker with confirmed daily presence and an actual assignment to that campaign. A planned assignment or presence without campaign context is insufficient. Participation may be known even when worked hours remain incomplete.
BR-083	A working day is identified by the start date of actual work. A period crossing midnight belongs to its start date. The same date is counted once within the relevant scope.
BR-084	Worked hours follow the approved actual-period and break rules. A worker's machinery-operation capability does not determine the task used to classify worked hours.
BR-085	A delivery count includes confirmed deliveries, independently of how many tickets or analyses may be associated with them. Weights, analyses and monetary amounts use their own approved eligibility and aggregation rules.
BR-086	Missing information, unknown allocation and unavailable calculations are not zero values. Indicators must disclose partial coverage and unresolved calculation dependencies.

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ID	Business Rule
BR-087	Campaign totals and area breakdowns follow the mixed-source allocation rule in BR-077. An eligible delivery contributes once to the campaign total.
BR-088	A filter applies only where the source data supports the requested relationship. A worker or task filter must not implicitly allocate deliveries or reported amounts to that worker or task.
BR-089	Campaign closure does not freeze indicators. Accepted late information and permitted corrections update the current campaign view. A generated PDF remains a snapshot of the information available at its generation time.
BR-090	Offline indicators use the locally available accepted dataset and identifiable pending local changes. They cannot guarantee knowledge of subsequent activity on other devices. Pending updates must replace, rather than duplicate, the cached contribution of the same record.
BR-091	Indicator access does not grant additional access to underlying records. A restricted view must identify its partial scope without disclosing restricted data or deriving restricted values from differences between totals.
BR-092	Indicator displays, supporting records and PDF exports apply the online financial-access restriction in BR-078. Offline views and reports exclude monetary information.

6.2 Record States and Transitions

6.2.1 Account Activation

The account activation states are:

Pending verification
: Registration has been recorded, but the email has not been verified. Active workspace access is unavailable.

Active
: Email verification has been completed and the account can authenticate, subject to its current business access.

The normal transition is:

$$\text{Pending verification} \longrightarrow \text{Active}$$

An expired verification link does not itself delete the pending registration. A replacement verification message may be requested.

Account suspension, account deletion and business ownership transfer require separate lifecycle rules and are not implicitly defined by this activation model.

6.2.2 Manager Invitations

Manager invitations have the following states:

Pending
: The invitation has been issued but not accepted.

Accepted
: The intended account has accepted the valid invitation and the business membership has been activated.

Cancelled
: The Owner cancelled the invitation before acceptance.

Expired
: The invitation's acceptance period has elapsed.

Allowed transitions are:

$$\text{Pending} \longrightarrow \text{Accepted}$$
$$\text{Pending} \longrightarrow \text{Cancelled}$$
$$\text{Pending} \longrightarrow \text{Expired}$$

A replacement invitation may be issued after cancellation or expiry. It must not activate a duplicate membership.

6.2.3 Manager Business Access

An accepted manager membership may be:

Active
: The account may act within the configured scope and permissions.

Revoked
: The account no longer has access to that business. The account and existing business records remain.

The defined transition is:

$$\text{Active} \longrightarrow \text{Revoked}$$

Restoring access requires a new explicit Owner authorisation through the invitation workflow. Existing records are not recreated or reassigned.

6.2.4 Session and Offline Authorisation

Session state is separate from account activation and business membership.

A session may be authenticated, expired or terminated. An authenticated session does not override a revoked business membership.

Offline availability additionally depends on:

- previous successful online authentication;
- locally available authorised data;
- an unexpired offline authorisation;
- an operation explicitly supported offline.

Offline work does not renew its own authorisation. Renewal requires online validation.

A password reset or permission change may not be known immediately to a disconnected device. The offline authorisation limit bounds that period.

6.2.5 Properties and Parcels

Properties and parcels distinguish an active record from an archived record.

An active parcel is operationally available only when its parent property is also active.

Current	Action	Result	Conditions
Active	Archive	Archived	Authorised operation with connectivity. History is preserved.
Archived	Reactivate	Active	Authorised operation with connectivity. Parcel availability also depends on the parent property.
Existing	Delete	Removed	No associated records, relevant permission and explicit confirmation.

Archiving a property does not delete or silently rewrite its parcel records. It makes the parent context unavailable for new operational use.

Reactivating the property does not automatically reactivate parcels that were independently archived.

Deletion is a terminal operation, not a reversible archival state.

6.2.6 Campaign Lifecycle

Campaign lifecycle distinguishes open and closed campaigns.

A reopened campaign is open again; reopened is a transition outcome, not a separate permanent lifecycle state.

Current	Action	Result	Conditions
Not created	Create	Open	Required campaign details, valid coverage and authorisation.
Open	Close	Closed	Actual end date confirmed and closure validation completed.
Closed	Reopen	Open	Reopening permission, date review and conflict validation.
Existing	Delete	Removed	No operational records, deletion permission and confirmation.

The distinction between a campaign that has not yet started and one with actual work is represented through its planned and actual information.

A separate planned or in-progress lifecycle state is not introduced by this specification.

6.2.7 Campaign Archival

Campaign archival is independent of the open or closed lifecycle state.

Therefore, a campaign may be:

- Open and visible in the usual lists.
- Open and archived.
- Closed and visible in the usual lists.
- Closed and archived.

Archiving or unarchiving changes presentation and discoverability in the usual lists. It does not itself permit or prohibit work that would otherwise be governed by lifecycle, area availability or permission rules.

Archived campaigns remain accessible through authorised historical consultation. Their coverage remains relevant to overlap validation.

6.2.8 Worker Availability

Worker availability distinguishes active and inactive records.

Current	Action	Result	Conditions
Active	Inactivate	Inactive	Authorised connected operation. History is preserved.
Inactive	Reactivate	Active	Authorised connected operation. The existing identity is reused.

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Current	Action	Result	Conditions
Existing	Delete	Removed	No associated records, permission and explicit confirmation.

Not being planned or present on a date does not make a worker inactive.

Existing future assignments are not silently removed by inactivation.

6.2.9 Daily Attendance

Attendance has three business states:

Awaiting confirmation. No attendance decision has been confirmed. This is not equivalent to absence.

Present. The worker attended. Actual hours and tasks may still be incomplete.

Absent. Absence was explicitly confirmed. Contradictory actual work must not remain.

Initial confirmation can change awaiting confirmation to present or absent.

Changing an already confirmed decision follows the correction workflow. A present-to-absent correction requires explicit review of the affected work removal.

Attendance is recorded once per worker and working date within the business. It is not duplicated for each campaign, property, parcel or task.

6.2.10 Work Information Completeness

Completeness is separate from attendance.

Incomplete. Required actual-work information for a complete result is still missing.

Complete. Required actual-work information has been supplied and validated.

These descriptions do not independently introduce a mandatory end-of-day closing action.

Completeness must be evaluated for the relevant record or calculation. Known confirmed periods may contribute to a clearly labelled partial total. They must not imply that all daily hours are known.

6.2.11 Synchronisation

Operations distinguish the following states:

Pending. Saved locally and awaiting submission or server acknowledgement.

Synchronising. A submission is in progress.

Accepted. The server has accepted the operation.

Failed. The operation did not complete. The interface identifies the failure and the appropriate retry or correction action.

Conflicted. Concurrent values require authorised resolution.

A failed transmission with an uncertain outcome must not be treated as proof that the server made no change. Retry handling must check the operation outcome and avoid duplicate application.

A server validation rejection must be distinguished from a temporary connection failure.

A worker may simultaneously be present, have incomplete work information and have a locally saved operation pending synchronisation. These dimensions are not a single combined state.

6.2.12 Document Categories and Version Currency

Document category determines its version-retention policy.

Within a document, a retained version is identified as current or previous.

For contracts, older retained versions remain individually consultable.

For insurance and other documents, versions older than the immediately previous are deleted after a new version is accepted.

Current-version status does not determine signature completion, insurance validity or eligibility for operational use.

6.2.13 Withdrawal

A retained version may be available for use or withdrawn.

Withdrawal prevents the version from being presented as operationally usable. It does not change its signature information to unsigned.

Withdrawing the current version does not silently restore an earlier version as current. The restoration procedure remains an implementation and workflow decision to finalise.

A withdrawn insurance or other-document version remains subject to rolling retention.

6.2.14 Insurance Validity

Insurance consultation distinguishes:

Valid. A recorded expiry date has not passed.

Expired. The recorded expiry date has passed.

Validity unconfirmed. Expiry information is missing.

These states describe the recorded expiry information. They do not establish independent verification of the insurance coverage.

An expired file may still be consulted by an authorised user.

Withdrawal is separate from expiry: a withdrawn file must not appear operationally usable merely because its recorded expiry date is in the future.

6.2.15 External Signature Information

An uploaded version may have no external-signature declaration or may be declared signed externally.

That declaration is not a completed application signing operation.

Where the external source does not establish individual signer information, the system must not infer that all associated workers signed it.

6.2.16 Application Signing

Signing is tracked per worker and document version.

No confirmed application signature. No drawing has been explicitly confirmed through the application for that pair.

Locally confirmed and pending. The worker confirmed the result, which is stored locally but not yet accepted by the server.

Accepted. The server accepted the signing result for the reviewed worker and version.

Resolution required. The submission cannot be completed without authorised intervention, for example because the original version was withdrawn or deleted.

Transmission progress and failure are synchronisation conditions, not alternative document identities.

An uncertain submission outcome must be checked before another signing event is initiated.

Cancelling before confirmation creates no completed signing result.

6.2.17 Transport and Delivery States

A transport and its associated delivery use the following operational states:

In transport
Departure has been recorded, but delivery has not yet been confirmed.

Delivered
Delivery has been explicitly confirmed and the delivery date and time have been recorded.

The normal transition is:

$$\text{In transport} \longrightarrow \text{Delivered}$$

A retrospective registration may create a record directly in the *Delivered* state, provided that both departure and delivery information are supplied and validated.

Correcting a delivery timestamp does not automatically change its operational state. Reversing an incorrect delivery confirmation is not defined by this state model and requires a specific correction procedure before implementation.

Weighing, analysis and financial completeness are separate from delivery state. A delivered load may still be awaiting a ticket, analysis results or a reported amount.

6.2.18 Weighing and Analysis Completeness

The availability of weighing and analysis information is represented separately:

Not recorded
: No corresponding ticket or analysis information has been entered.

Incomplete
: A record exists, but some information remains unavailable or its interpretation is unresolved.

Available
: The expected information for the record has been supplied. Eligibility for a particular calculation still depends on the required meanings, units and selection rules.

These are completeness indicators rather than approval states. An attachment alone does not establish that all structured information is available or suitable for calculation.

Required completeness fields remain pending under Q-DA-08.

6.2.19 Reported Amount States

A ticket's financial information is represented as follows:

Not reported
: No monetary amount has been supplied. This condition is distinct from a reported zero amount.

Provisional
: The current amount has been communicated as provisional.

Final
: The current amount has been communicated as final.

The expected progression is:

$$\text{Not reported} \longrightarrow \text{Provisional} \longrightarrow \text{Final}$$

An amount may also be recorded directly as final:

$$\text{Not reported} \longrightarrow \text{Final}$$

When a final amount replaces a provisional amount, only the final amount remains current.

Correcting an amount does not create a second financial record to be included in totals. Reclassification of a final amount as provisional is not assumed to be a normal transition and remains pending under Q-DA-09.

6.2.20 Synchronisation Status

Operational state, information completeness and synchronisation status are independent dimensions.

A locally confirmed delivery may therefore be *Delivered* while still awaiting server acceptance. Likewise, a locally recorded ticket or analysis may be available on the device but absent from the accepted central dataset.

The synchronisation states defined for other offline operations also apply here. Pending or conflicted changes must remain identifiable, and a failed transmission must not be treated as proof that the server rejected the record.

6.2.21 Indicator Availability and Freshness

Indicators are derived views rather than independently editable business records. They do not have a separate approval or campaign-closure state.

The interface must distinguish the following conditions:

Available
: The indicator can be calculated from eligible information within the displayed scope.

Partial
: A result is available, but some relevant information is missing, unallocated or outside the user's authorised or locally available scope.

Unavailable
: The required information or approved calculation rule is absent. No numerical result is asserted.

Provisional due to local changes
: The displayed result incorporates operational changes awaiting server acceptance.

Based on last synchronisation
: The result uses locally available information and may exclude subsequent changes made elsewhere.

These conditions may coexist. For example, an offline result may be partial and include pending local changes.

A provisional financial amount is a classification of the reported source amount. It is distinct from an indicator being provisional because it includes unsynchronised changes.

A zero result may be displayed only when it is supported by the available scope and source information. Absence of locally available records does not establish that the campaign has no records.

6.2.22 PDF Report Snapshot

A generated PDF represents the authorised information available at a specific generation date and time.

Its content does not change when source records are subsequently updated. A new export is required to obtain an updated snapshot.

An offline report must identify its synchronisation limitations and any pending local changes included in operational indicators. Financial information is excluded.

6.3 Corrections and Change History

6.3.1 Changes to Account Access

Business permission changes take effect for subsequent online requests after they are saved.

Permission changes do not rewrite previously accepted operational records or change their recorded responsible account.

Revoking business access is distinct from deleting an account or deleting business information.

Pending offline operations. Pending operations retain their originating account and business context.

On reconnection, the server revalidates current authorisation and the applicable business rules before accepting them.

A proposal that is no longer authorised must be identified as unresolved or rejected. It must not be accepted solely because it was created before revocation.

The application must preserve enough protected local information to avoid silent loss while the disposition of the proposal is resolved. A manual resolution procedure for proposals left after revocation must be agreed before this workflow is deployed.

Shared devices and logout. Logout must prevent further access through the local session. It must not silently delete unsynchronised work.

Retained local information must remain associated with the originating account and inaccessible to other accounts.

Logging back in does not automatically authorise submission: the account must still hold the required business permissions.

Account information changes. Changing a registered email, changing business ownership and deleting an account are not covered by an ordinary password reset.

These operations require separate identity and data-lifecycle rules if included in the project scope.

Security records. The decision not to retain permanent field-change histories for ordinary business corrections does not prohibit security event records.

Authentication attempts, password resets, invitation acceptance, permission changes and revocation should be addressed by the security and retention requirements. Passwords, session secrets and reset tokens must not be stored in event logs.

6.3.2 Identity and Associations

Property, parcel and campaign identities must remain stable when names or descriptive information are corrected.

A name change must not create a new identity, reassign historical work or change the provenance of existing deliveries.

Creating parcels within an existing property must not automatically redistribute historical property-level records.

6.3.3 Campaign Coverage Changes

Valid coverage may be added after campaign creation, subject to permissions and conflict checks.

Direct removal is permitted only when the area has no operational records within that campaign.

If an association with existing activity is wrong, the correction must explicitly address the affected records. It must not be implemented as an unreviewed deletion of coverage.

The detailed exceptional correction process remains pending under Q-PC-03.

6.3.4 Dates and Closure

The planned start is an expectation and may be revised.

Actual dates represent events that occurred and must remain distinguishable from planning information.

Closing preserves operational history and does not require all external results to have arrived.

Correcting confirmed work after closure requires reopening. Late external information follows BR-026.

Unarchiving alone does not provide the effect of reopening.

6.3.5 History Boundaries

Stable record associations and preserved historical operational records do not, by themselves, require a permanent log of every descriptive field change.

This group does not introduce a detailed property or campaign field-change log.

Document versions, signatures and other domain-specific historical records follow their own requirements.

6.3.6 Worker Profile Changes

Worker-profile edits retain current values and the last-modified date.

Previous telephone numbers, descriptions and capability values do not require a detailed field-change history.

Profile edits preserve the identity and existing record associations.

Changing machinery capability must not modify historical task attribution. Task-specific hours are derived from the recorded actual periods.

6.3.7 Plans and Actual Work

Planning and actual work remain separate.

A reused plan or actual-work record is a proposal for the target date. It does not confirm attendance, periods, tasks or breaks.

Once actual work exists for the affected daily area context, the corresponding plan remains the reference for what was expected. Differences belong to actual work or the correction workflow.

6.3.8 Correction Permissions and Campaign Status

Work-entry permission does not automatically grant correction permission.

Corrections require authorisation for the affected records and contexts.

Confirmed work in a closed campaign cannot be corrected until authorised reopening. No additional correction deadline applies.

Initial completion of missing earlier work is distinct from replacement of confirmed values. It follows the campaign-entry rules without silently overwriting accepted information.

Completion versus correction. Completion supplies a previously unrecorded or explicitly unknown field without replacing a confirmed value or association. Correction changes or removes confirmed information, including the worker, date, campaign or work-area association.

For a closed campaign, permitted initial completion concerns work performed within its valid dates. An initial work record for a date with no confirmed work in that campaign may be entered under those rules. Once work has been confirmed for a date in that campaign, adding previously unrecorded periods requires authorised reopening. Filling an unknown field must not indirectly replace confirmed information or bypass this rule.

Correction of confirmed work requires authorised reopening under BR-025. The decision on additional periods was confirmed on 23 October 2026; see D-WF-01 in Chapter 10.

6.3.9 Correction Scope

A correction may change:

- Attendance.
- Actual start and end information.
- Task type.
- Break timing and counting treatment.
- Worker association.
- Working date.
- Campaign, property or parcel association.

Origin and destination permissions, campaign coverage, lifecycle status, dates and time conflicts must be validated.

Moving one period must not silently remove or duplicate daily attendance. Other work performed by the original worker on that date remains relevant.

6.3.10 Presence Changed to Absence

The system identifies actual campaign-area assignments, including those without hours, and all periods and breaks associated with the worker and working date.

The user must explicitly confirm their removal before the attendance decision becomes absent.

The operation must satisfy permissions and campaign rules for all affected work. It must not remove only an authorised subset and leave an absent worker with contradictory confirmed assignments, periods or breaks elsewhere.

Cancelling preserves the existing attendance and work records.

6.3.11 Current Values and Conflict Resolution

Accepted corrections retain current values only, without a permanent field-change history.

Offline corrections remain pending until server validation.

When concurrent values exist, an authorised user compares the current server values and pending correction.

The user may retain the server values or review and confirm a correction. The resulting operation is validated again.

The comparison is temporary workflow information, not a permanent correction log.

Local totals identify pending contributions. Server-rejected proposals are not presented as part of accepted central totals.

6.3.12 Metadata Changes and New Versions

Correcting document metadata is distinct from replacing the document contents.

A different source file is handled as a new document or a new version, according to the reviewed user action.

Accepted signed contents must not be overwritten through metadata editing.

A new version requires its own signature information. Neither a prior application signature nor an external-signature declaration is inherited automatically.

6.3.13 Shared Documents

A shared document uses one versioned source with several worker associations.

A version update therefore applies to the shared document rather than creating independent versions in each worker profile.

Application-captured signatures remain individual. Consultation presents the selected worker's signature result.

The source file itself may already contain personal information or external signatures. Hiding application-captured signatures does not redact the source file. Its access must remain consistent with the approved sharing scope.

6.3.14 Incorrect Uploads

An authorised user may delete an incorrect unsigned upload without dependent records, after explicit confirmation.

Signed incorrect versions are withdrawn from operational use while retained.

Withdrawal does not erase or relabel the existing signature.

The rolling deletion policy for insurance and other documents remains applicable, including to signed or withdrawn older versions.

6.3.15 Retention Cleanup

Retention cleanup runs only after a replacement version is successfully accepted.

Cancellation, failed validation, failed upload and duplicate retry must not trigger extra version deletion.

Cleanup applies to the old version, its stored source file, application-produced derivative files and version-specific signature data.

Shared associations do not preserve independent copies of a deleted version.

Offline copies receive the deletion through synchronisation. An offline operation must not recreate a deliberately deleted version automatically.

Storage and backup implementation must be aligned with the approved deletion policy.

6.3.16 Confirmed Signing Results

Before confirmation, the worker can clear, redraw or cancel.

After confirmation, the result is not edited by replacing the drawing.

A new document version requires a new review and signing interaction.

If a pending result has an uncertain outcome, the system checks or resumes the existing operation. It must not request repeat signing merely because an acknowledgement was lost.

6.3.17 Version Changes During Offline Signing

A pending result remains linked to the version the worker reviewed.

If that version remains available and eligible, the server may accept the result while identifying that a newer version exists.

If the original version is withdrawn or deleted, the operation requires explicit resolution.

The signature is never transferred automatically to the new version. Any new signing must follow a separate review and confirmation process.

6.3.18 Corrections to Delivery-Related Information

The Owner and appropriately authorised Field Managers may correct delivery-related information within their permitted business scope. Financial corrections additionally require financial permissions.

Corrections update the current record. The application does not maintain a permanent history of previous field values for these operations. Supporting source documents are distinct from a field-change history.

Transport and delivery corrections. Corrections must preserve chronological consistency between departure and delivery and maintain valid associations with the campaign, source areas, driver and vehicle.

Ordinary corrections preserve the campaign and operational state. Changes to source areas within that campaign validate the affected context and dependent ticket, analysis and financial information. They must not invalidate associations or duplicate allocation.

Reversing confirmation, cancellation and campaign reassignment require the exceptional procedure pending under Q-DA-06; ordinary correction permission does not authorise them.

Weighing corrections. A correction may replace an incorrectly entered weight, ticket reference, adjustment or other supported field.

The corrected value must retain its meaning and unit. A correction must not cause an adjustment to be applied twice or convert an unknown value into an assumed zero.

Replacing incorrect ticket information is distinct from recording a genuinely additional ticket. Until ticket multiplicity and replacement rules are confirmed, the system must not automatically treat multiple ticket records as independent quantities to be added.

Analysis corrections. Correcting an analysis updates its current information. A genuinely new analysis represents a separate business event, subject to the final multiplicity and selection rules.

A changed yield percentage or reference weight must trigger recalculation of affected indicators only when the record is eligible under the approved calculation rules.

Monetary corrections. A corrected reported amount replaces the current amount. When a final amount becomes available, it replaces the provisional amount rather than being added to it.

Previously entered amounts are not retained as a permanent revision history. A supporting cooperative document may be associated with the current record, but it does not create an additional amount for aggregation.

Monetary corrections require an internet connection.

Closed campaigns. Campaign closure does not prevent the completion or correction of weighing information, analysis results or reported amounts for existing deliveries and tickets.

This exception does not extend to corrections of actual work, which follow the campaign reopening rule established for work records. It also does not permit new operational activity outside the campaign's authorised period.

Offline corrections and conflicts. Offline corrections to available operational records remain local proposals until accepted by the server.

If the accepted record has changed, the application must present the conflict for review rather than silently overwrite it. The user may retain the accepted information or submit a revised proposal, subject to renewed validation.

Transient information needed to resolve a conflict does not constitute a permanent business correction history.

Effect on indicators. Accepted corrections must update affected indicators. Where offline indicators are supported, local changes must be identified as provisional and applied without duplicating the underlying record.

An update replaces the previous contribution of that record. A rejected proposal must not contribute to accepted central totals.

6.3.19 Corrections Affecting Campaign Indicators

Indicator values cannot be edited directly. An authorised user must correct the underlying attendance, work, delivery, ticket, analysis or reported-amount record through its corresponding use case.

The correction rules of that source record continue to apply. In particular:

- correcting actual work in a closed campaign requires campaign reopening;
- completing or correcting weighing information, analysis results or reported amounts for existing deliveries and tickets does not require reopening;
- monetary corrections require online access and financial permissions.

After an accepted source change, affected indicators must be recalculated using the current eligible information.

Changes to filters or viewing scope do not modify source records.

Pending local changes. Where supported offline, a local proposal modifies the displayed operational result provisionally:

- a new record contributes once;
- an updated record replaces its cached contribution;
- a record pending removal no longer contributes to the local proposed view.

If a proposal is rejected, its contribution must not be presented as accepted central information. Any unresolved local proposal must remain clearly distinguishable.

Synchronisation conflicts are resolved through the relevant source-record workflow. The indicator view does not silently choose which conflicting value becomes authoritative.

Previously exported reports. Correcting a source record does not alter a previously generated PDF. Users must generate a new report to reflect the updated information.

6.4 Calculation Rules

6.4.1 Access Validity and Configuration

Account access does not introduce operational or financial calculation rules.

Time-based validity applies to verification links, password reset links, invitations, sessions and offline authorisations.

Their exact lifetimes must be specified in the non-functional requirements rather than assumed by individual use cases.

An offline authorisation must not remain valid indefinitely. Its renewal requires successful online authentication or session validation.

Configuration and account-scope decisions are tracked under Q-AA-01 to Q-AA-07 in Chapter 10.

6.4.2 Area Information

Declared property and parcel areas are optional values expressed in hectares.

The application must distinguish an unknown area from a recorded numerical value.

A future map-derived area is an estimate with a different source. It must remain separate from the owner's declared reference area.

This specification does not assume that all of a property is divided into parcels or that parcel-area totals must equal the declared property area.

No automatic redistribution of operational quantities is derived from parcel areas.

6.4.3 Coverage Overlap

For periods whose overlap can be determined:

- The same property or parcel cannot have conflicting campaign coverage.
- A whole property conflicts with its own parcels during the same period.
- Different parcels of the same property may belong to simultaneous campaigns.
- Duplicate whole-property and parcel selection within one campaign is prevented.
- Changing parcel definitions must not bypass existing coverage.

The handling of campaigns without a known end date remains pending under Q-PC-01.

No default end date or unrestricted overlap is assumed merely to complete the validation logic.

Calculation dependencies. Campaign overlap, actual-start recording and coverage corrections remain dependent on Q-PC-01, Q-PC-02 and Q-PC-03 respectively.

6.4.4 Interval Duration

A period's duration is calculated from its complete end and start timestamps.

If the timestamp difference is expressed in seconds, duration in hours is:

$$D_i = \frac{t_{\text{end},i} - t_{\text{start},i}}{3600}$$

The end must be later than the start. An earlier clock time is not sufficient to establish the end date automatically.

A period from 22:00 to 02:00 on the following day has a duration of four hours and belongs to its start date.

Overlap validation considers full intervals, including periods beginning on adjacent dates.

6.4.5 Counted Working Time

Let W_i be an eligible actual work period and let $w(i)$ identify the worker to whom that period belongs.

For each worker w, let B_w be the union of that worker's applicable break intervals explicitly marked as non-counted.

The counted duration of period i is:

$$H_i = \text{duration}(W_i) - \text{duration}\left(W_i \cap B_{w(i)}\right)$$

Total counted hours within the selected scope are:

$$H_{\text{total}} = \sum_i H_i$$

Only breaks belonging to the same worker may reduce a work period's counted duration. Using the union prevents duplicate deduction of the same time.

Breaks marked as counted do not reduce hours. Time outside recorded work periods is already excluded and must not be deducted again.

Actual periods for the same worker must not overlap. Validation uses complete start and end timestamps, including periods crossing midnight.

6.4.6 Task and Area Attribution

Each actual work period identifies its campaign and work area. Where the actual task is known, it is associated with the period. A task that remains unknown is explicitly identified as unclassified.

A valid period with an unknown task may contribute to the overall counted-hours total, but its hours remain separate from named task totals. Task totals use the counted duration of the relevant periods.

A non-counted break affecting consecutive periods reduces each period only by the overlapping duration applicable to the same worker. Property and parcel breakdowns use the same eligible periods without duplicating their contribution.

6.4.7 Acceptance Examples

Confirmed Information	Expected Result
08:00–18:00, with no non-counted break.	10 counted hours.
08:00–18:00, with 12:00–13:00 marked as a counted break.	10 counted hours.
08:00–18:00, with 12:00–13:00 marked as a non-counted break.	9 counted hours.
08:00–12:00 and 13:00–18:00.	9 counted hours. The gap is not deducted again.

Continued on next page

Confirmed Information	Expected Result
08:00–11:00 machinery operation; 11:00–13:00 general work.	3 machinery hours and 2 general-work hours.
22:00–02:00 the following day, without deductions.	4 hours, associated with the start date.
Workers A and B both work 08:00–12:00. Only A has a non-counted break, 10:00–10:30.	A: 3.5 hours; B: 4 hours. Total: 7.5 hours.
08:00–10:00 with a valid campaign-area association and an unknown task.	2 total hours, presented as unclassified; no allocation to a named task.

6.4.8 Planned and Incomplete Information

Planned periods must not enter actual-hour totals without explicit confirmation.

Attendance without completed periods does not establish zero worked hours.

Known confirmed periods may support a partial total, provided missing information is identified.

A continuous planned period does not, by itself, establish actual counted hours.

6.4.9 Recalculation and Synchronisation

Accepted local entries or corrections update the affected local totals.

Updates replace the corresponding local baseline; they are not added as duplicate contributions.

Where records move between workers, dates, tasks, campaigns or areas, both origin and destination totals are recalculated.

Pending results remain distinguishable from server-accepted results.

If the server rejects a proposal, accepted totals must be recalculated from accepted information. Any unresolved proposal remains separately identified.

Other devices receive accepted changes through synchronisation.

No wage calculation is derived from these working-time formulas.

Configuration dependencies. Precision, timestamps, break validation and offline recovery are tracked under Q-WF-01 to Q-WF-04.

6.4.10 Insurance Expiry Status

Insurance status uses the recorded expiry date and the current date.

- If no expiry date is recorded, validity is unconfirmed.
- If the expiry date has passed, the insurance is displayed as expired.
- If the expiry date has not passed, the recorded expiry status is valid.

The application must use consistent date handling for online and offline display.

The latest-uploaded or current version is not automatically the valid version.

6.4.11 Version Retention Examples

Category	Accepted Versions	Retained Result
Contract	v1, v2, v3	v1, v2 and v3 retained.
Insurance	v1, v2	v1 and v2 retained.
Insurance	v1, v2, then accepted v3	v2 and v3 retained; v1 deleted even if signed.
Other document	v1, v2, then accepted v3	v2 and v3 retained; v1 deleted even if signed.
Insurance	v1, v2, failed v3 upload	v1 and v2 unchanged.
Shared insurance	Accepted v3 replaces current v2	All associated worker views refer to the same retained v2 and v3.

The table assumes ordinary sequential version acceptance. Withdrawal must not silently reorder which version is current.

6.4.12 File Size Validation

Each uploaded file is checked individually against the proposed 10 MB limit.

The implementation must define the exact byte threshold consistently across the client and server.

The limit applies to supported uploaded files. Limits for generated signing outputs must be specified separately if required.

6.4.13 Signing Completion

Signing completion is evaluated for the pair of worker and document version.

A shared document must not use one global signed flag to imply that every worker signed.

A local confirmation is not included as a server-accepted result before acknowledgement.

No monetary amount, wage or payment is calculated by document signing.

Document dependencies. Operational signing, withdrawal recovery, offline protection, pending signatures and generation scope remain subject to Q-DOC-01 to Q-DOC-06.

6.4.14 Weight Interpretation and Aggregation

All recorded weights must have an identified unit and meaning. Values must be converted to a common unit before any permitted aggregation.

Let E_w denote the set of current weighing records that are eligible for a particular indicator. A total may only be calculated as:

$$W_{\text{total}} = \sum_{i \in E_w} w_i$$

when all included values:

- represent the same weight concept;
- use compatible units;
- refer to distinct, non-duplicated quantities;
- satisfy the approved ticket selection rules.

The existence of several ticket records does not by itself establish that their weights are additive.

A gross weight must not be combined indiscriminately with a net or final weight. Missing weights are excluded from the relevant calculation and reported as missing coverage; they are not treated as zero.

Tare and adjustments. The application shall not assume that every reported final weight must be recalculated from gross weight and tare.

Likewise, no automatic deduction formula is established until the business confirms:

- the meaning and unit of each adjustment;
- the value to which it applies;
- whether it is already included in the reported result;
- the order of application when several adjustments exist;
- the applicable rounding rules.

Until these decisions are approved, the system records the reported values and their known interpretation without inventing a derived final weight.

6.4.15 Oil Yield Aggregation

A campaign-level oil yield indicator must not be calculated as an unqualified arithmetic average when deliveries represent different quantities.

The candidate weighted-average model is:

$$\bar{r} = \frac{\sum_{i \in E_a} w_i r_i}{\sum_{i \in E_a} w_i}$$

where:

- E_a is the set of eligible, non-duplicated analysis results;
- r_i is the reported oil yield percentage for analysis i;
- w_i is the approved reference weight associated with that result.

This formula is a proposed calculation model, not an approved business rule. Its use depends on confirmation of the reference weight and analysis selection rules.

The indicator is unavailable when no eligible reference weight exists or the denominator is zero.

If only part of the campaign has eligible analysis information, the application must identify the result as partial and disclose the available coverage. Missing results must not be represented as a zero yield.

Any theoretical oil quantity derived from weight and yield must be labelled as an estimate and must not be presented as oil actually received. Its inclusion and calculation remain subject to approval.

6.4.16 Reported Monetary Totals

Let E_f denote eligible tickets whose current reported amount is final, and E_p eligible tickets whose current reported amount is provisional.

The corresponding totals are:

$$V_{\text{final}} = \sum_{t \in E_f} a_t$$

$$V_{\text{provisional}} = \sum_{t \in E_p} a_t$$

where a_t is the single current amount reported for ticket t.

The two totals must be identified separately. A provisional amount and the final amount that replaces it must never contribute simultaneously.

Eligibility depends on confirming whether multiple tickets for one delivery represent separate amounts, replacements or another business relationship. Until that relationship is known, the application must not assume that all ticket amounts can be summed.

Tickets with no reported amount are identified separately and excluded from monetary totals. An explicitly reported zero amount is a known value.

Amounts are recorded in EUR with two decimal places under the current scope. Financial calculations must preserve decimal monetary precision.

No weight adjustment, quality adjustment or other deduction is automatically reapplied to a cooperative-reported amount.

6.4.17 Mixed-Source Deliveries

When a delivery contains olives from multiple properties or parcels, the known delivery-level quantity may contribute once to an eligible campaign-level total.

It must not be attributed in full to every source property or parcel.

Property-level or parcel-level allocation requires either:

- quantities explicitly reported for each source; or
- an allocation method approved by the stakeholders.

Where no allocation is available, the application must retain the delivery-level information and identify the quantity or amount as unallocated in the relevant breakdown.

An unallocated value is not equivalent to a zero value.

Calculation dependencies. Eligibility and calculation rules remain subject to Q-DA-01 to Q-DA-08. Resolve the applicable decisions before implementing calculations or transitions that depend on them.

6.4.18 Distinct Campaign Participants

Let P_C be the set of workers who have confirmed daily presence and an actual assignment to campaign C, within the applicable viewing scope.

The participant count is:

$$N_{\text{workers}}(C) = |P_C|$$

A worker participating on several days or in several properties of the same campaign is counted once in the campaign total.

Participation does not require complete work-hour information. Where hours are incomplete, the worker may contribute to the participant count while the hour-related limitation is disclosed.

A driver association alone does not establish participation: attendance and actual campaign assignment must be recorded through the corresponding work-recording process.

6.4.19 Actual Working Days

Let E_C be the set of eligible actual work periods for campaign C, and let d_i be the start date of period i.

The number of actual working days is:

$$N_{\text{days}}(C) = |\{d_i \mid i \in E_C\}|$$

Several workers or periods on the same date do not increase the campaign working-day count beyond one for that date.

A period crossing midnight belongs to its start date. Date filters for work-day indicators use this same convention.

An actual assignment with no recorded work period may establish participation, but does not establish a complete duration or an additional measured working day.

6.4.20 Worked Hours by Worker and Task

Let h_i be the counted working duration of actual work period i, calculated using the work-period and break rules in Section 6.4.5.

Total campaign hours are:

$$H_C = \sum_{i \in E_C} h_i$$

For worker w:

$$H_{C,w} = \sum_{\substack{i \in E_C \\ \text{worker}(i)=w}} h_i$$

For task f:

$$H_{C,f} = \sum_{\substack{i \in E_C \\ \text{task}(i)=f}} h_i$$

These formulas reuse the work-period calculation rules defined earlier in this chapter. They do not introduce a second break deduction.

Hours are attributed to the task actually recorded for each period, rather than to the worker's general capability.

Hours without a recorded task remain identified as unclassified in the task breakdown. They must not be silently omitted from an otherwise eligible overall total.

Incomplete intervals are excluded from duration calculations and disclosed as missing coverage.

Worked-hour indicators do not calculate wages or payroll.

6.4.21 Confirmed Delivery Count

Let D_C be the set of distinct confirmed deliveries for campaign C, within the applicable scope.

The delivery count is:

$$N_{\text{deliveries}}(C) = |D_C|$$

Associated tickets, analyses and attachments do not increase the number of deliveries.

A registered departure that has not been confirmed as delivered does not contribute to this count.

6.4.22 Weight, Yield and Monetary Indicators

Weight totals, oil yield aggregation and reported monetary totals follow the eligibility and calculation rules defined in the preceding delivery-related subsections.

The indicator view must not introduce alternative formulas or assumptions to bypass unresolved business decisions.

In particular:

- weights with different meanings must not be combined as though they represented the same quantity;
- multiple tickets or analyses must not be assumed to be additive;
- a yield indicator requires an approved reference weight and selection rule;
- current provisional and final monetary amounts are presented separately;
- a replaced amount does not contribute again;
- a theoretical oil quantity is not presented as oil actually received.

6.4.23 Breakdowns and Non-Additive Counts

Property and parcel breakdowns must preserve the identity of the underlying records.

Distinct-worker and distinct-day counts are not necessarily additive across properties or parcels. The same worker or date may occur in several breakdowns but must be counted once in the campaign total.

Mixed-source quantities and amounts without an approved allocation remain unallocated at the relevant breakdown level. They must not be duplicated to force detailed values to match a campaign total.

The interface must explain any difference caused by unallocated information or non-additive distinct counts.

6.4.24 Filters and Reporting Scope

Each indicator must identify the filters that actually apply to it.

Work indicators use actual work dates and applicable worker, task and area associations. Confirmed-delivery counts use delivery dates.

The date basis for ticket, analysis and financial indicators must be labelled explicitly. The choice between delivery date, source-document date and reporting date remains subject to confirmation before date-filtered versions of those indicators are implemented.

A filter without a supported relationship to an indicator must not silently alter that indicator.

Each PDF report must identify:

- the selected campaign;
- the authorised reporting scope;
- the applicable filters and date basis;
- the generation date and time;
- relevant missing or unallocated information;
- unresolved calculation limitations;
- the last successful synchronisation, where relevant;
- any included pending local changes.

Detailed spreadsheet export and cross-campaign comparison are outside the current baseline.

7 Non-Functional Requirements

7.1 Performance

7.2 Usability and Accessibility

7.3 Security

Password policy, authentication request limits, session expiry and verification, invitation and password-reset lifetimes remain to be specified. Security-event retention must also be defined. No numerical values are assumed at this stage.

7.4 Privacy

7.5 Reliability and Recovery

7.6 Availability and Offline Continuity

The maximum offline authorisation duration and the handling of expiry with pending changes remain to be specified. Renewal requires online validation. Reconnection revalidates current business access and permissions before pending operations are accepted.

8 Solution Architecture

8.1 System Context

8.2 Components and Responsibilities

8.3 Main Interaction Flows

8.4 Offline Operation and Synchronisation

8.5 Deployment and Operation

8.6 Architecture Decisions

9 Data Architecture

9.1 Conceptual Data Model

9.2 Logical Data Model

9.3 Data Dictionary

9.4 Integrity Constraints

9.5 Data Lifecycle and History

9.6 Local and Central Data

9.7 Physical Data Model

10 Validation, Decisions and Open Questions

This chapter records the confirmed work-entry clarification and centralises outstanding decisions. Non-functional requirements, solution and data architecture and the detailed validation plan remain pending development and agreement.

10.1 Acceptance Criteria

The functional catalogue supplies observable acceptance criteria. Where a criterion cites a pending decision, that dependent aspect remains provisional until validation. Catalogue inclusion does not establish MVP approval or implementation completion.

10.2 Requirements Traceability

Section 5.2 maps every functional requirement to its originating use cases. Each developed operational use case also identifies the relevant business-rule IDs. Decision IDs link local dependencies to the register below. The detailed acceptance-evidence plan remains to be developed.

10.3 Decision Log

D-WF-01 — Additional periods in a closed campaign

Status: confirmed on 23 October 2026 by the report requester. When work is already confirmed for a working date in a campaign, adding previously unrecorded periods for that date requires authorised reopening of the closed campaign. The definition in Section 6.3.8 distinguishes this from filling an unknown field without replacing confirmed information.

Traceability: UC-06, UC-10 and UC-11; FR-062 and FR-063; BR-025 and BR-042. Other pending decisions are not approved by this clarification.

10.4 Open Questions

All entries in this register have status *Pending confirmation.* The validation roles stated for each group are proposed responsibilities; named owners and outcomes remain to be recorded after agreement. Referencing a pending decision does not approve a capability or formula.

10.4.1 Accounts and Access

Proposed validation responsibility: Project team and client for role, membership and lifecycle scope; technical team for session and security configuration.

Q-AA-01 — Field Manager role

Confirm whether the role is included in the MVP and which operational responsibilities it covers.

Q-AA-02 — Responsibility scope

Confirm whether a manager's scope is assigned by business, property, campaign or an approved combination.

Q-AA-03 — Multiple business memberships

Confirm whether one account may own or participate in several businesses. The initial

Owner registration creates one business workspace. In UC-04, explicitly confirm how an invitation is handled when its recipient already owns or belongs to another business; a second membership must not be assumed from account creation.

Q-AA-04 — Offline authorisation
Define the maximum offline authorisation duration and the behaviour when it expires with pending changes.

Q-AA-05 — Session and token lifetimes
Define session duration and the lifetimes of verification, invitation and password reset authorisations. Also define simultaneous-device limits, password policy, request limits, email identity normalisation and security-event retention in the non-functional requirements.

Q-AA-06 — Revoked users with pending work
Define how an Owner or authorised support process resolves operational proposals that cannot be submitted after access revocation.

Q-AA-07 — Account lifecycle
Confirm whether email changes, account deletion, business deletion and ownership transfer belong to the project scope. Confirm whether assisted recovery without access to the registered inbox is included; it is not covered by ordinary email recovery.

10.4.2 Properties and Campaigns

Proposed validation responsibility: Project team and client.

Q-PC-01 — Open-ended campaign overlap
Define how temporal coverage is checked when the actual end date is not yet known.

Q-PC-02 — Actual start recording
Decide whether actual start is entered manually or established through another confirmed event. It must not be silently inferred from an unconfirmed plan.

Q-PC-03 — Coverage correction
Define the authorised process for correcting campaign-area associations when operational records already exist.

10.4.3 Workforce and Working Time

Proposed validation responsibility: Project team with the client for work semantics; technical team for precision, timestamps and recovery.

Q-WF-01 — Working-time precision
Define display precision and rounding without changing the underlying interval duration or introducing wage calculations.

Q-WF-02 — Timestamps and time zones
Define consistent timestamp and time-zone handling across devices and the server, including overnight periods and date filters.

Q-WF-03 — Break validation
Define validation and presentation of overlapping, duplicate or out-of-period break entries. The counted-hours formula already limits deductions to the same worker and to actual-period intersections.

Q-WF-04 — Offline work and planning recovery
Finalise the interaction for existing target plans, conflict comparison, rejected changes and recovery of pending work. Preserve the defined manual resolution, account isolation and server revalidation rules.

10.4.4 Documents and Signing

Proposed validation responsibility: Project team and client for document scope and operational validation; technical team for file and offline policies.

Q-DOC-01 — Operational document and signing requirements
Confirm the requirements of each intended document type and validate the selected in-person drawn-signature procedure before operational use. Define document eligibility without assuming an automatic compliance check.

Q-DOC-02 — Restoration after withdrawal
Define the permitted procedure after a current document version is withdrawn. An earlier version must not silently become current.

Q-DOC-03 — Prepared document protection and cleanup
Define offline document access expiry, revocation, storage and deletion propagation in conjunction with Q-AA-04. Confirm the sharing scope of source files that may contain personal information or external signatures; hiding application-captured signatures does not redact the source.

Q-DOC-04 — Pending signatures on removed versions
Define authorised resolution when a locally confirmed signature refers to a withdrawn or deleted original version. It must not be transferred to another version or recreate a deleted version automatically.

Q-DOC-05 — Document generation scope
Confirm whether UC-20 is included, which documents are required, who supplies and approves templates, the required fields and sources, additional employer/worker/area data, permitted document-specific edits, output format, offline need, single/shared-worker scope and when regeneration creates a version or a separate document. No definitive generation requirement is approved by reserving UC-20.

Q-DOC-06 — Worker-document upload limit
Confirm the proposed 10 MB per-file limit and its exact byte threshold. Generated signing outputs require a separately specified limit if needed.

10.4.5 Deliveries, Tickets and Analysis

Proposed validation responsibility: Project team and client, using explained cooperative source examples where needed.

Q-DA-01 — Ticket multiplicity
Can one delivery have several tickets? If so, do they represent additional quantities, replacements, split loads or another relationship?

Q-DA-02 — Weight meanings and adjustments
Which weight values are reported, which value is used operationally, and how are discounts or adjustments expressed and applied? Validate actual cooperative tickets, percentage

bases, application order, already-applied deductions, decimal precision and rounding before approving calculations.

Q-DA-03 — Analysis multiplicity
Can one delivery have several analyses? If so, which results are current and eligible for indicators? Distinguish replacement results from additional measurements; confirm valid ranges and precision using real source examples.

Q-DA-04 — Yield reference weight
Which weight must be used to aggregate oil yield percentages, and what rounding rules apply? Confirm the selection formula and rounding; inclusion of a theoretical oil quantity remains subject to approval and must not imply oil actually received.

Q-DA-05 — Source allocation
Will quantities or monetary amounts be reported separately for mixed-load properties or parcels? If not, should these values remain unallocated?

Q-DA-06 — Exceptional delivery corrections
How should an incorrect delivery confirmation or a delivery assigned to the wrong campaign be corrected when dependent records already exist?

Q-DA-07 — External drivers
Is restricting drivers to registered workers sufficient, or must external drivers also be supported?

Q-DA-08 — Record completeness
Which ticket and analysis fields are mandatory before a record can be considered complete?

Q-DA-09 — Reclassifying a final amount
Confirm whether, and under which authority, a current final valuation may be reclassified as provisional. This is not an ordinary transition assumed by the reported-amount state model.

Q-DA-10 — Transport baseline
Validate the proposed one-transport-per-delivery baseline. Independent trip management remains outside the current proposal unless explicitly approved.

10.4.6 Source Attachments

Proposed validation responsibility: Project team and client for information classification; technical team for access, storage and recovery.

Q-AT-01 — Tickets and attachments with mixed content
Define how ticket, analysis or operational source files containing both operational and monetary information are classified and authorised for offline use. Monetary content remains subject to FR-119 and BR-078. The document does not yet select a file-handling solution; automatic PDF editing, redaction or extraction is not assumed.

Q-AT-02 — Source-attachment lifecycle and recovery
Define file limits, retention, access and pending-upload recovery for ticket, analysis and monetary attachments. They are not worker documents and do not automatically inherit the two-version policy. Preserve the distinction between structured-record storage and attachment-upload status.

10.4.7 Indicators and Reports

Proposed validation responsibility: Project team and client for date semantics; technical team for offline preparation and reporting.

Q-IN-01 — Indicator date basis
Confirm whether ticket, analysis and financial filters use delivery date, source-document date or reporting date. Label each applicable basis explicitly and do not apply unsupported worker/task relationships to those indicators.

Q-IN-02 — Offline indicator and PDF preparation
Specify locally available operational detail, preparation coverage and PDF generation behaviour in the architecture. Reports identify their authorised scope, last synchronisation and included pending changes; offline output excludes financial information.

10.4.8 MVP Prioritisation

Proposed validation responsibility: Project team and client; priority remains a proposal until approved.

Q-MVP-01 — MVP priorities and optional enhancements
Agree mandatory demonstration and operational capabilities, responsibilities, effort and acceptance evidence with the project team and client. Maps, boundaries, area estimates, detailed spreadsheet export and cross-campaign comparisons remain candidate enhancements, separate from the proposed baseline.